Máster de Formación Permanente

Máster en Energías Renovables y Medio Ambiente

20 EDICIÓN 2025-2026

Módulo 6
Autoconsumo Fotovoltaico

Módulos FV de silicon cristalino. Fabricación, defectos, selección

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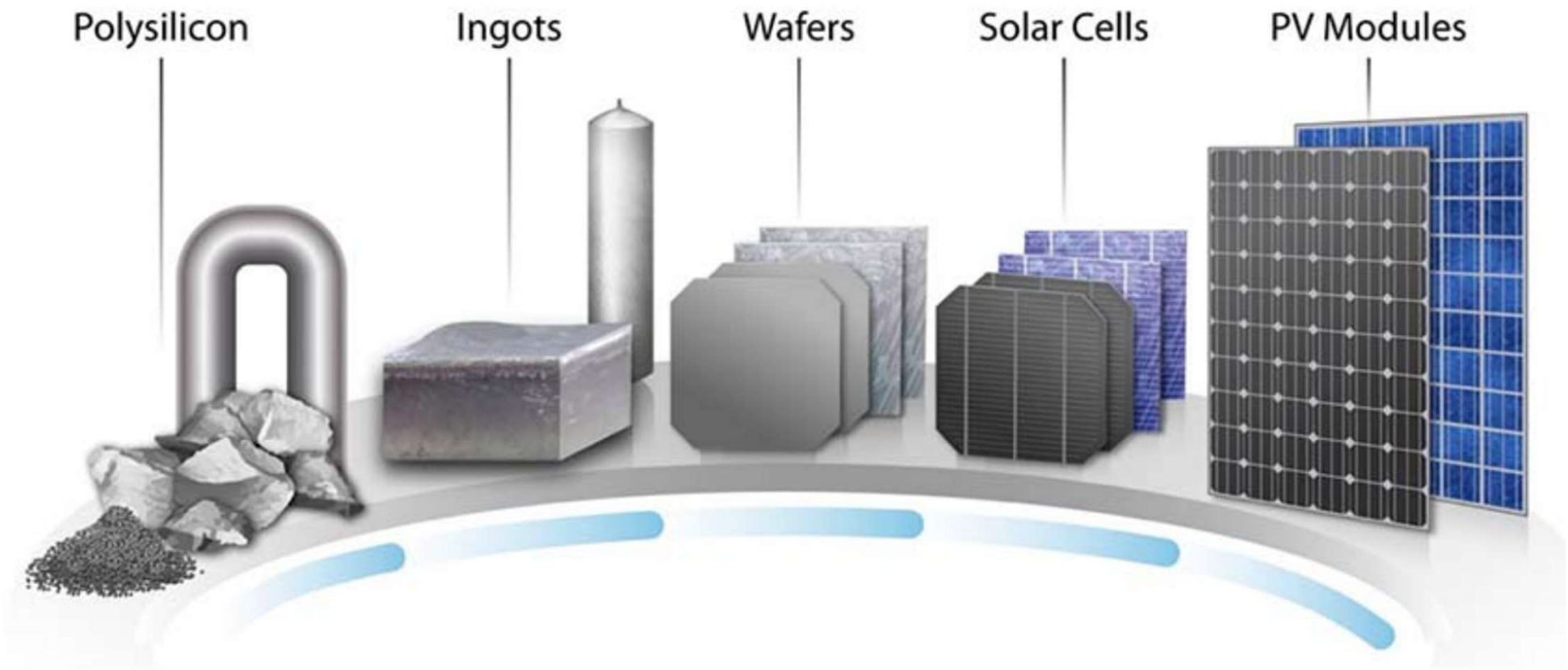
Quality Manager Sonnedix

1. INTRODUCTION. VALUE CHAIN



NREL analysis of manufacturing costs for silicon solar cells includes bottom-up cost modeling for all the steps in the silicon value chain.

1. INTRODUCTION. PRODUCTION COSTS

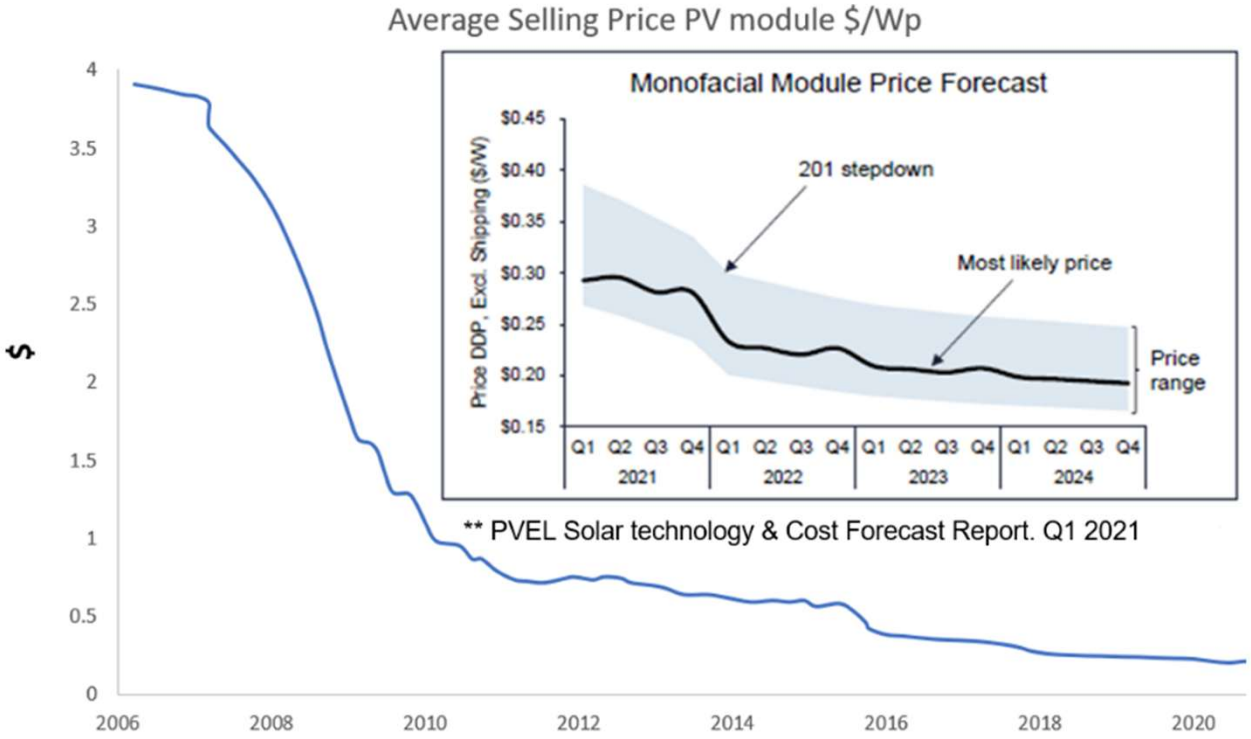
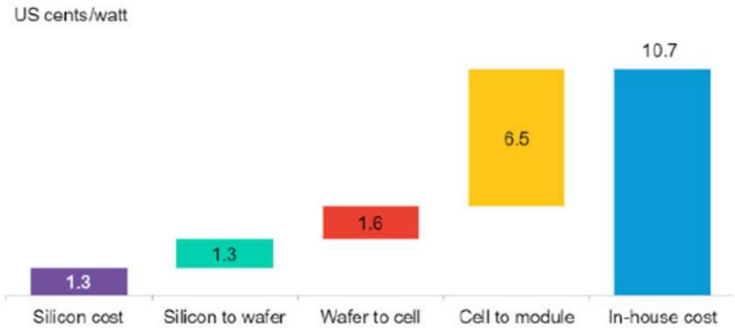


Figure 12: Estimated best-in-class cash cost for each step of the solar module manufacturing process by the end of 2024

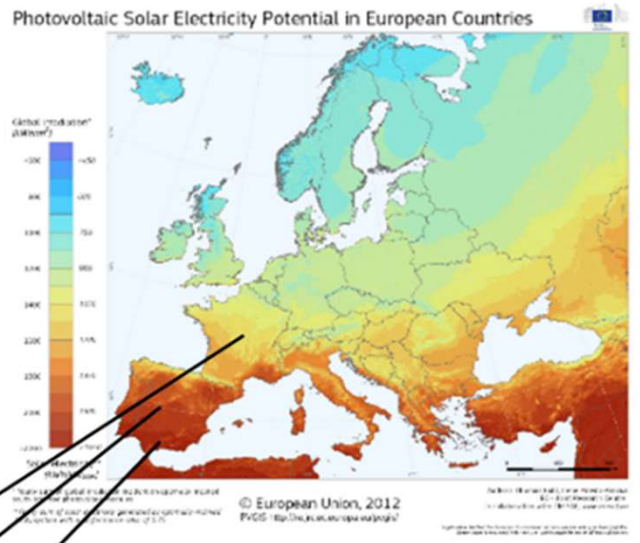


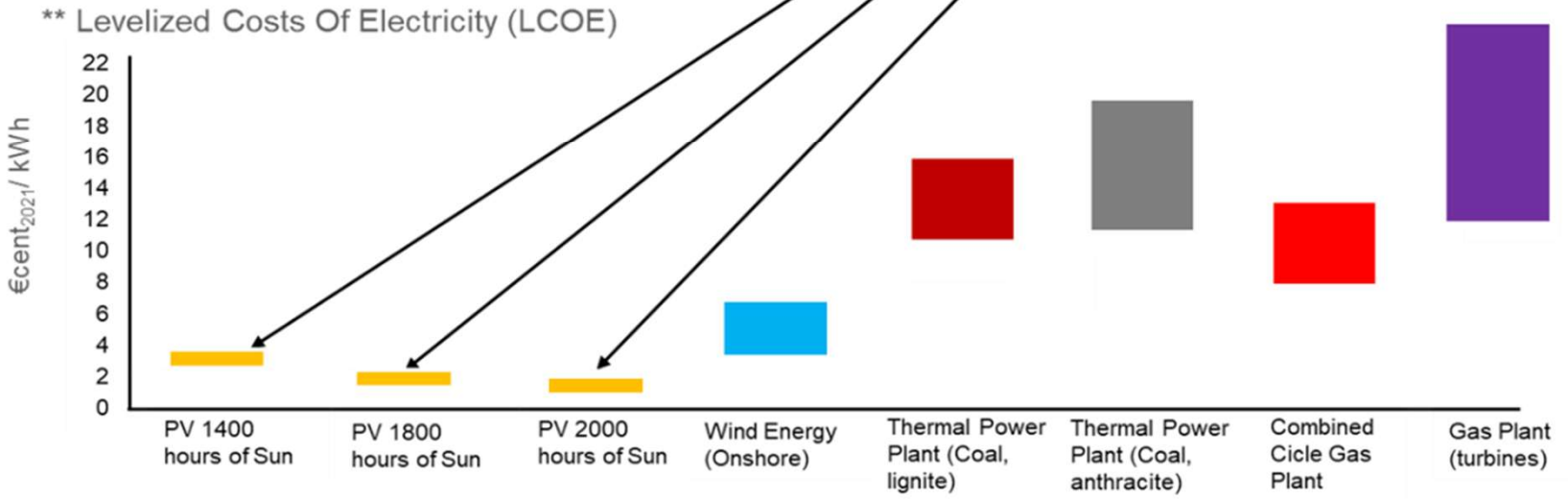
Source: BloombergNEF. Note: Cash cost assumes the in-house production from polysilicon to modules by integrated solar manufacturers. Depreciation and selling, general and administrative expenses are excluded. Polysilicon price assumed to be $6/kg. Assumes 2.14 grams/ polysilicon in 2024. $1=7 yuan when referring to cost level in mainland China factories.

1. INTRODUCTION. LCOE

$$LCOE = \frac{I_o + \sum_{t=1}^{n} \frac{A_t}{(1+i)^t}}{\sum_{t=1}^{n} \frac{M_{t,el}}{(1+i)^t}}$$

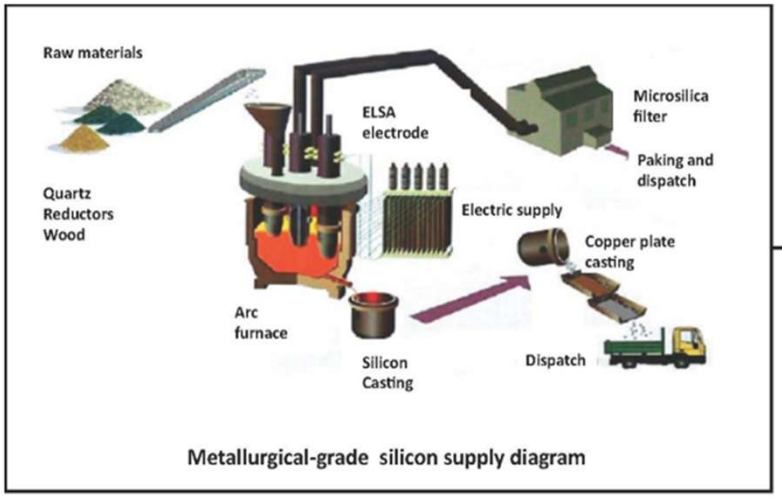
LCOE	Levelized Costs of Electricity in EUR/kWh
I_0	Investment expenditure
A_t	Annual total cost in EUR per year t
$M_{t,el}$	Produced amount of electricity in kWh per year
i	Real interest rate in %
n	Economic lifetime in years
t	Years of lifetime (1, 2, ⋯ n)





3.1. SILICON PURIFICATION. CHEMICAL ROUTE

Production of metallurgical grade silicon (MG-Si) from quartz



Metallurgical-grade silicon supply diagram

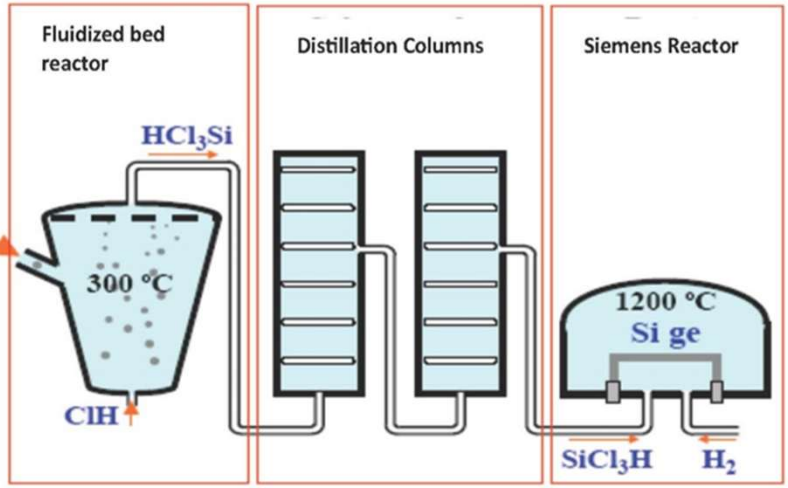
$SiO2 + C \rightarrow Si + CO2$

SiO_2 reduction with coal
2000 ºC
Metallurgical grade silicon (MG-Si)
98% purity

MG-Si

Purity: 98%-99,9%

Indirect/chemical purification → Electronic Grade Silicon (EG-Si)



$Si + 3HCl \rightarrow SiHCl3 + H2$

MG-Si purification
Chemical reaction with
Chloridric acid
300 ºC
SiHCl3

distillation

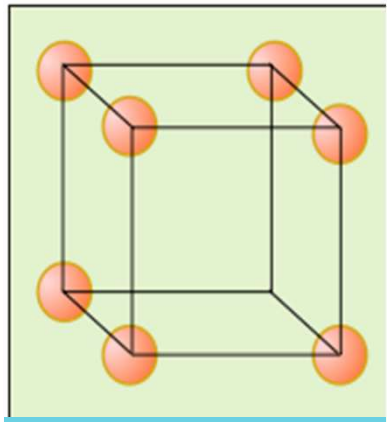
$2SiHCl3 + 2H2 \rightarrow 2Si + 6HCl$

Process Temp 1200 °C
High purity polycrystalline silicon
Ci<1 ppm ($1013\ cm^{-3}$)

EG-Si

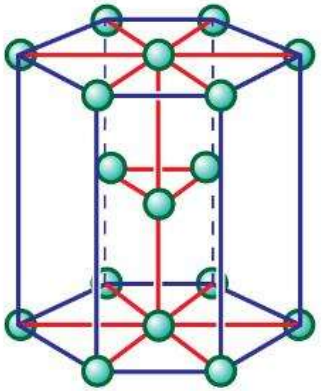
Purity: 99,9999%-99,999999%

3.2. CRYSTAL GROWTH. CRYSTAL



Simple Cubic

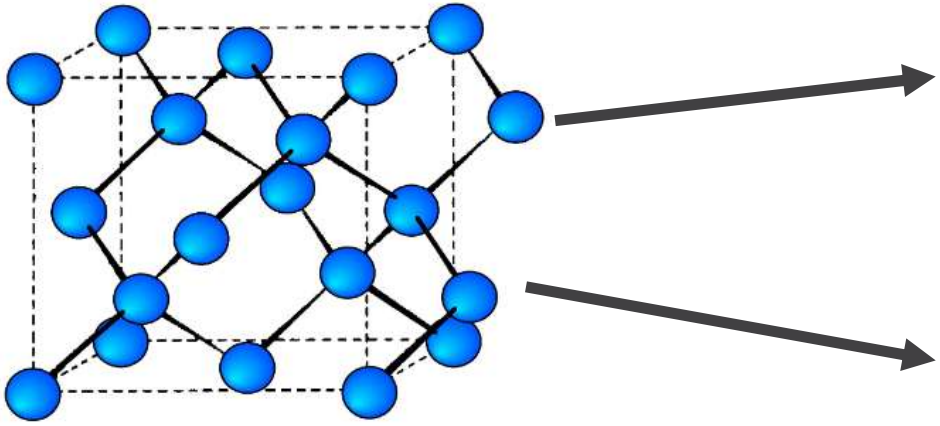
Pyrite



Hexagonal close-packed

Emerald

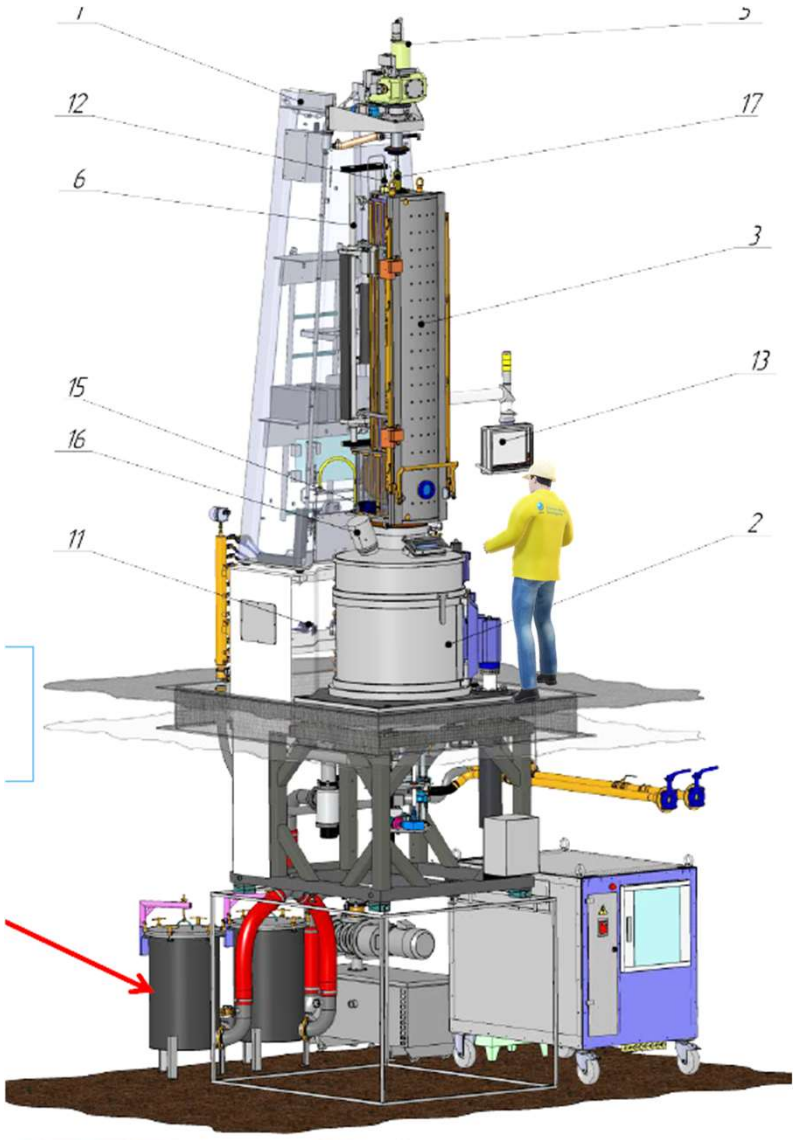
Diamond structure



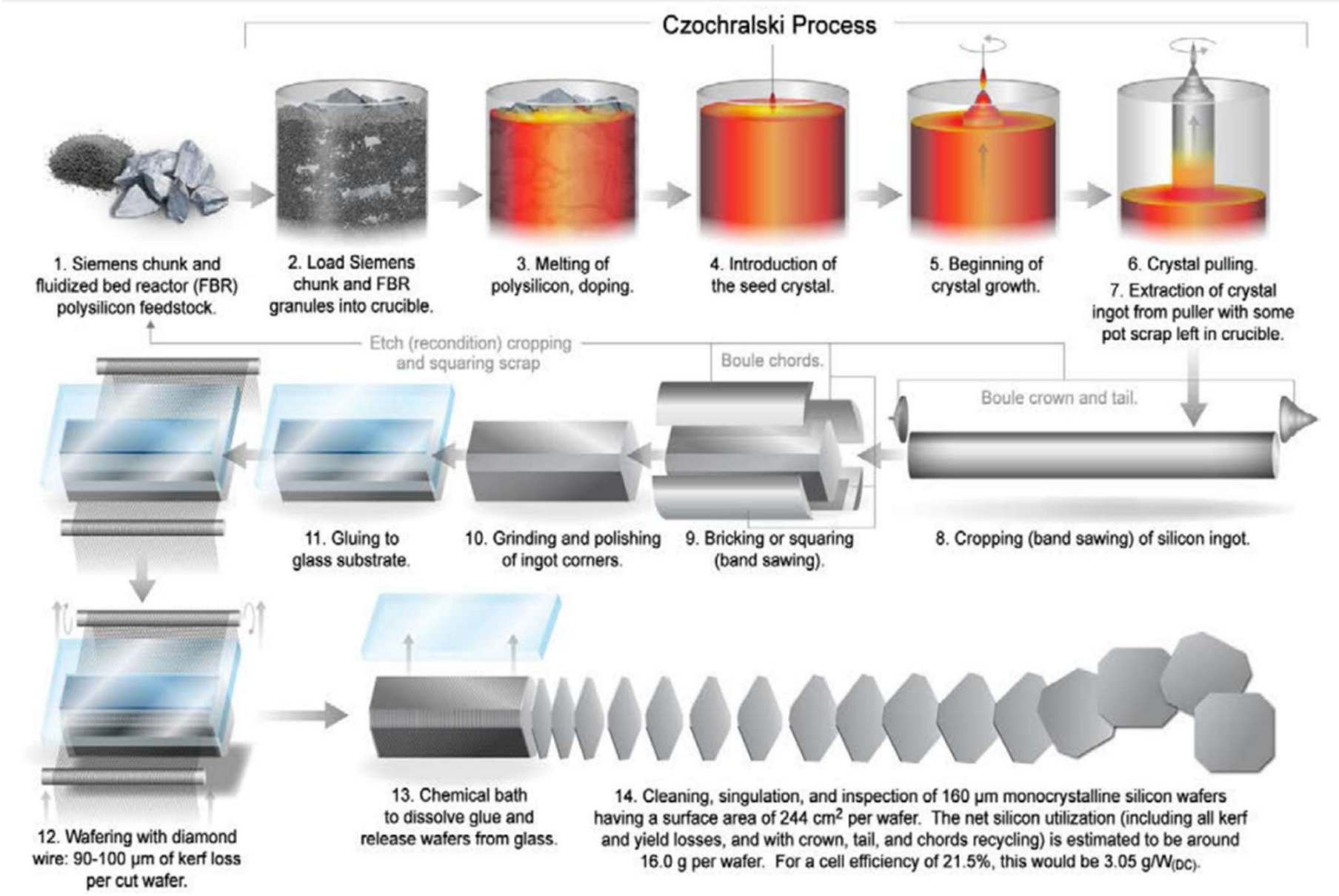
Diamond

Silicon

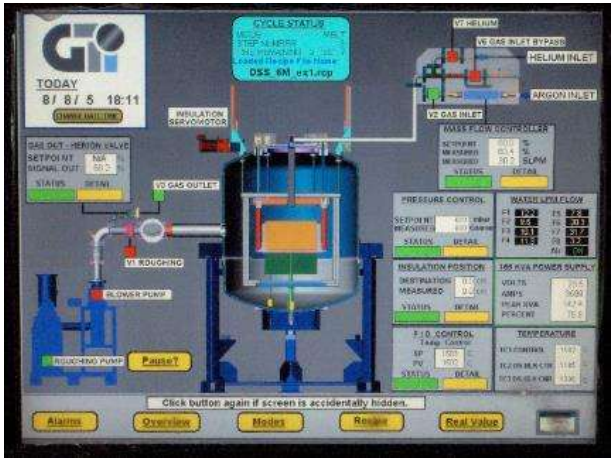
3.2. VALUE CHAIN : SINGLE CRYSTAL GROWTH. MONO



3.2. SINGLE CRYSTAL GROWTH. MONO.



3.2. VALUE CHAIN: MULTI CRYSTAL GROWTH. MULTI



3.3 VALUE CHAIN: SOLAR CELLS

As received

Cleaning and etching. Alkaline or acidic baths

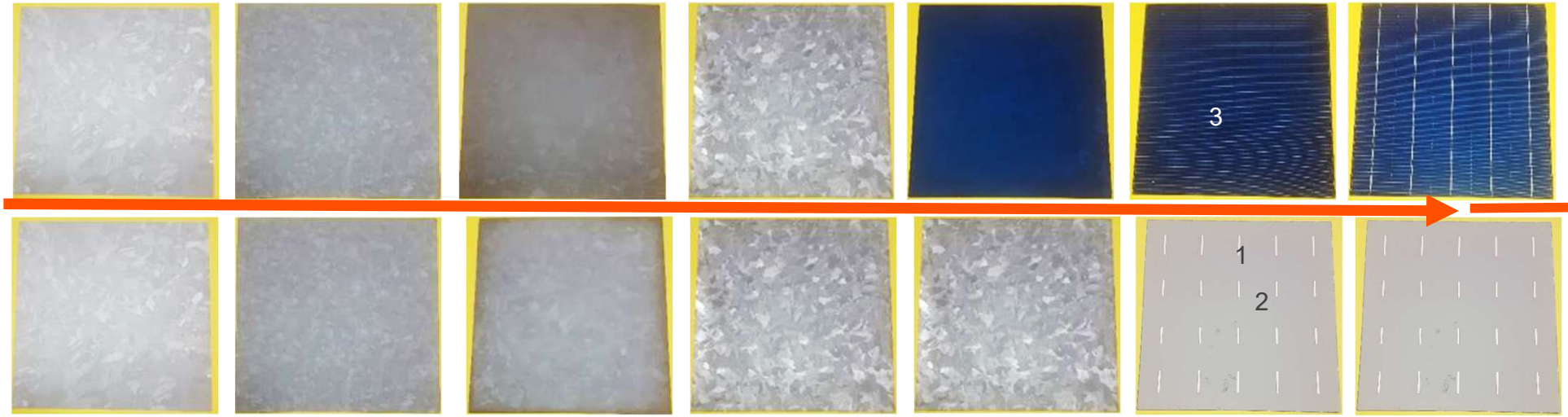
CVD. Phosphorous diffusion

Edge etching,

ARC coating deposition

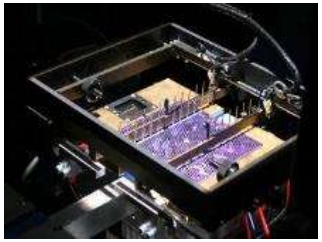
Screen print. Dual print. 1st front fingers

Screen print. Dual print. 2nd front buses



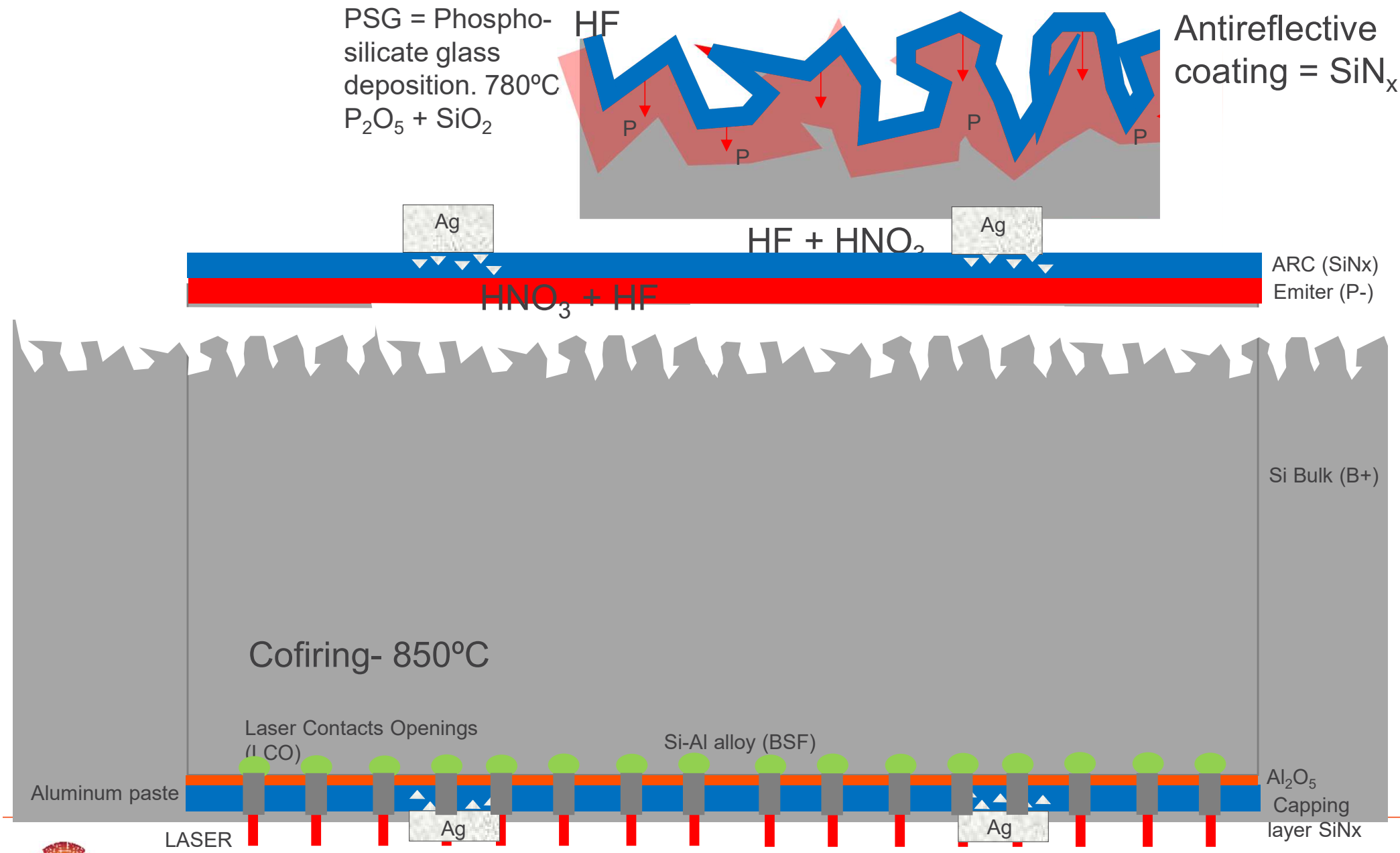
MONO

Cell tester and sorter. Measures the electrical parameters and classifies the cell according to them



Co-firing. Belt furnace . Makes the drive-through of the contacts

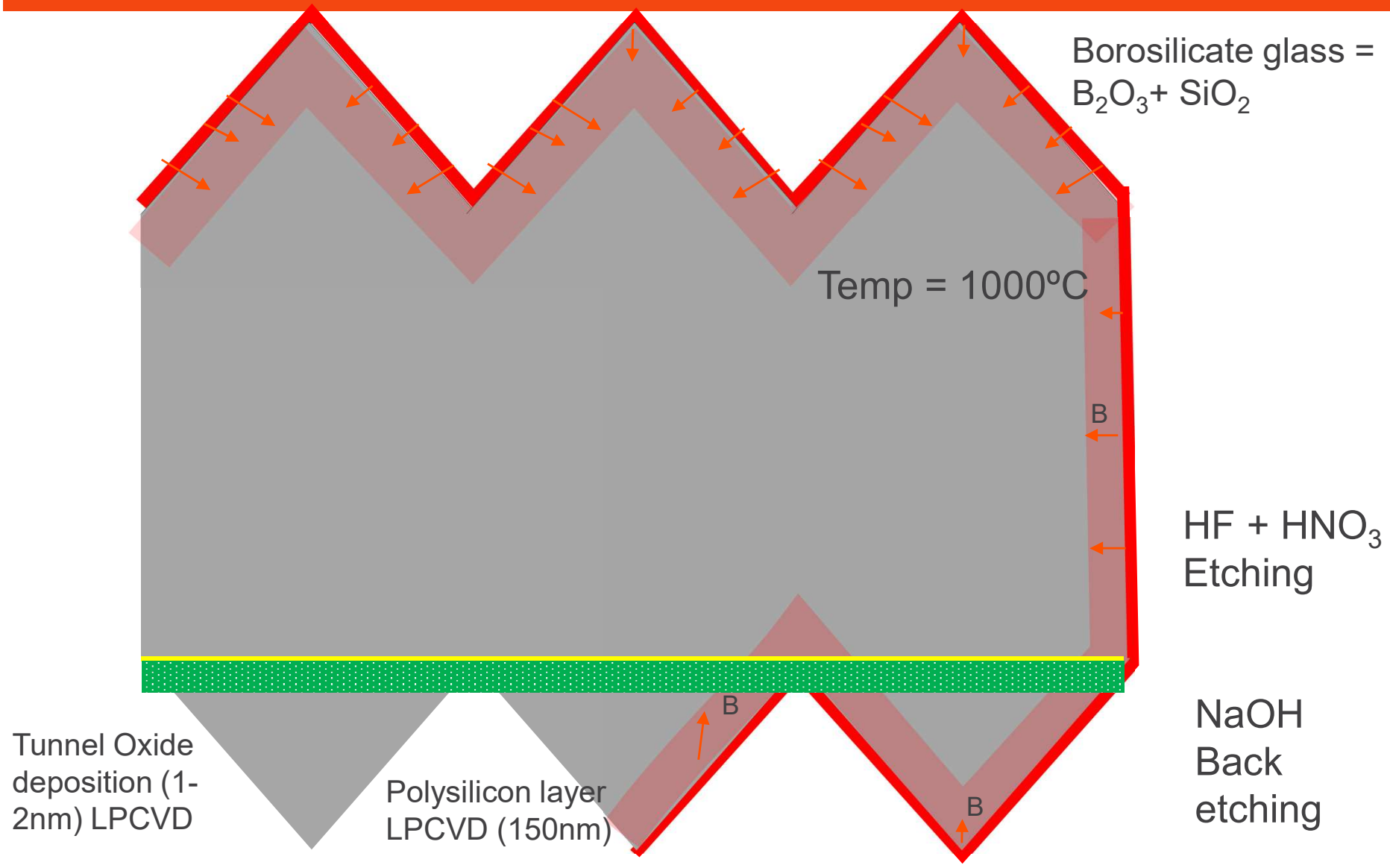
3) PERC SOLAR CELLS. MAIN STEPS



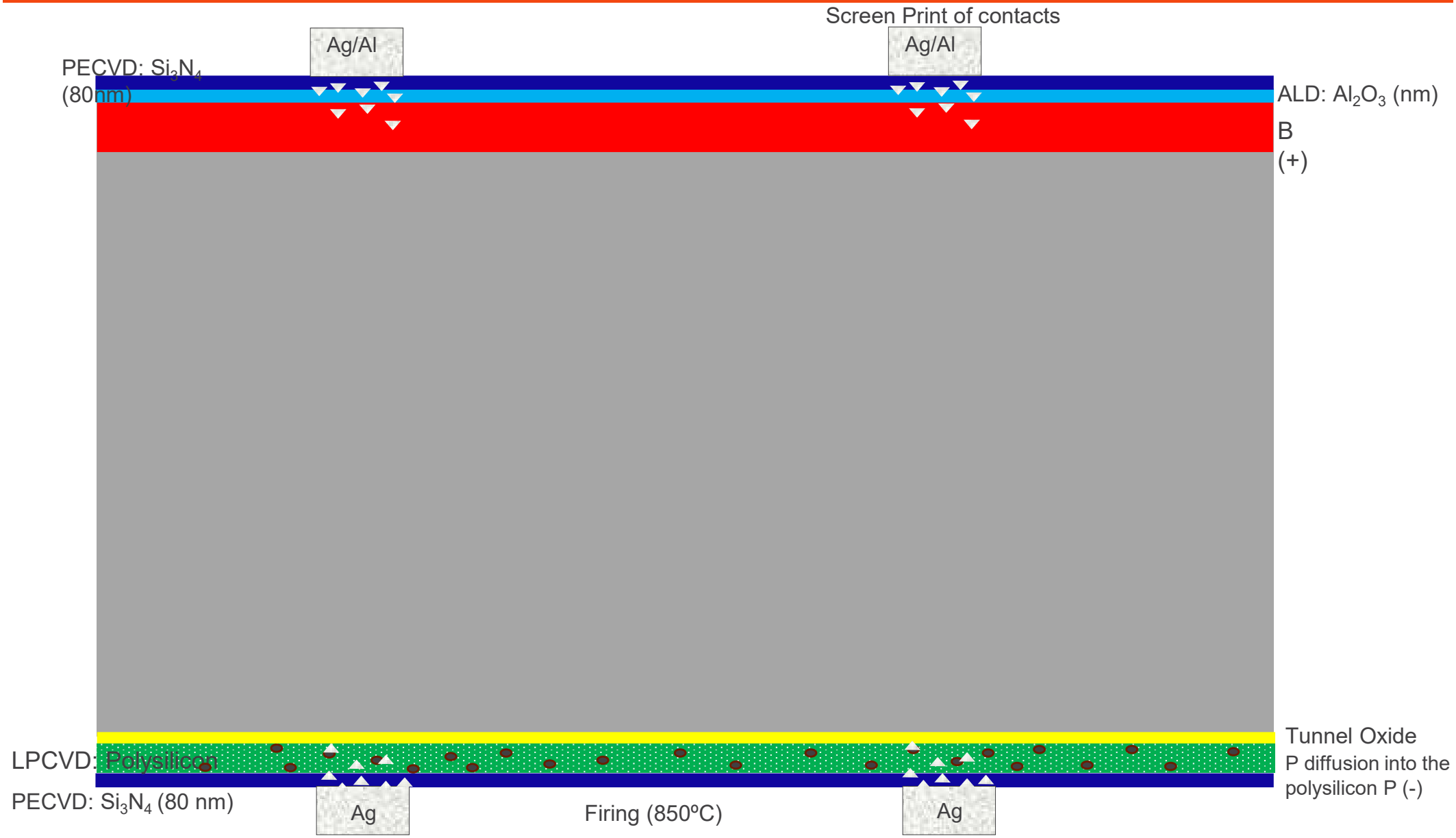
TOPCON

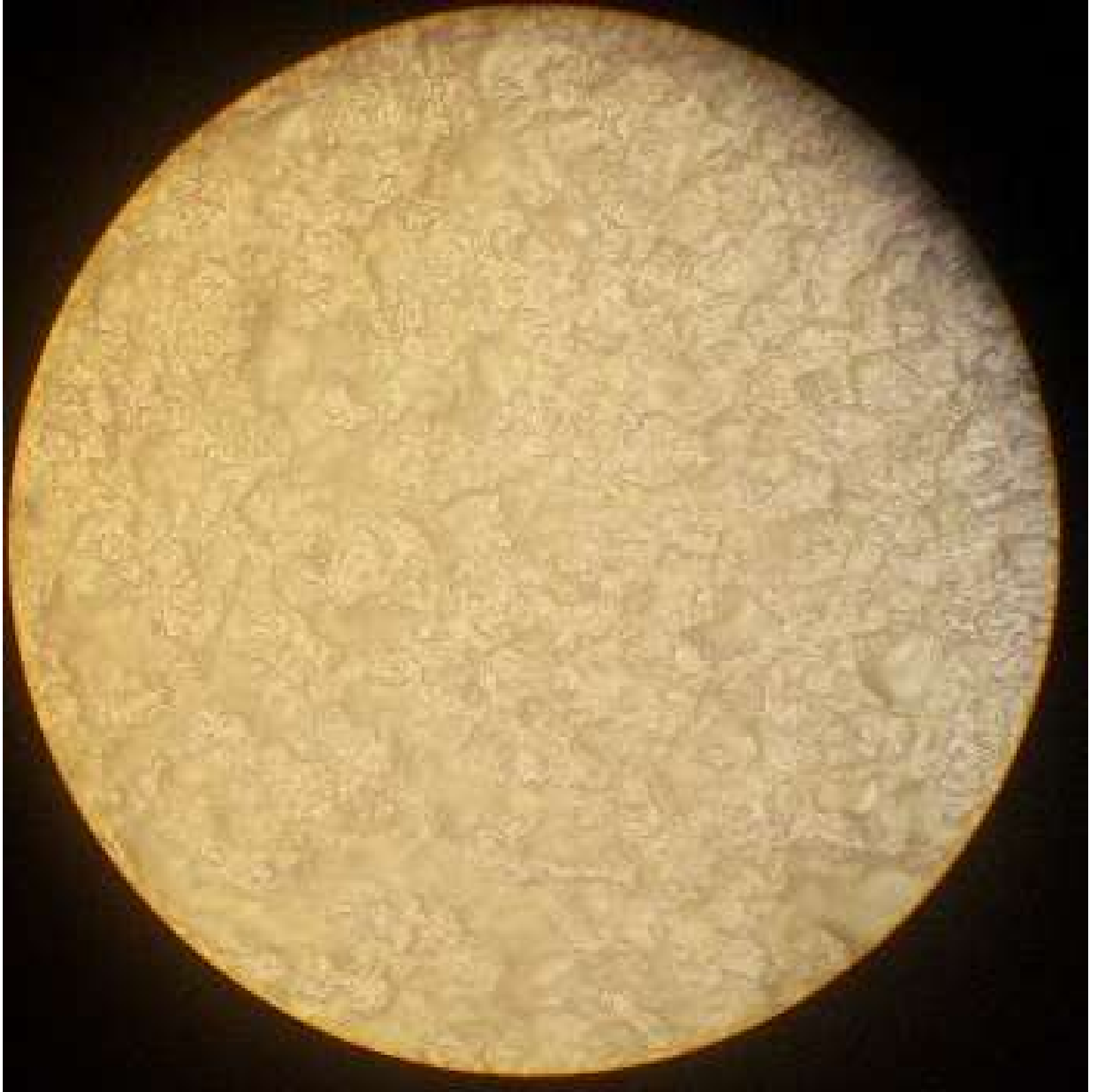
IPA + NaOH (87ºC)

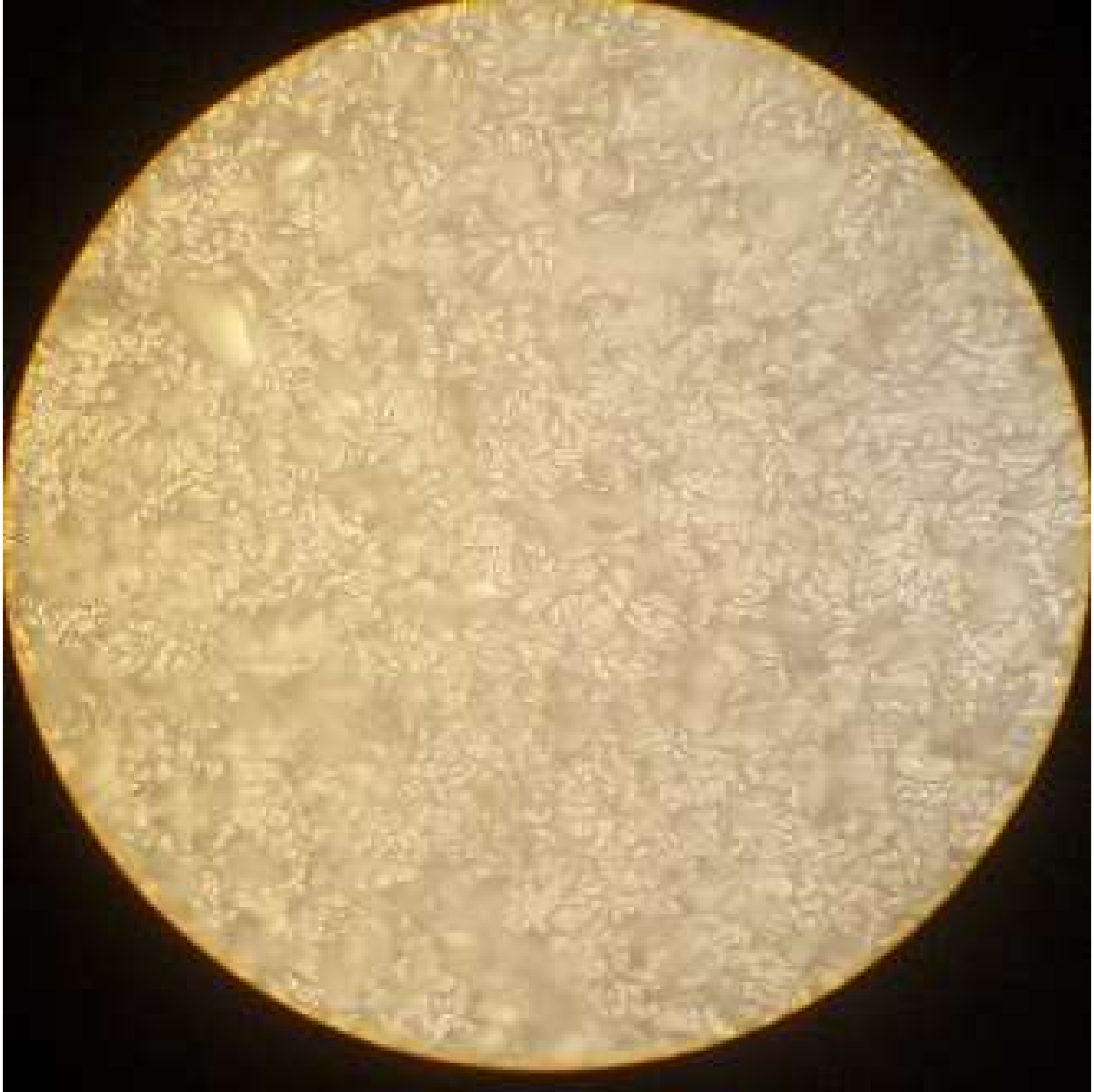
TOPCON

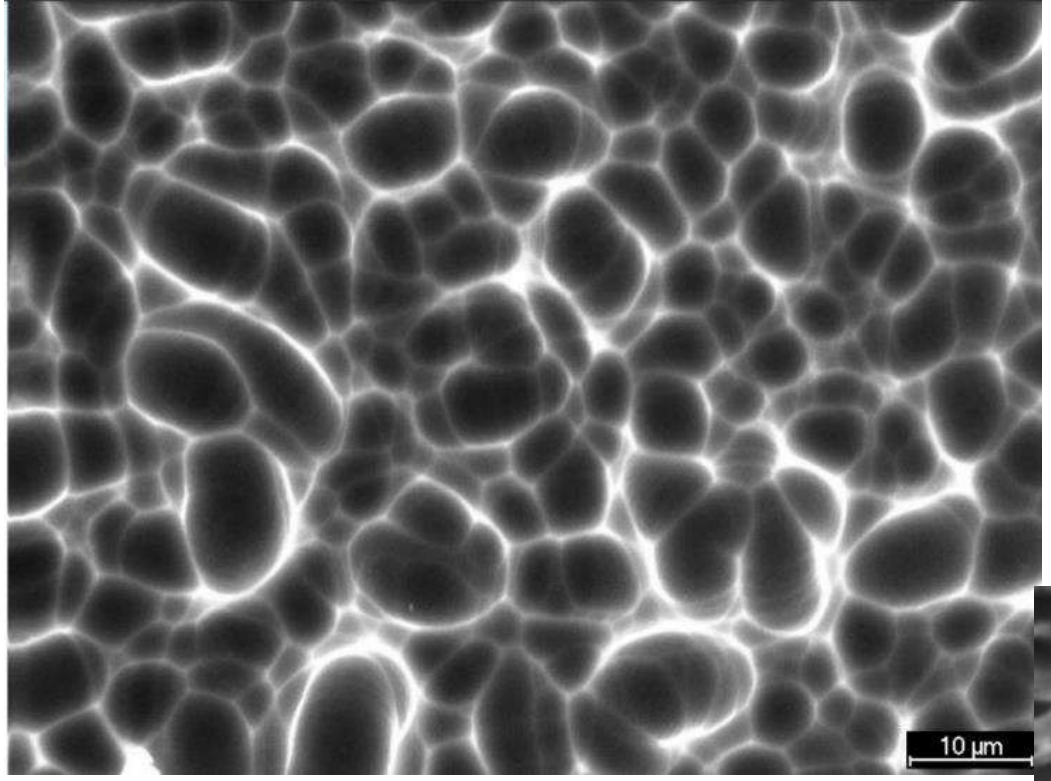

Borosilicate glass =
B_2O_3+ SiO_2
Temp = 1000ºC
B
HF + HNO_3
Etching
B
B
NaOH
Back
etching
Tunnel Oxide
deposition (1-
2nm) LPCVD
Polysilicon layer
LPCVD (150nm)

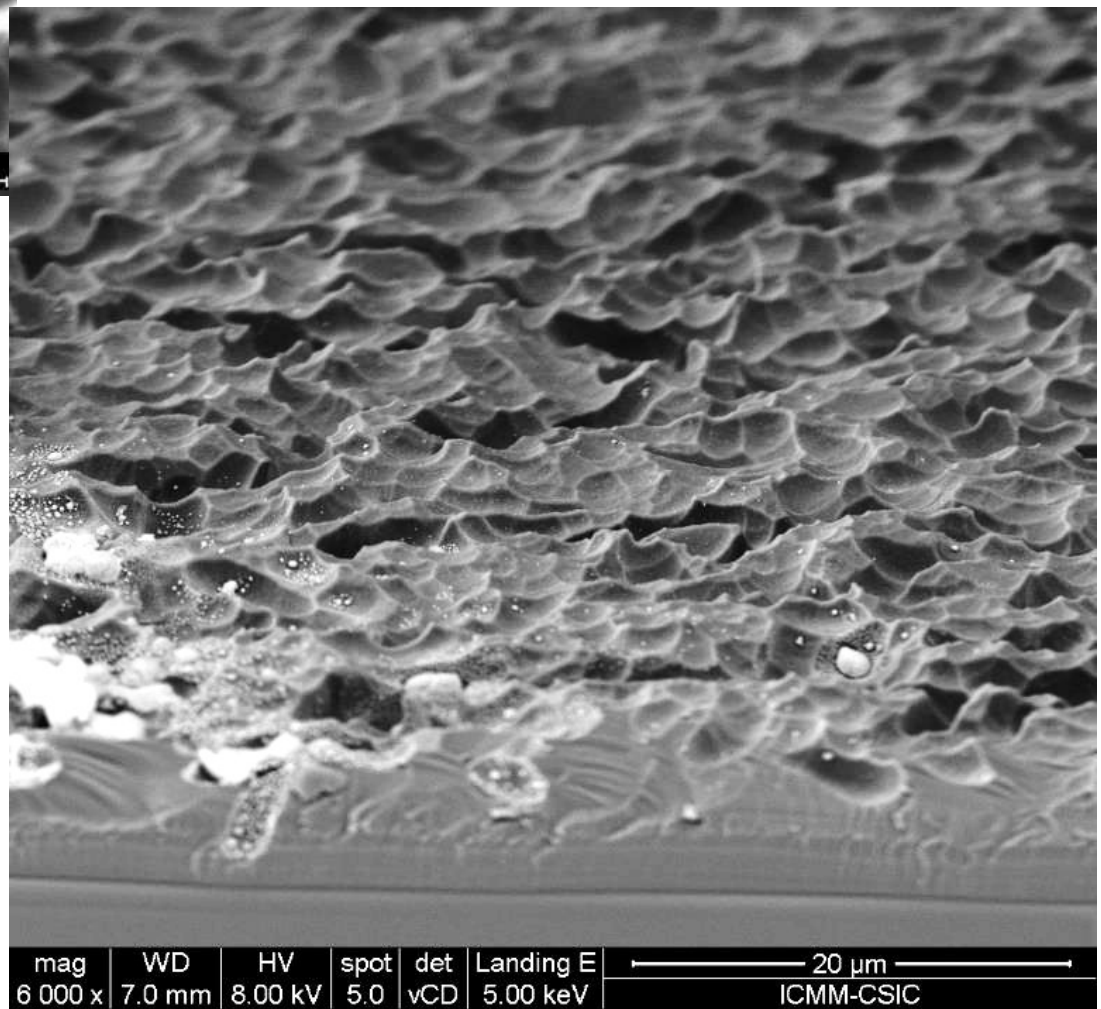
TOPCON

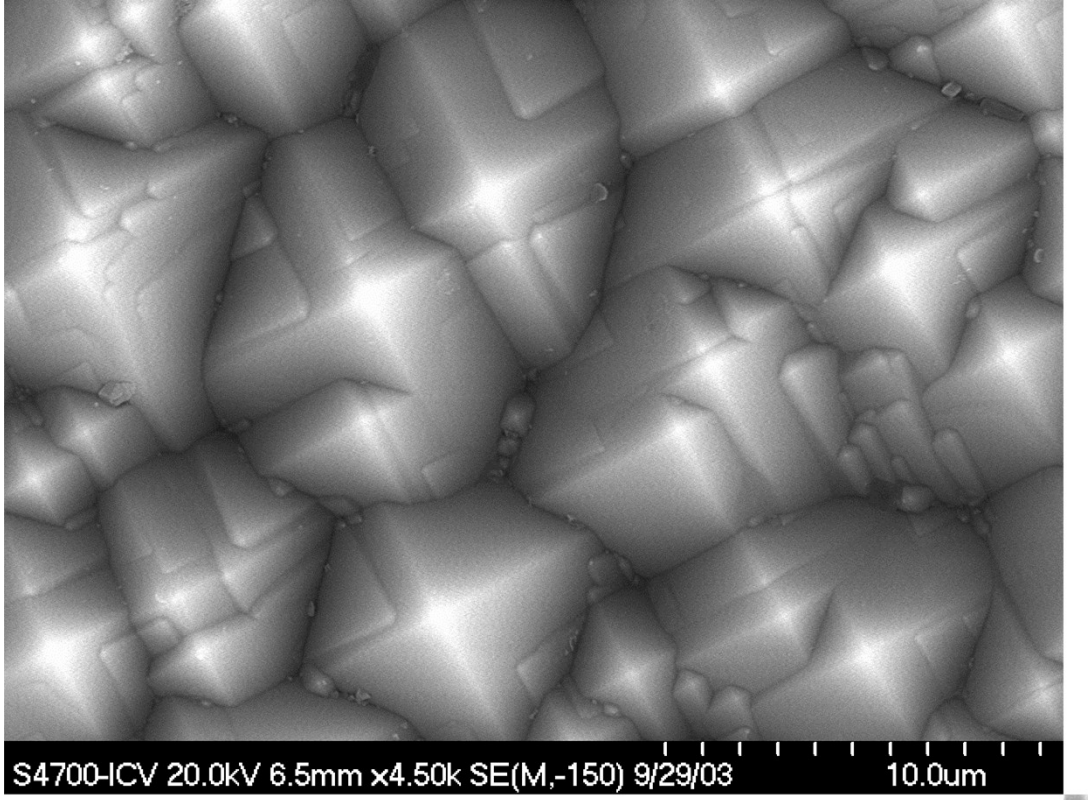

Screen Print of contacts
Ag/Al
Ag/Al
PECVD: Si_3N_4 (80nm)
ALD: Al_2O_3 (nm)
B
(+)
Tunnel Oxide
P diffusion into the polysilicon P (-)
LPCVD: Polysilicon
PECVD: Si_3N_4 (80 nm)
Ag
Firing (850ºC)
Ag

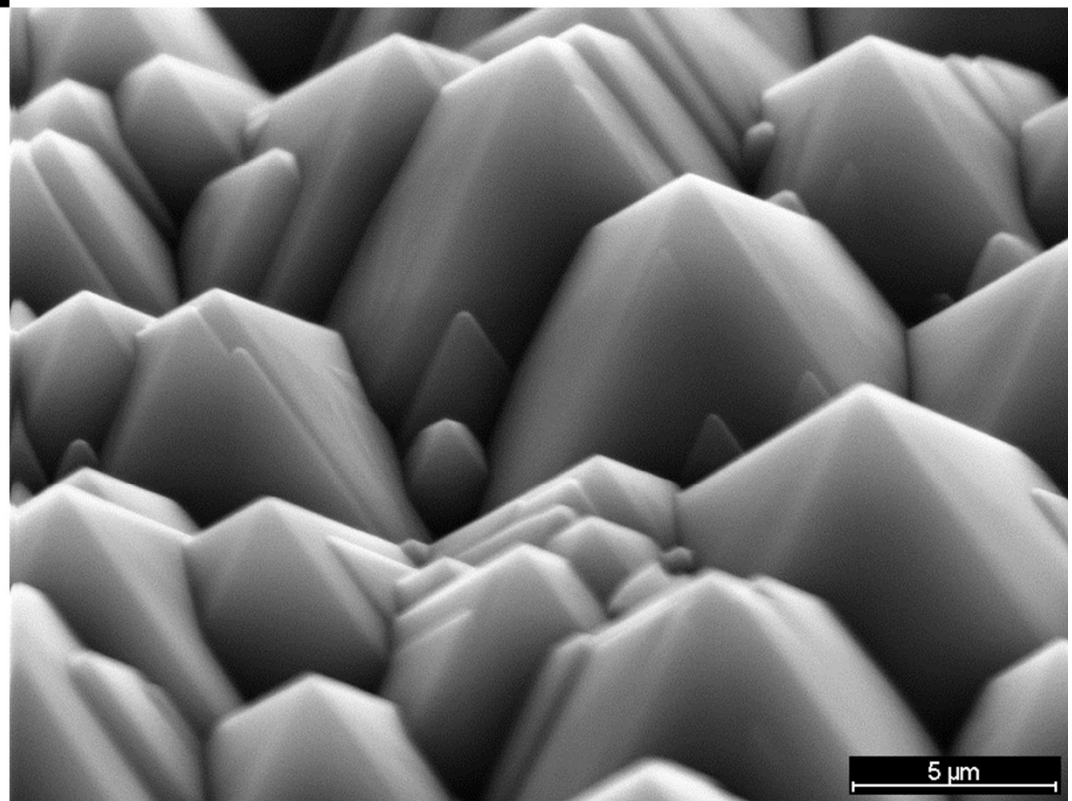


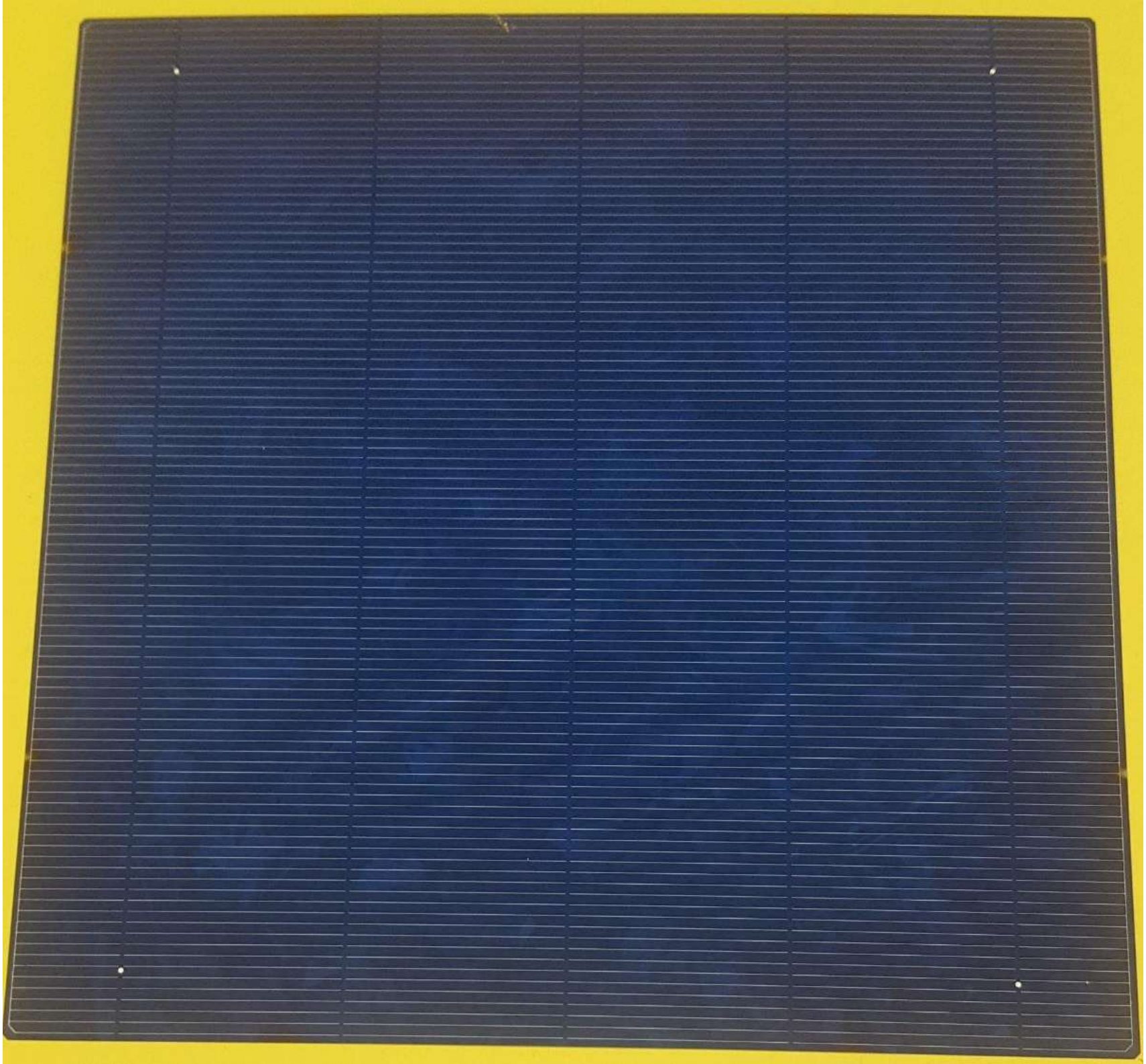


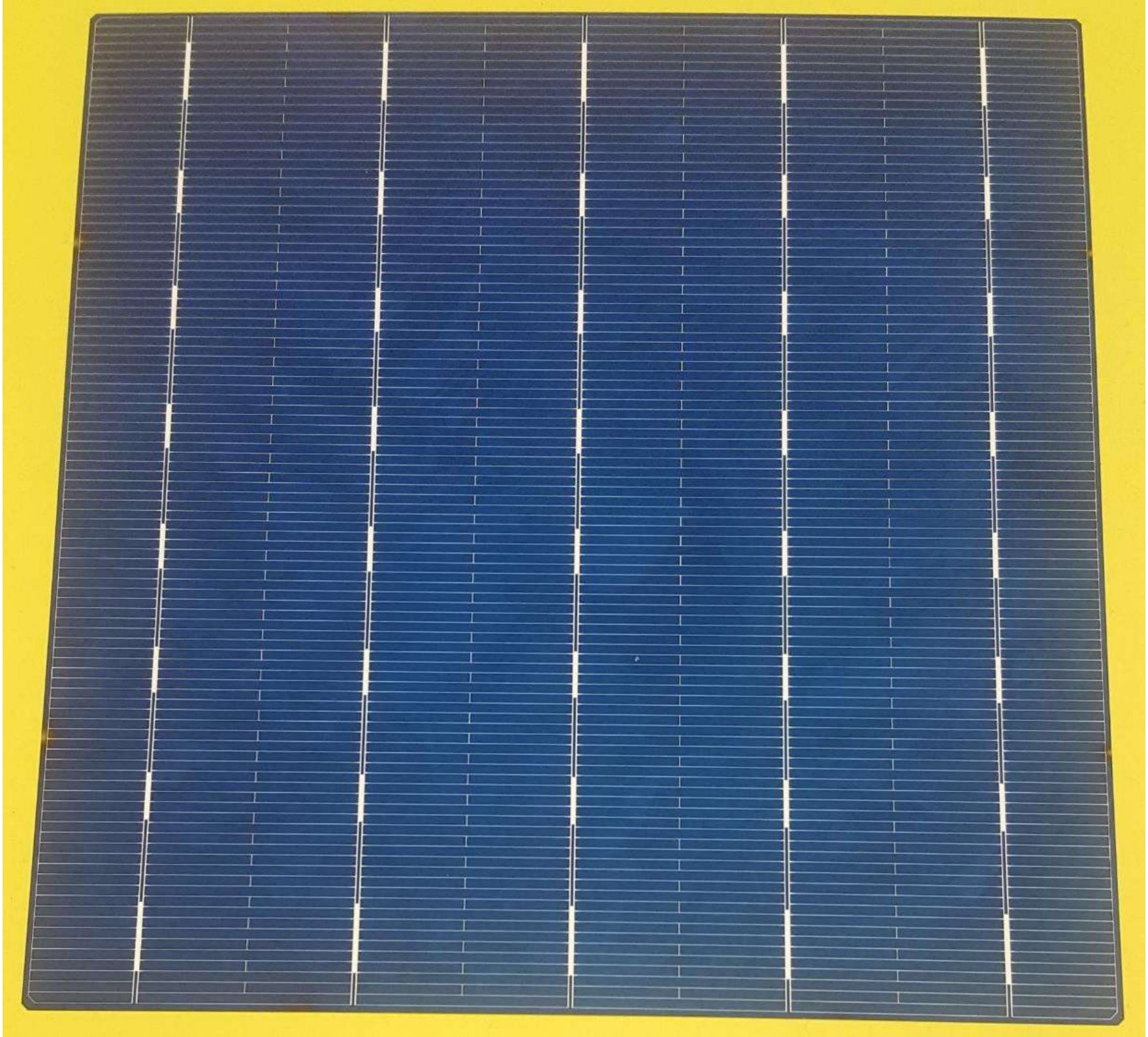

10 µm


mag
6 000 x
WD
7.0 mm
HV
8.00 kV
spot
5.0
det
vCD
Landing E
5.00 keV
20 µm
ICMM-CSIC

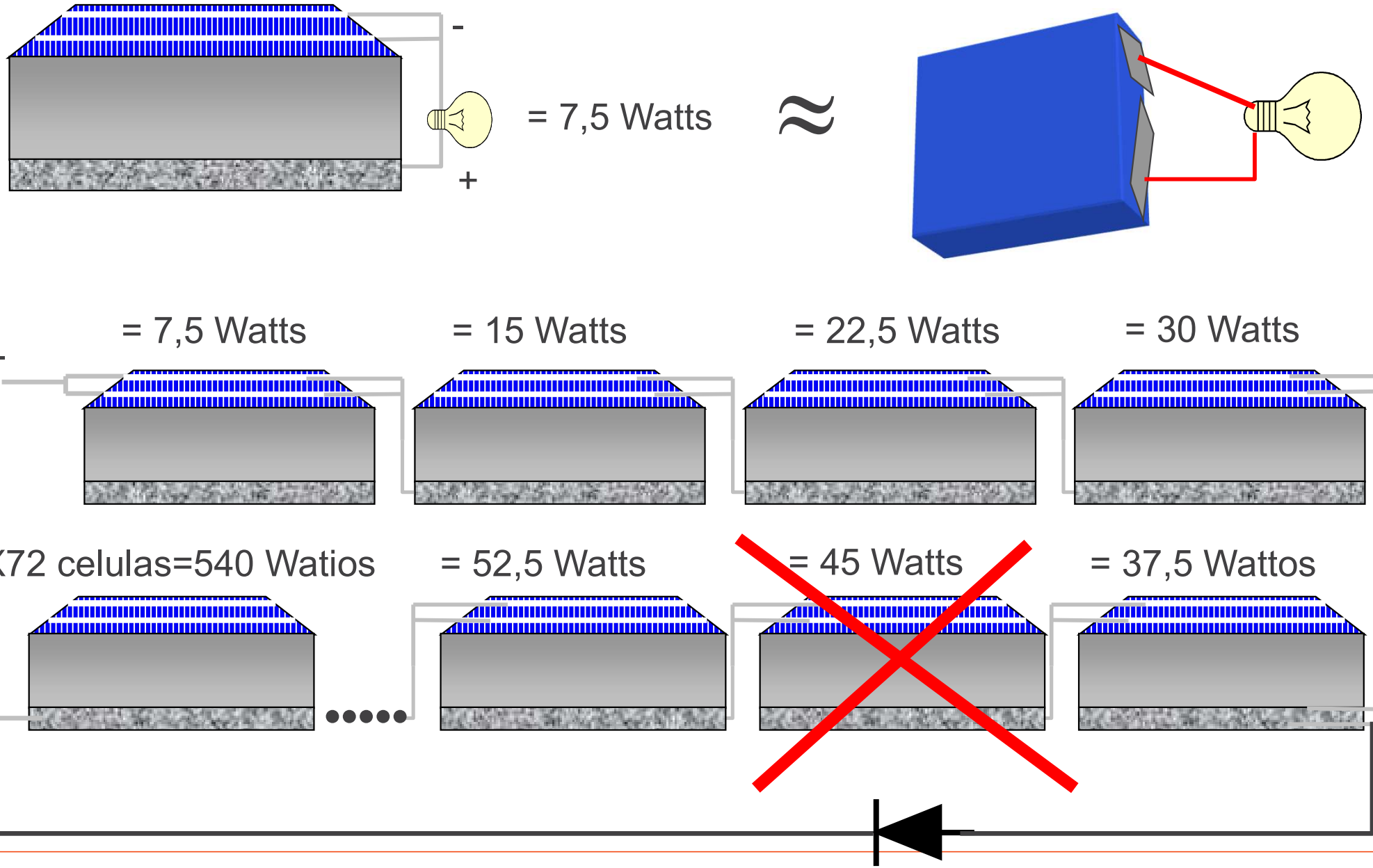

S4700-ICV 20.0kV 6.5mm x4.50k SE(M,-150) 9/29/03
10.0um


5 µm

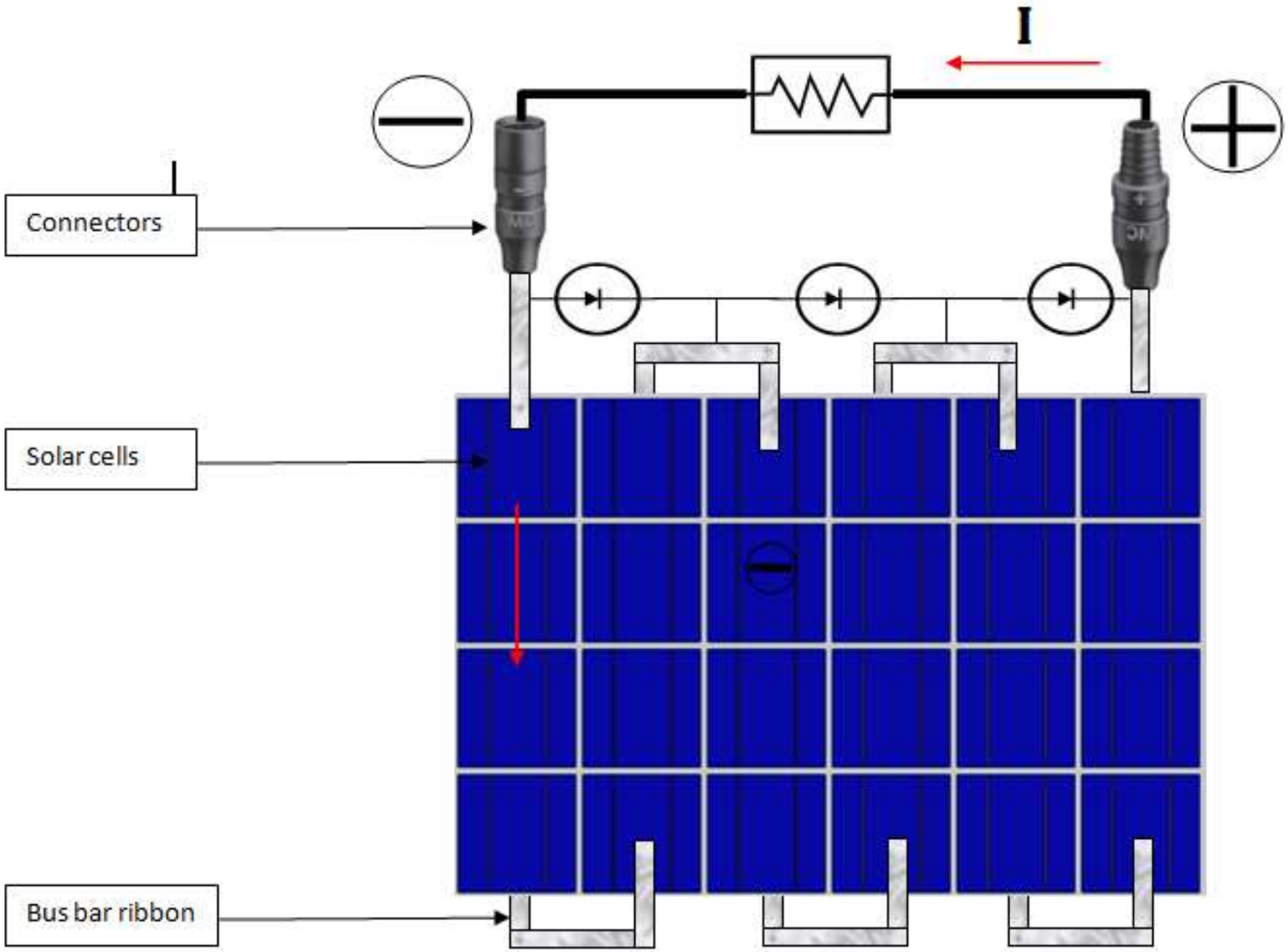




3.4 ASSOCIATION OF CELLS

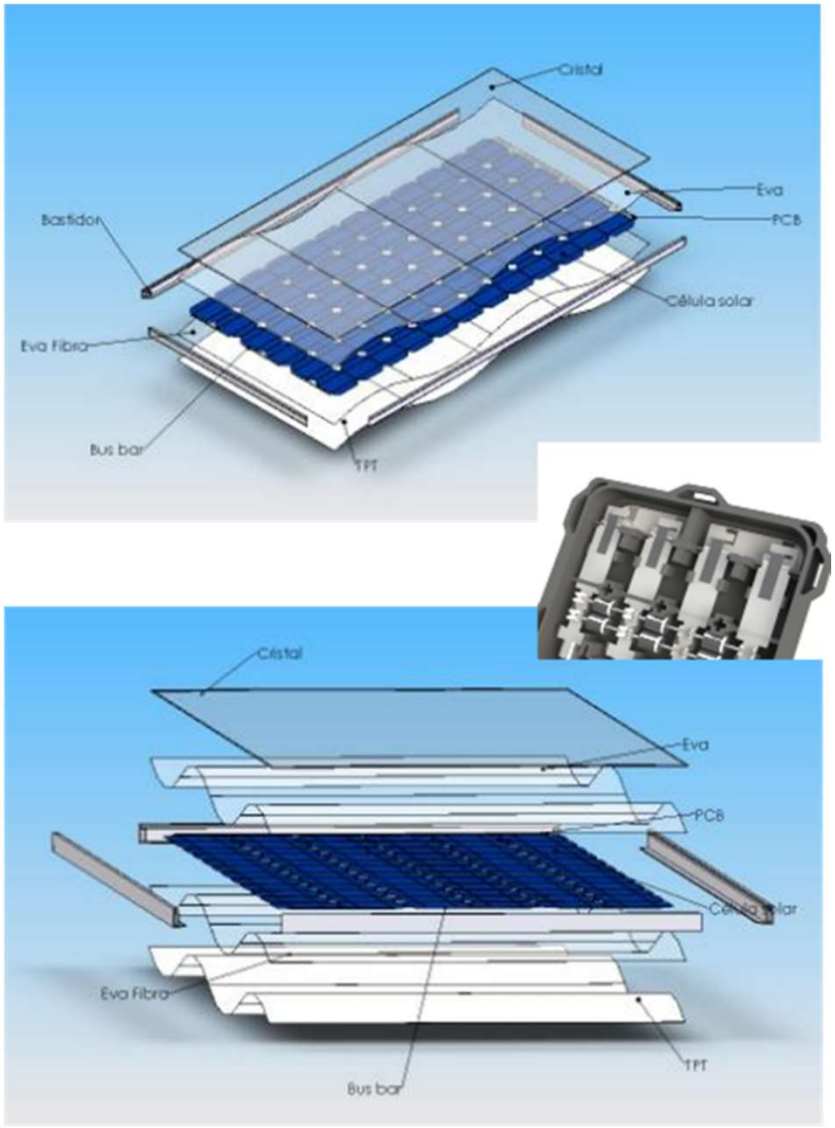


3.4. PV MODULE. STANDARD MATERIALS

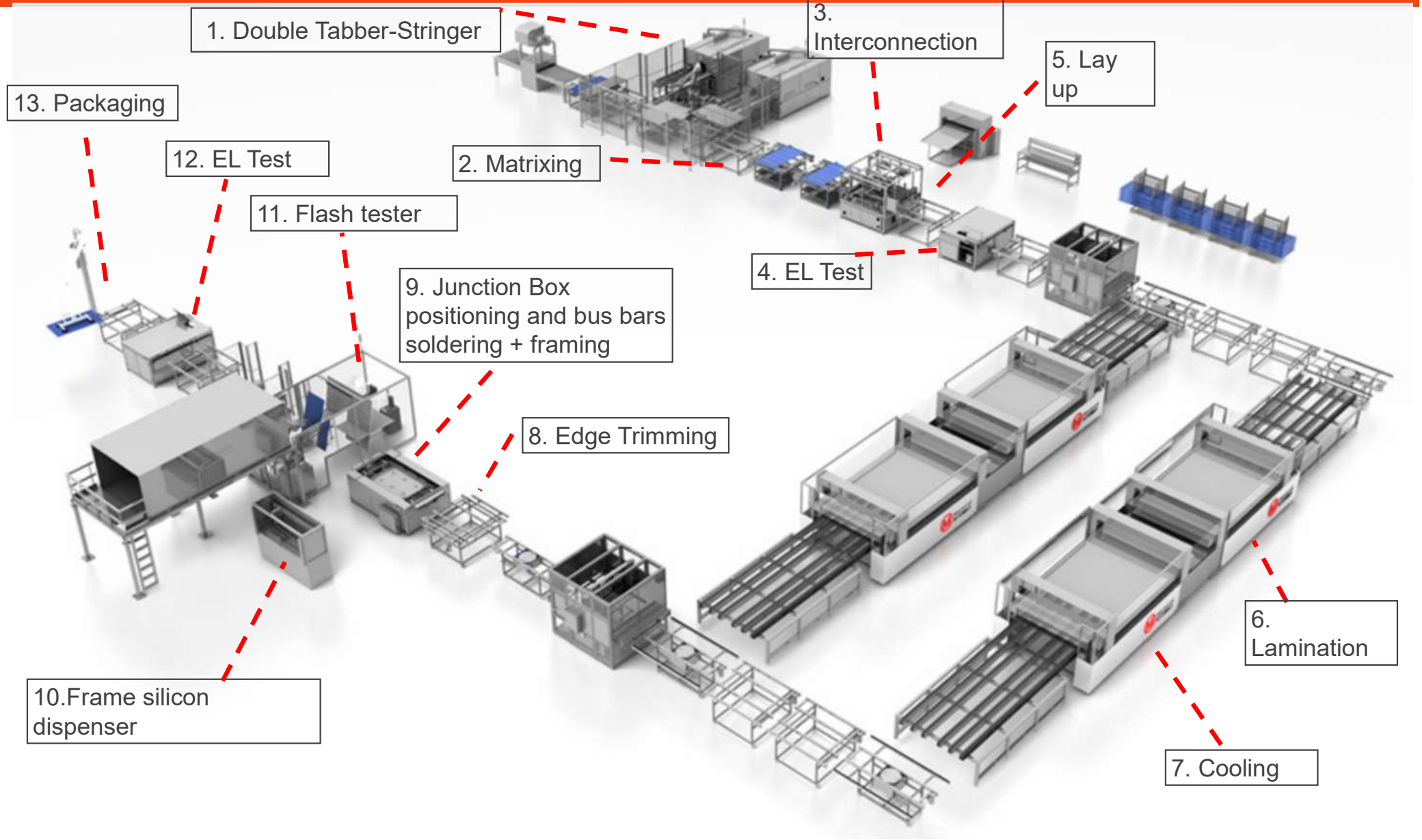


3.4. PV MODULE. STANDARD MATERIALS

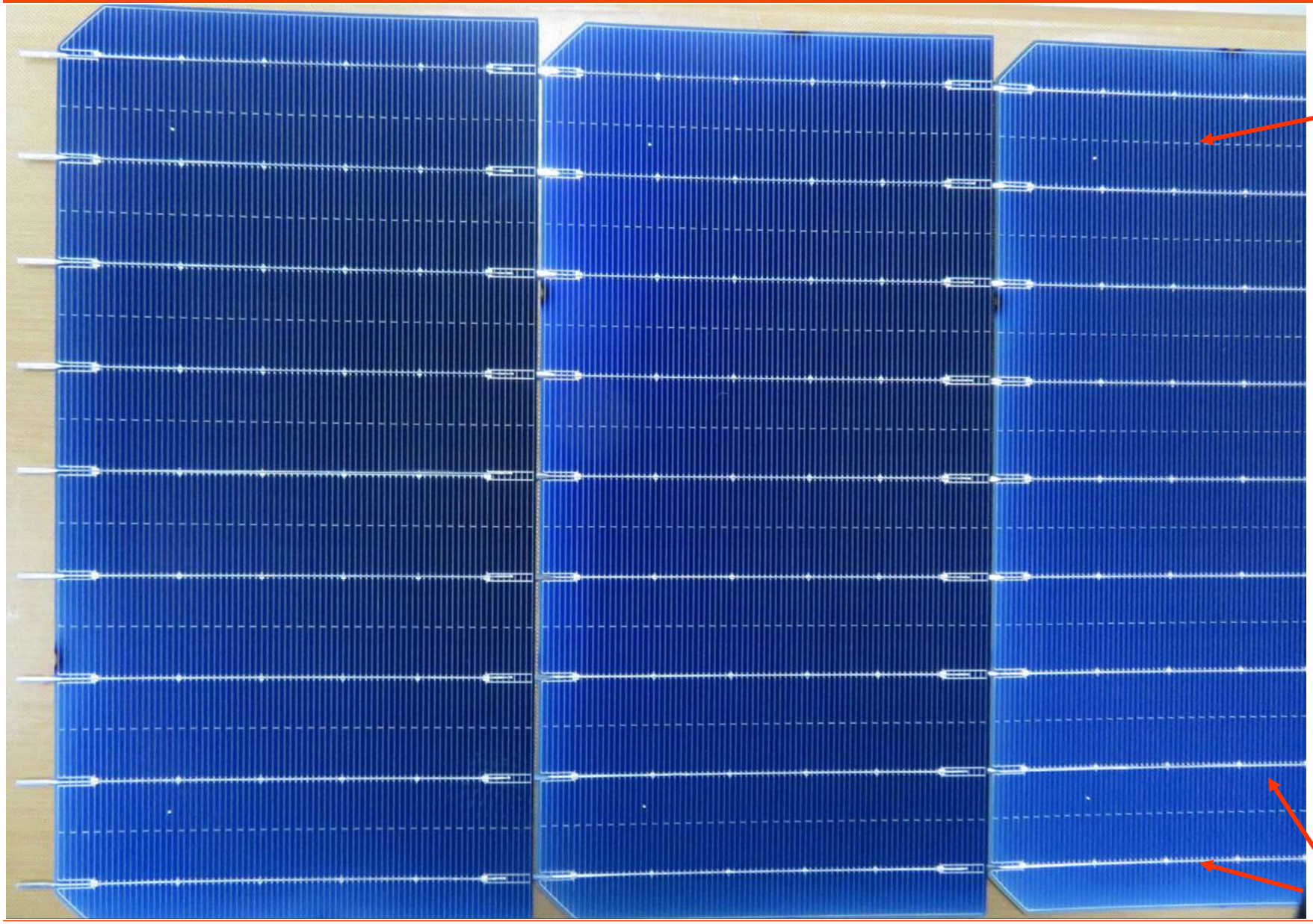
Material Description
Glass
Solar cells
Tab-ribbon
EVA
Fibra
Backsheet (TPT or Polyester)
Connection cables
Junction Box
Profiles/Frame
Labels
Screws
Bus bar ribbon
Silicone



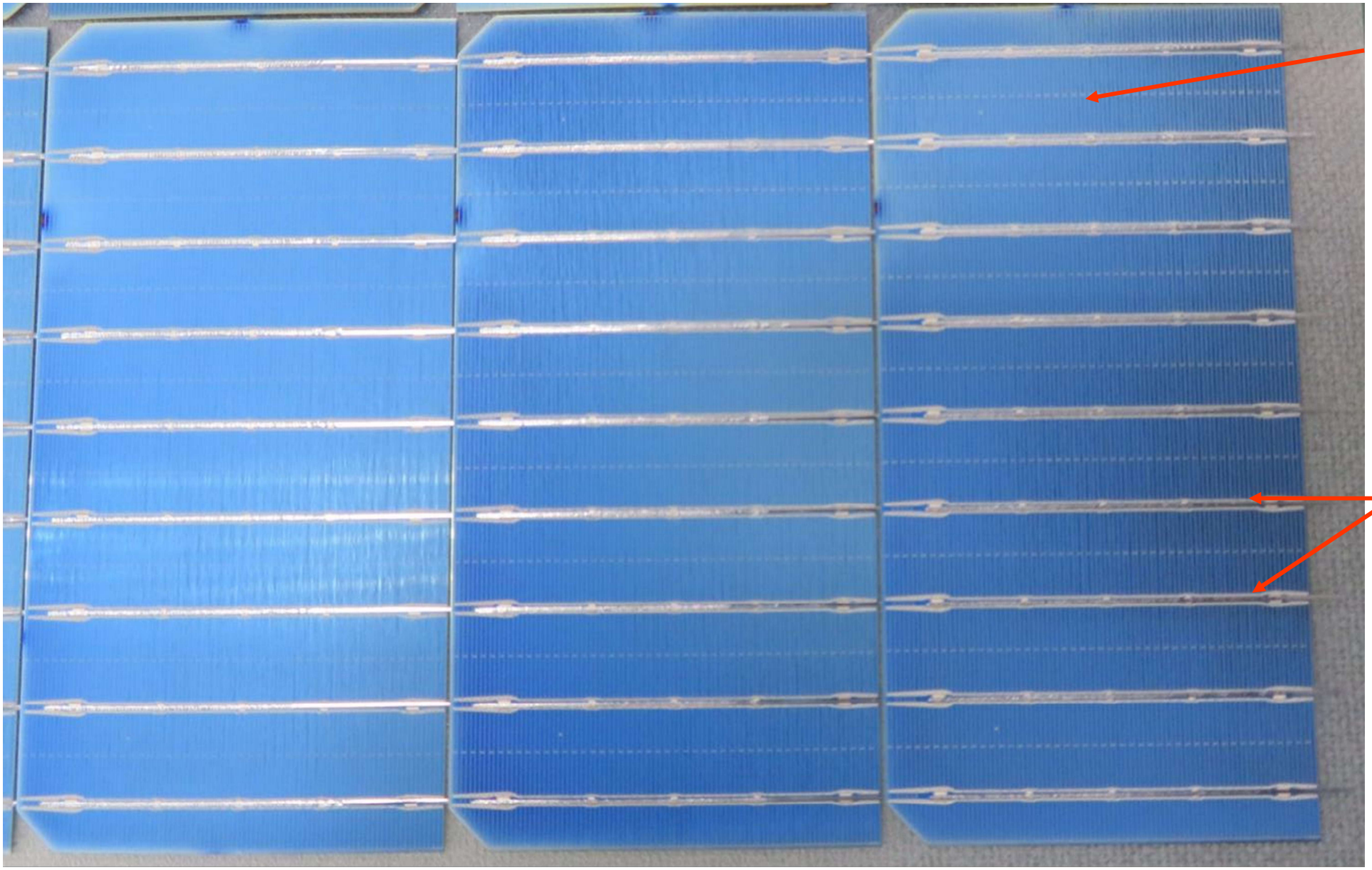
3.5. LINE LAY OUT. MANUFACTURING STAGES.



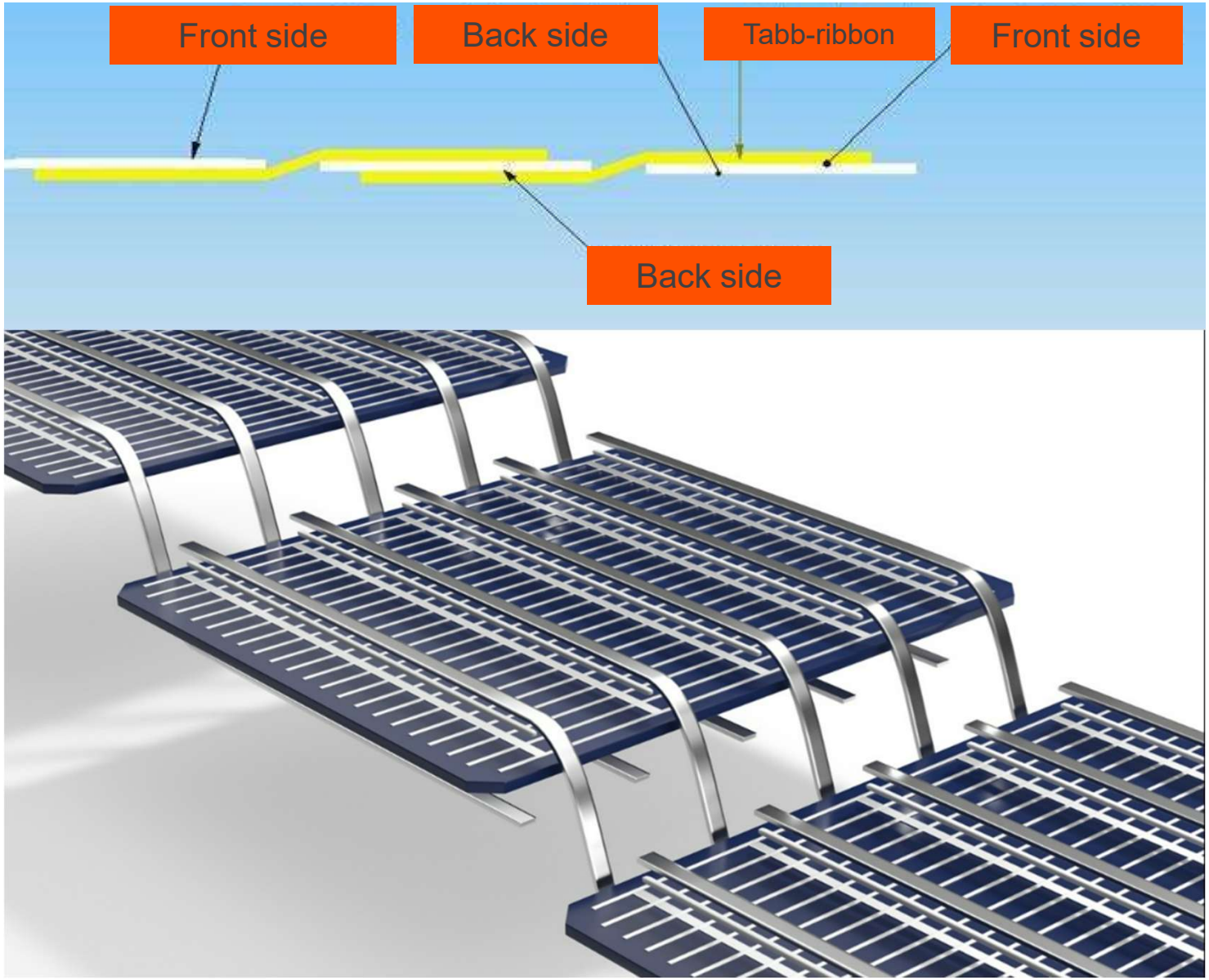
3.5. TABBER-STRINGER. SOLAR CELL FRONT SIDE

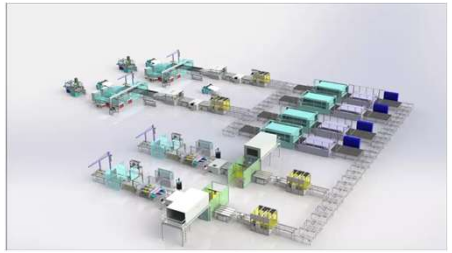


3.5. TABBER-STRINGER. SOLAR CELL BACK SIDE

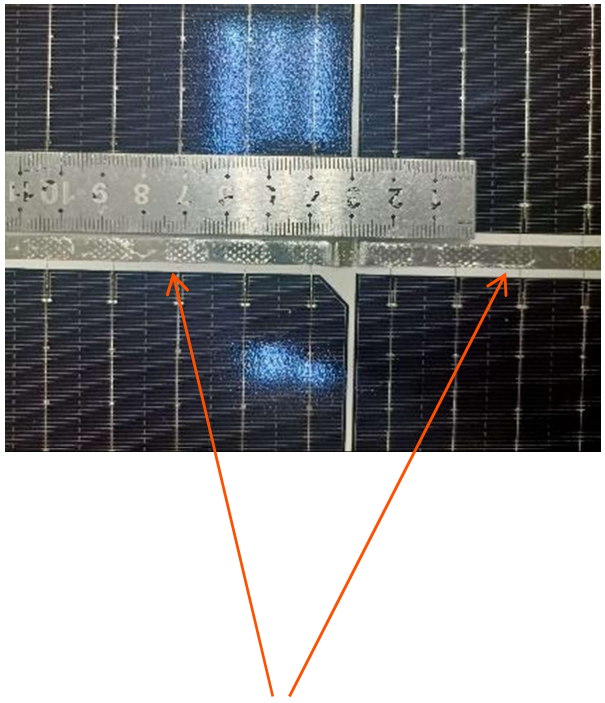


3.5. TABBER STRINGER-SOLAR CELLS SOLDERING





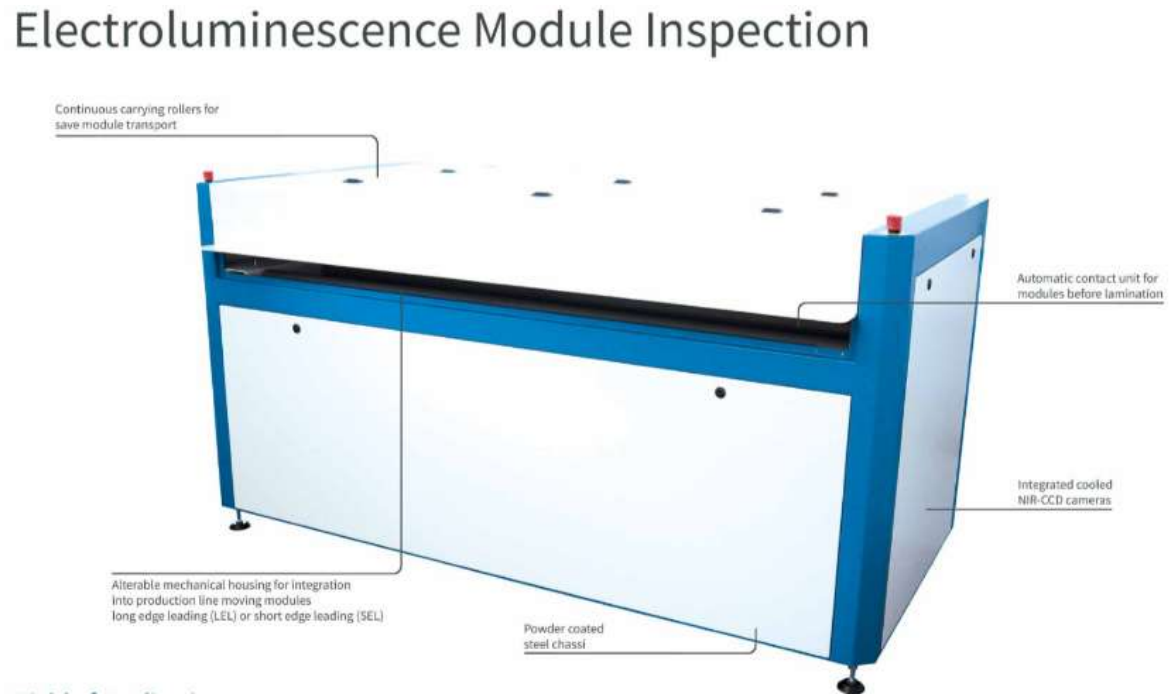
3.5. INTERCONNECTION

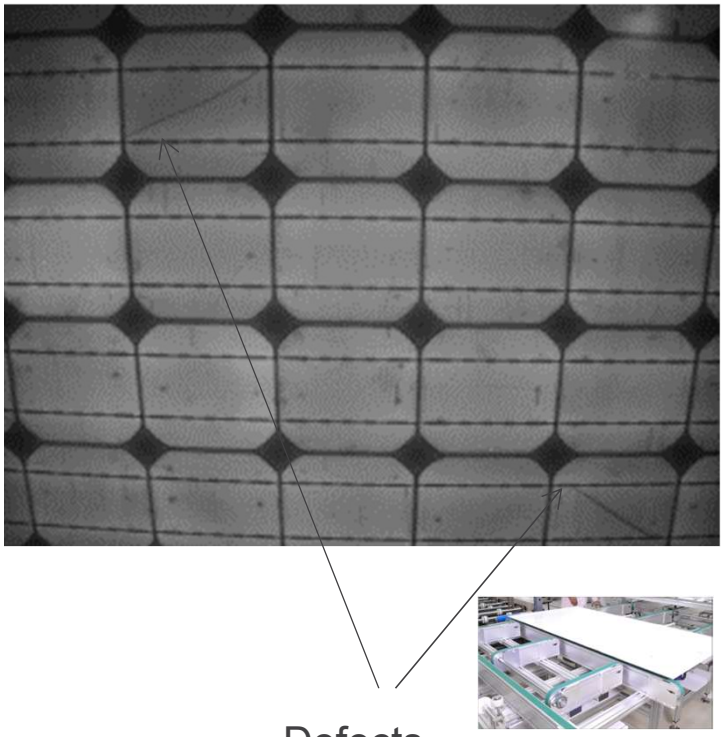


Soldering points

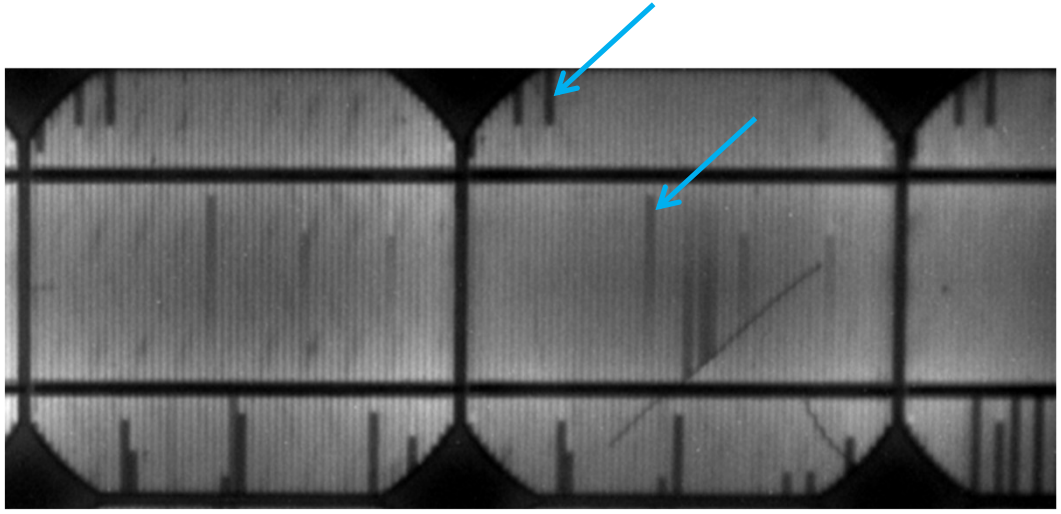
3.5. ELECTROLUMINESCENCE TEST. EL

- By electric excitation of cells (direct load) they emit light (luminescence) which is collected by a camera.
- Inactive areas like **breaks**, areas of high density of **recombination centers**, **unsoldered tab-ribbons**, **bad printed fingers,** etc are non-emission areas → they appear as dark areas.

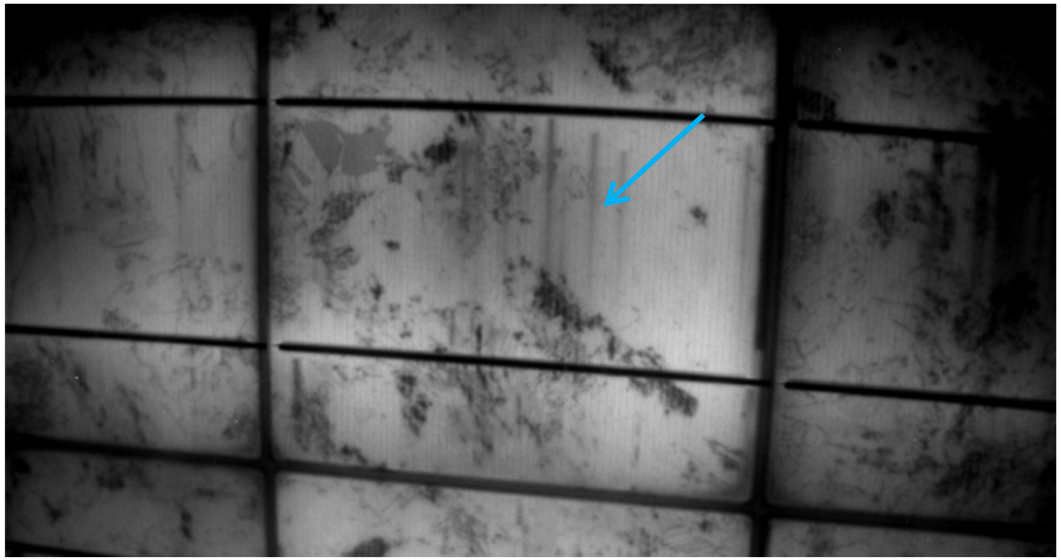




3.5. ELECTROLUMINESCENCE TEST. EXAMPLES

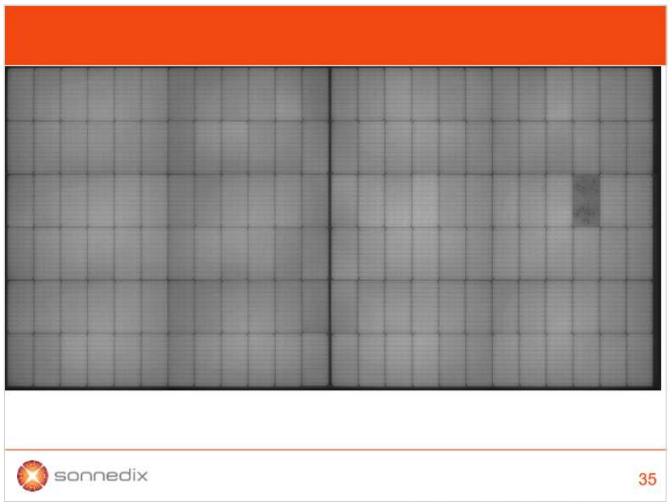


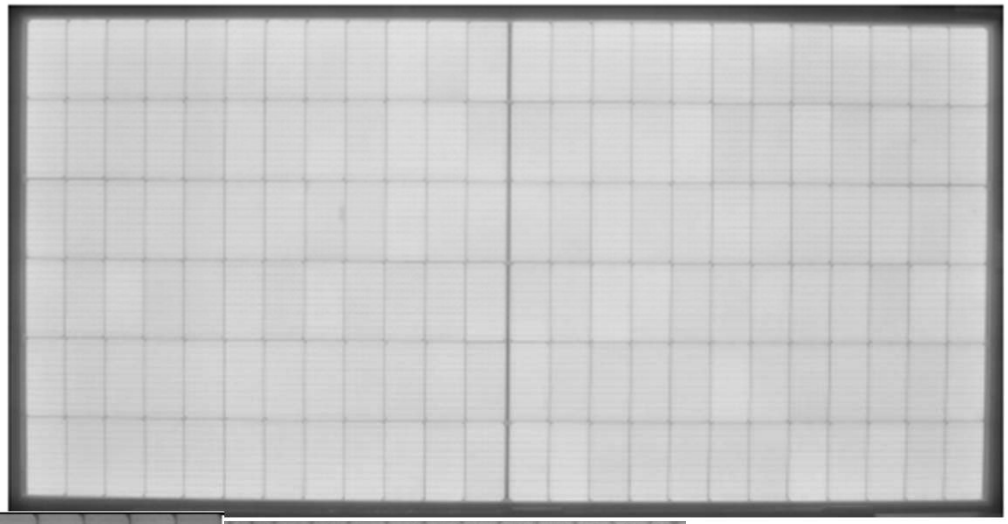
Finger interruptions
Cause 1 Single-crystal: Bad quality of the screen print (solar cell manufacturing). No reject

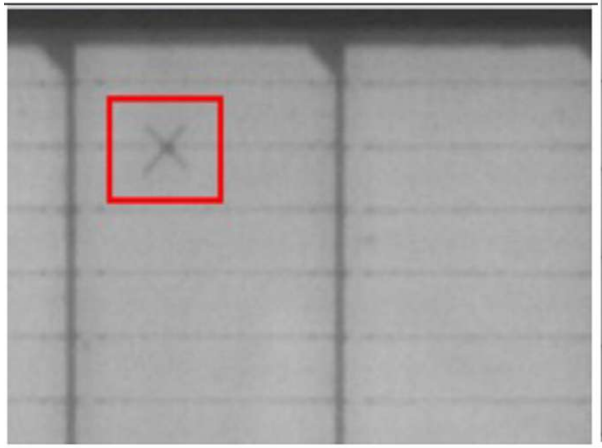


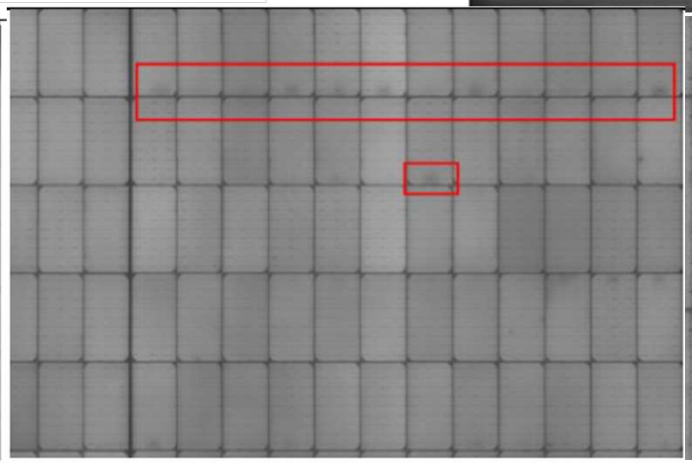
Finger interruptions
Cause 1 Multi-crystal: Bad quality of the screen print (solar cell manufacturing). No reject

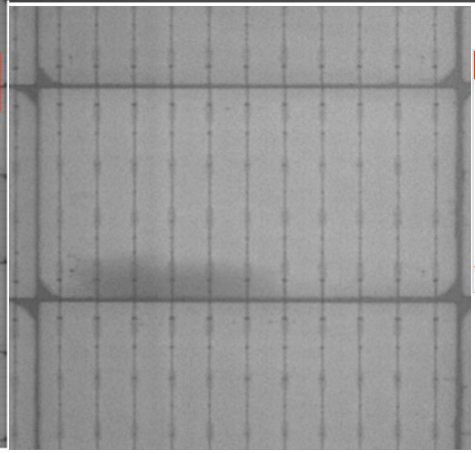
3.5. ELECTROLUMINESCENCE TEST. EXAMPLES

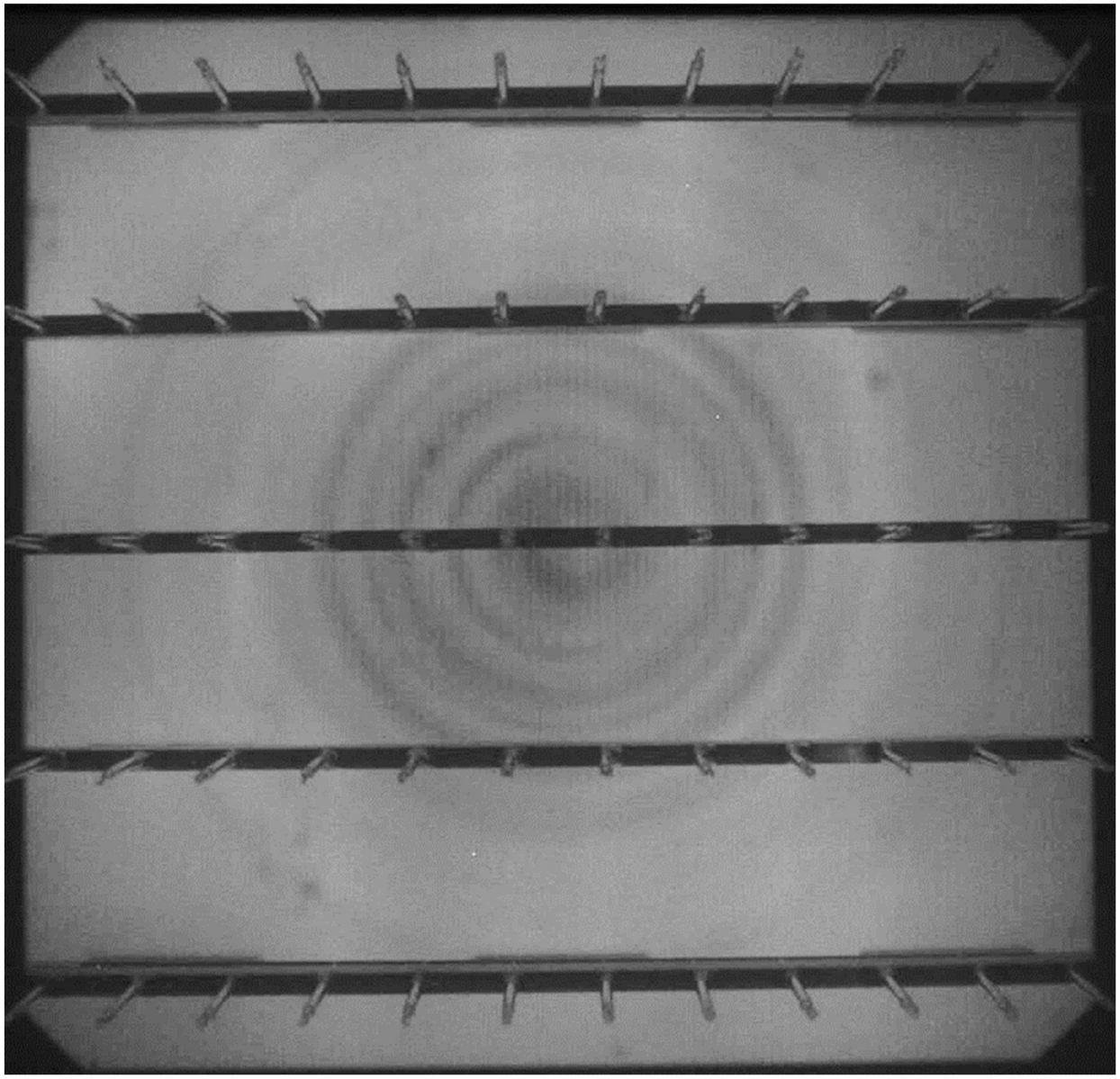


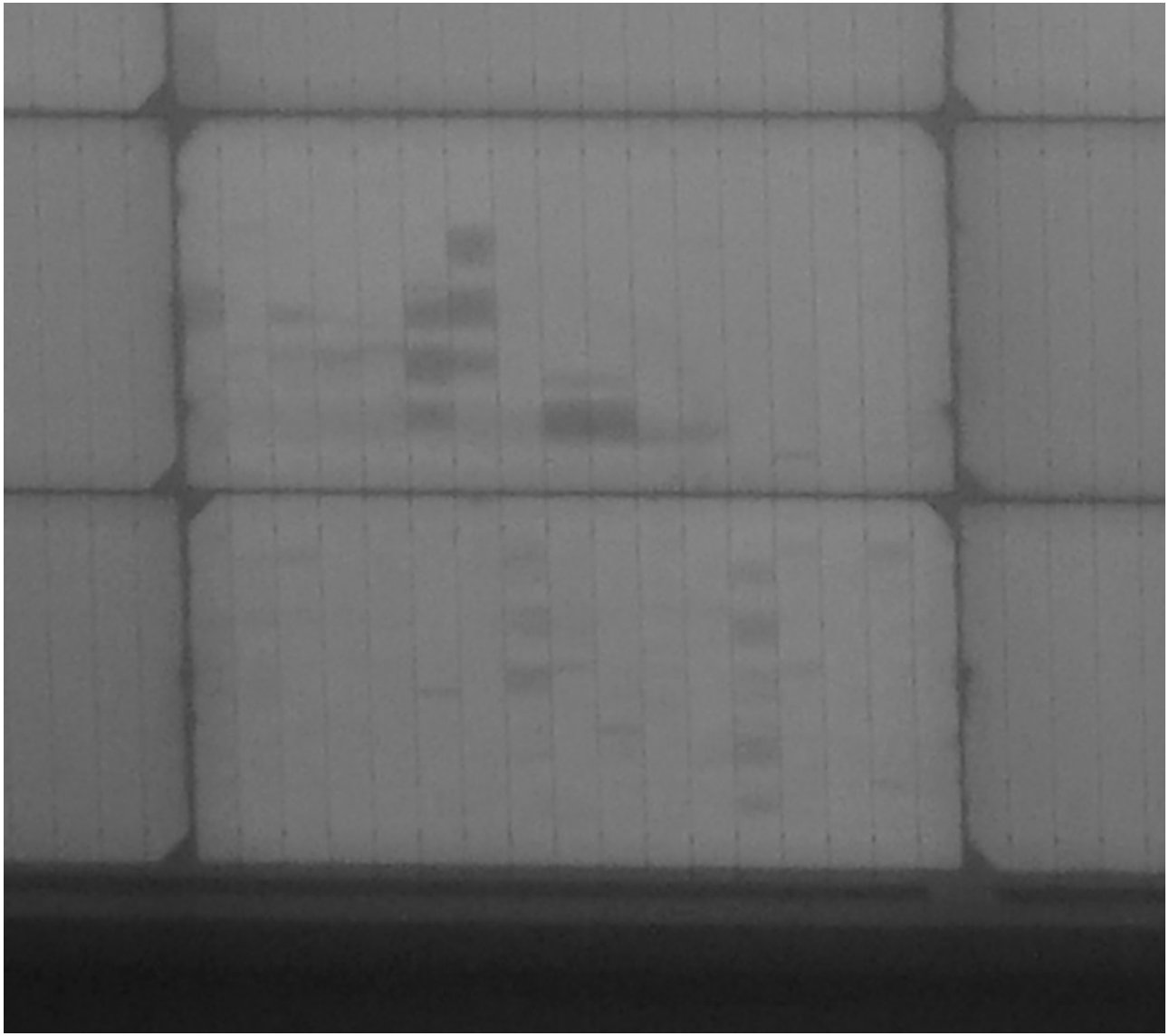


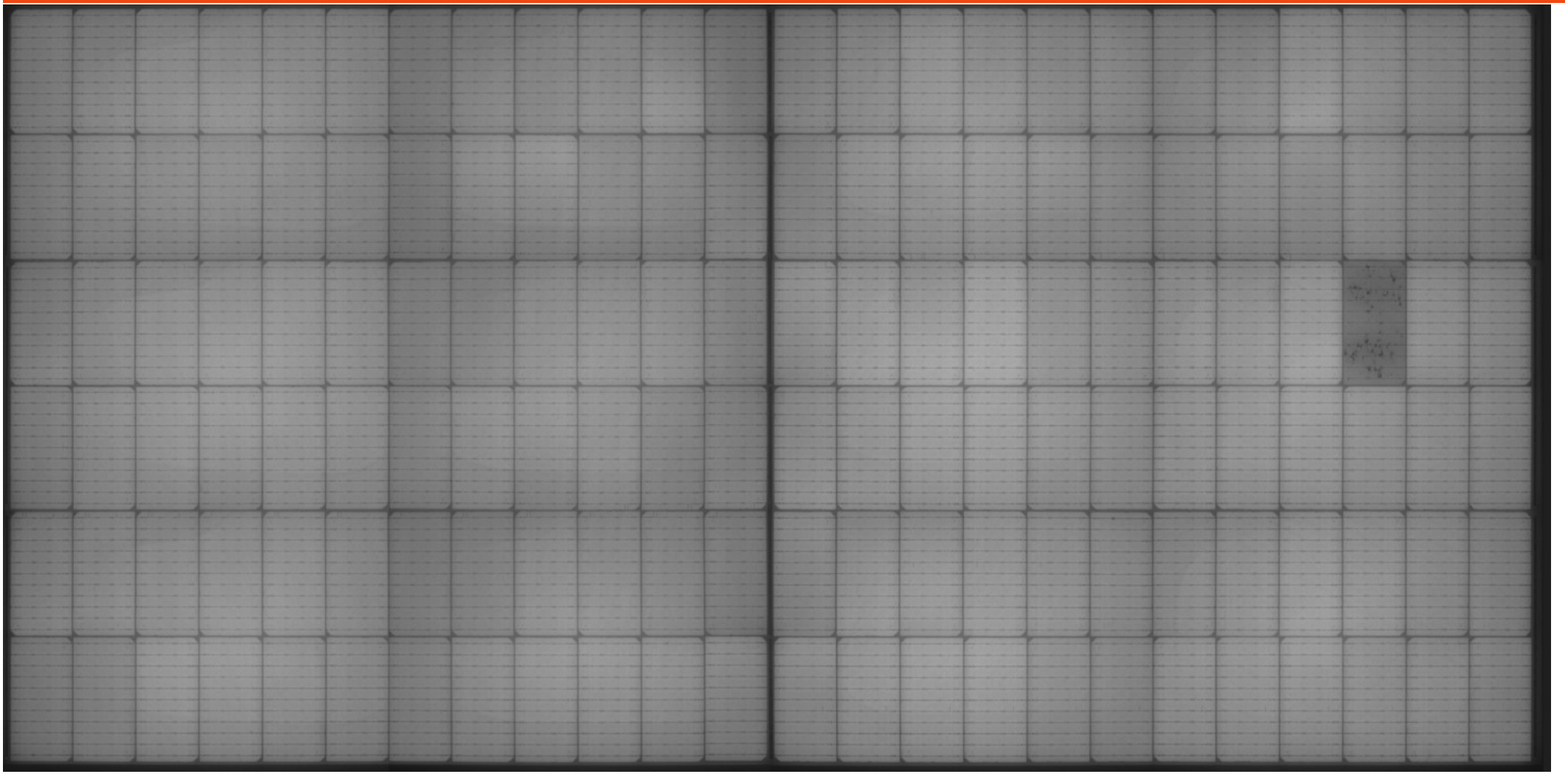




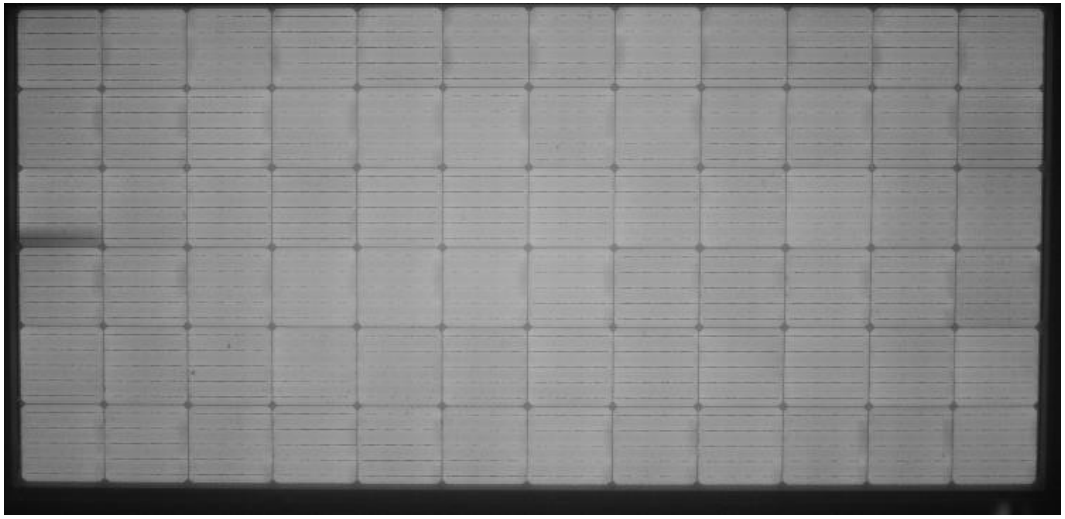


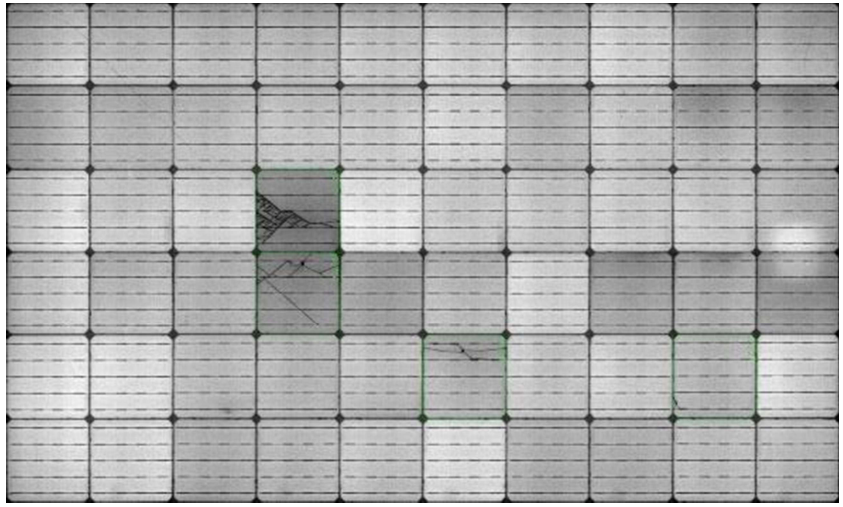


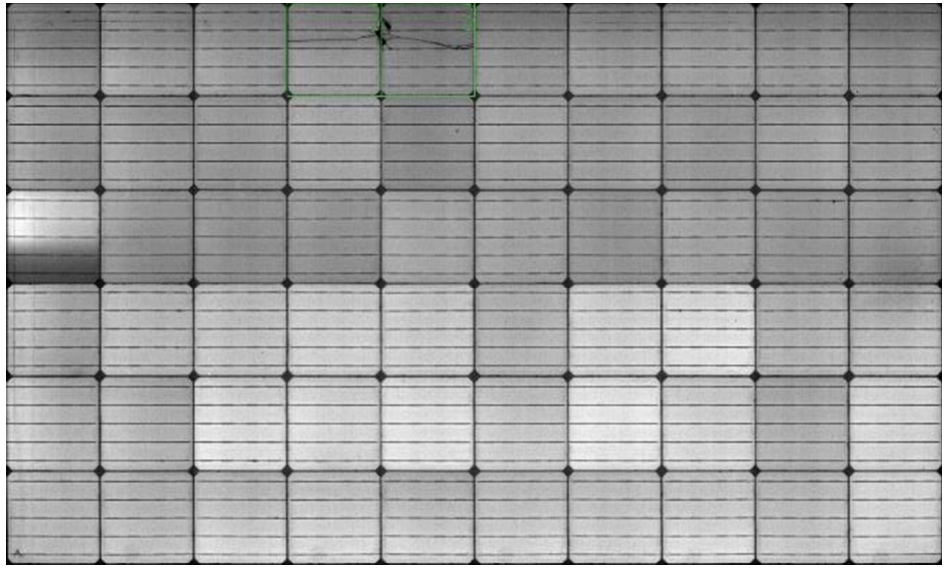


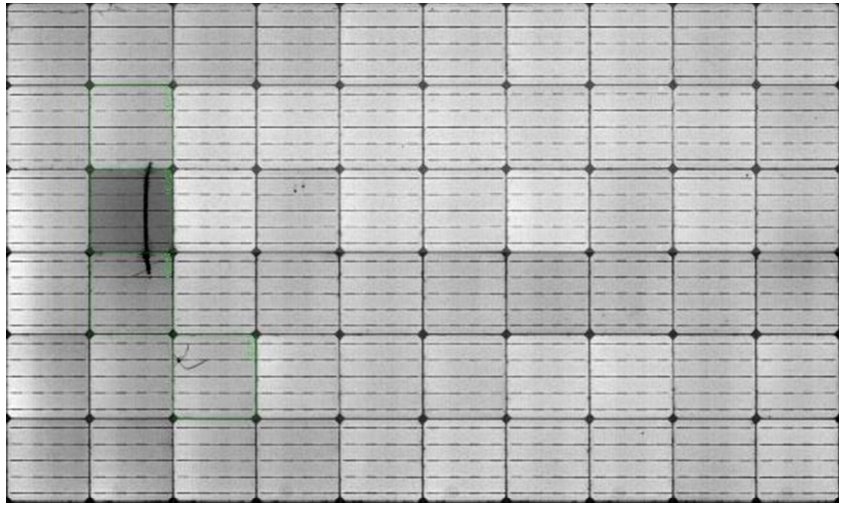


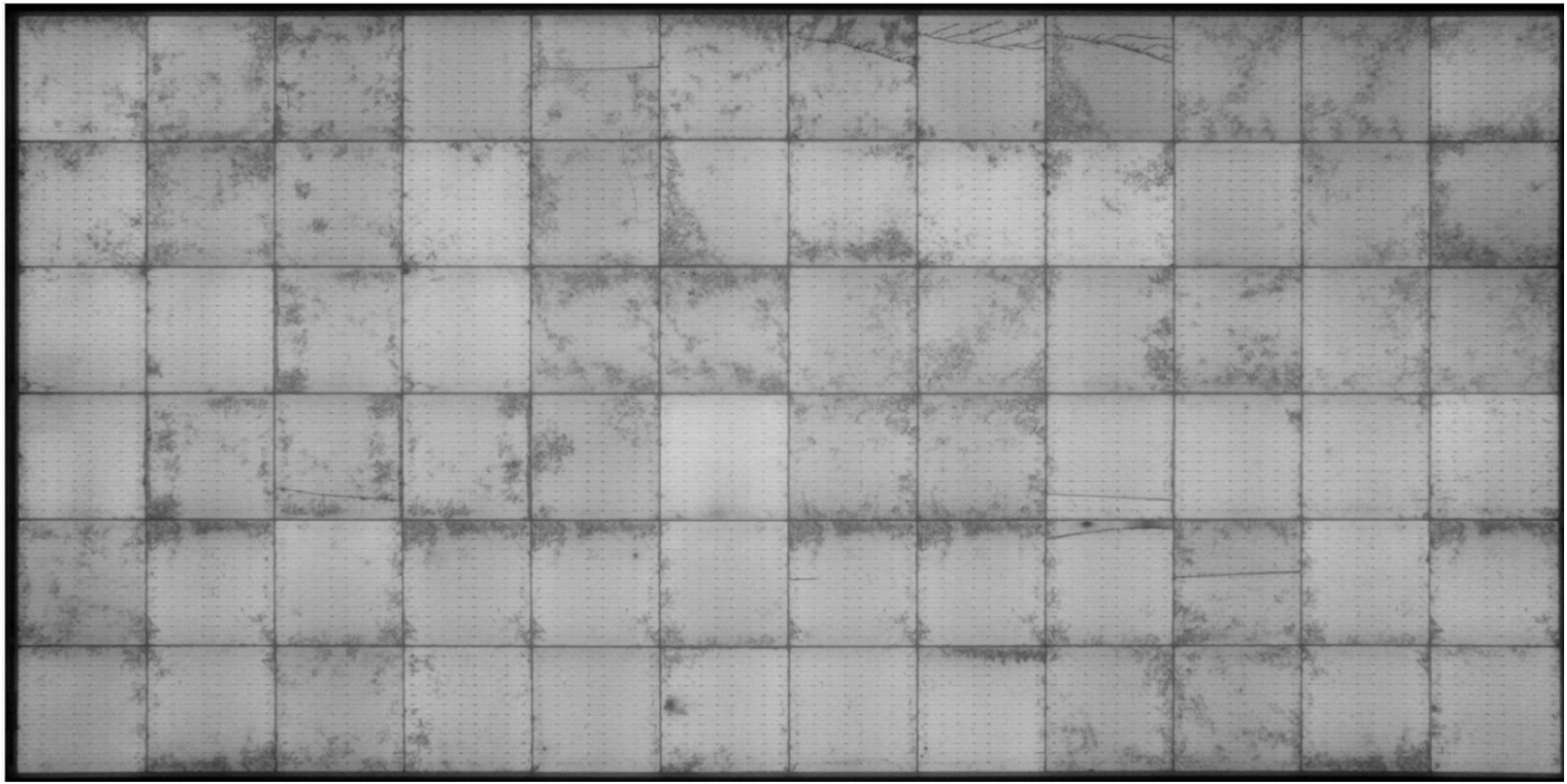
3.5. ELECTROLUMINESCENCE TEST. EXAMPLES





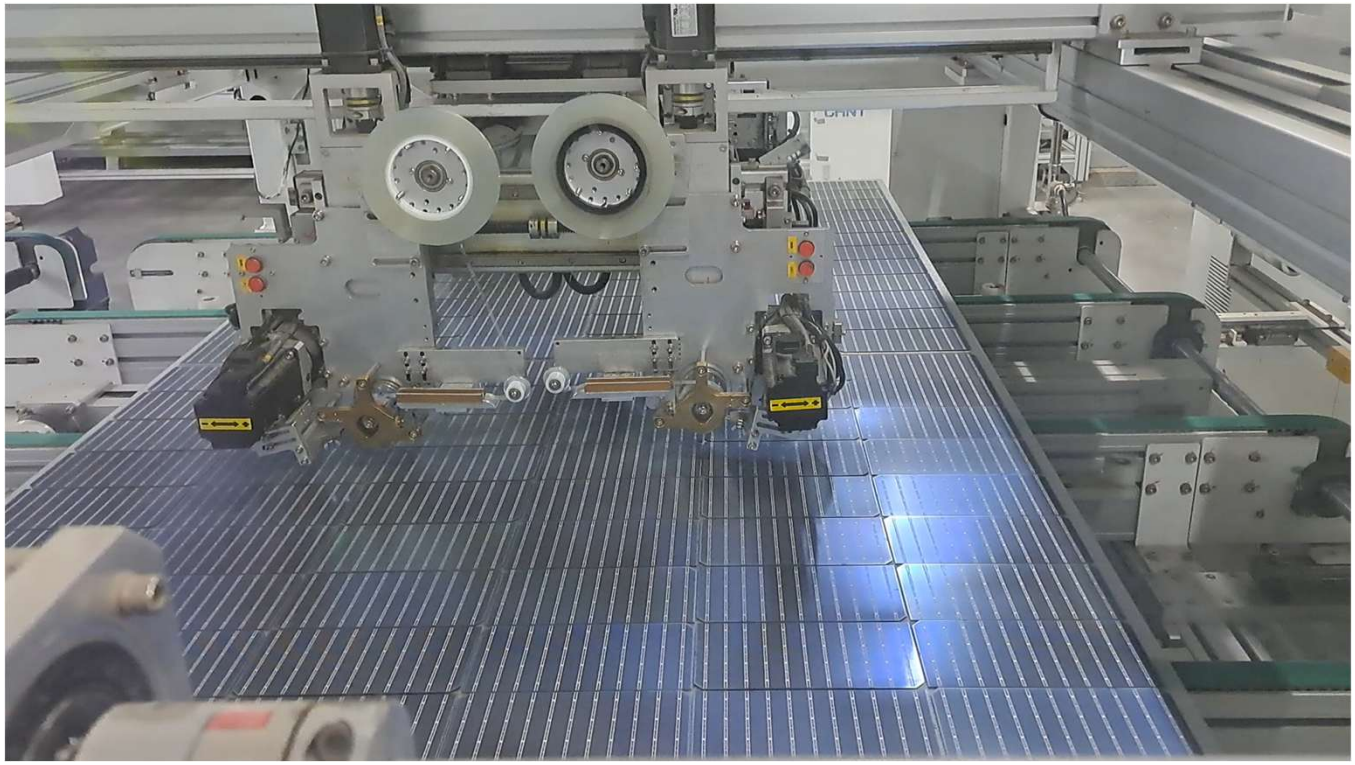


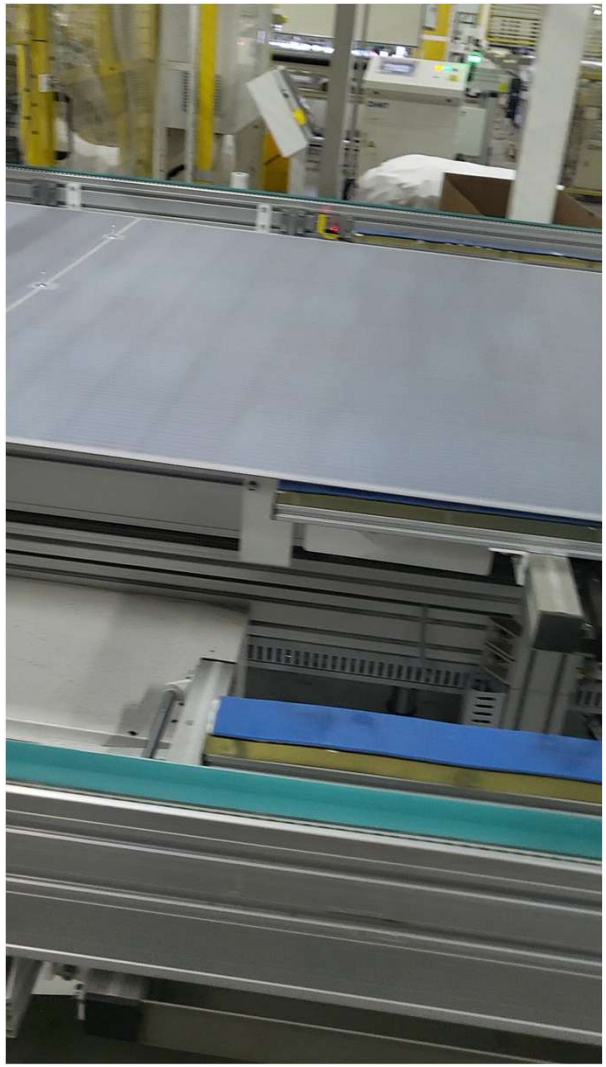




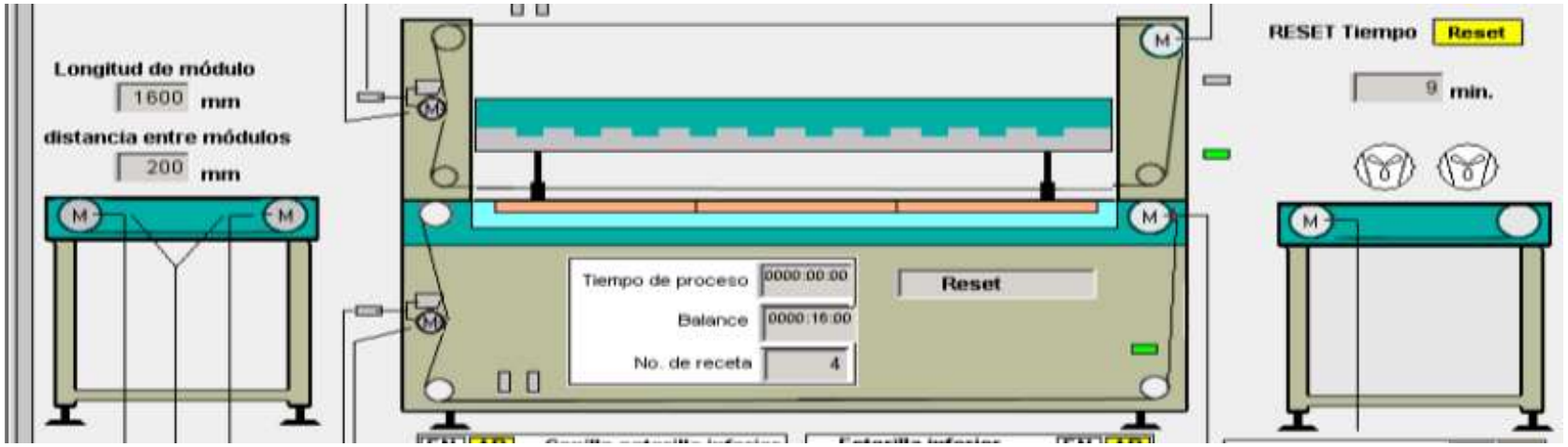
walk on solar panes el - YouTube

TAPE DISPENSER, BACK GLASS POISTIONING



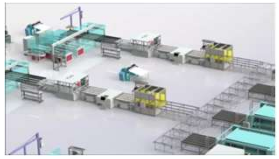


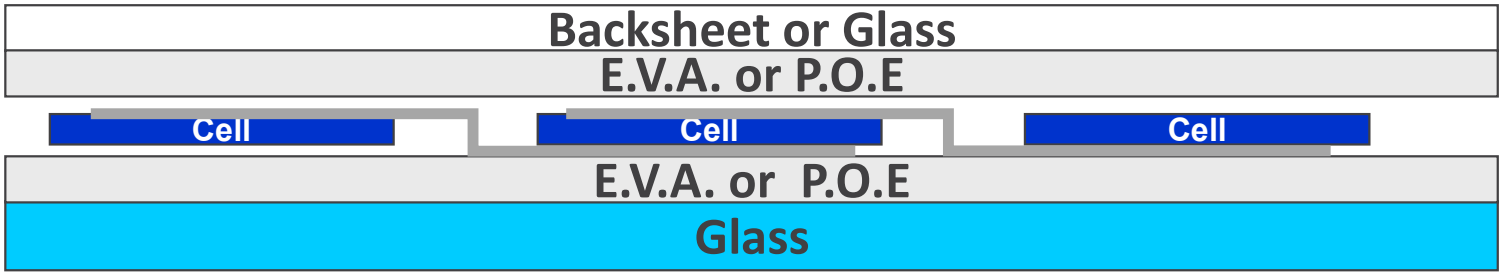
3.5. LAMINATION



3.5. LAMINATION

- A very good lamination process that permits a high percentage of cross-linking is essential.





3.5. EDGE TRIMMING

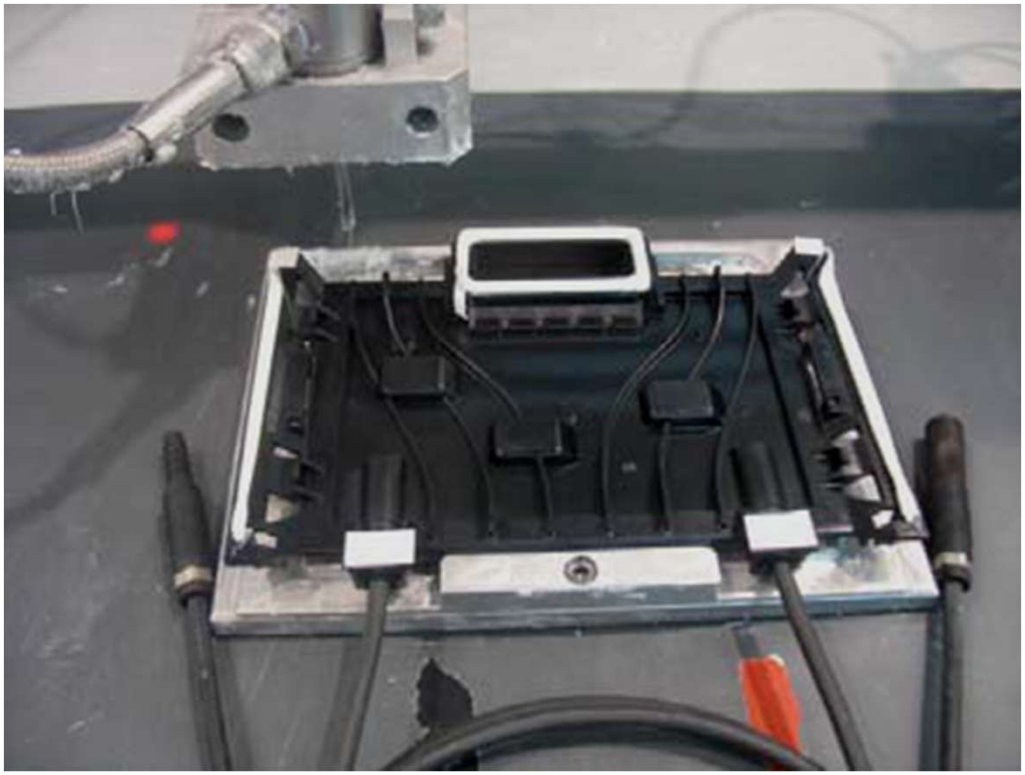
Manual

Automatic

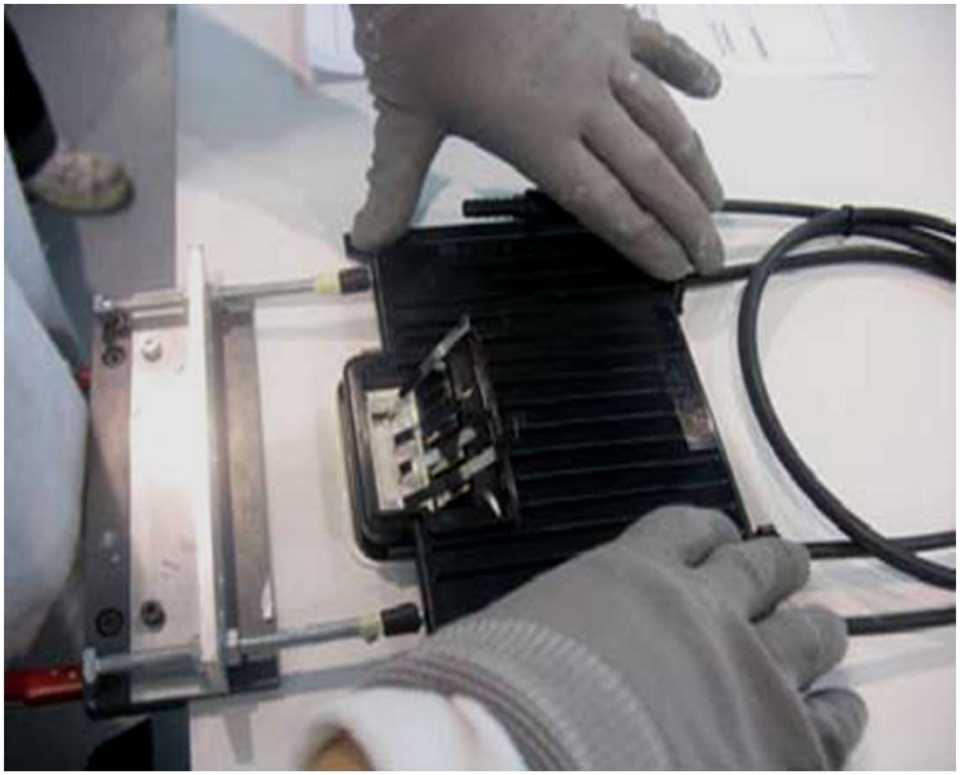
3.5. FRAMING

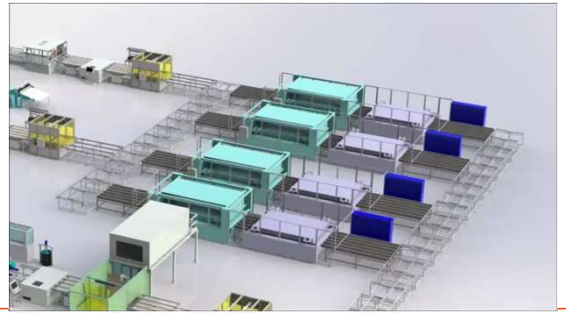
3.5. JUNCTION BOX (JB)

JB silicon dispenser

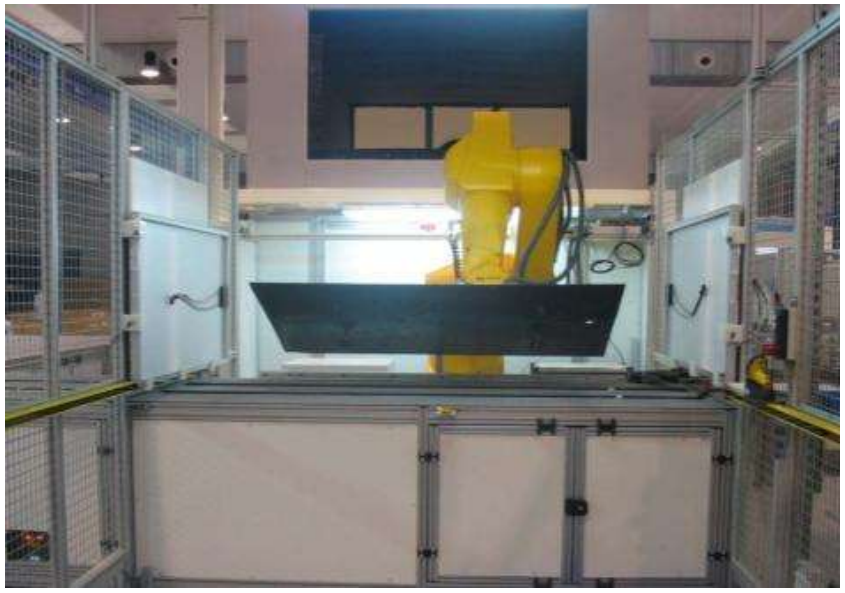


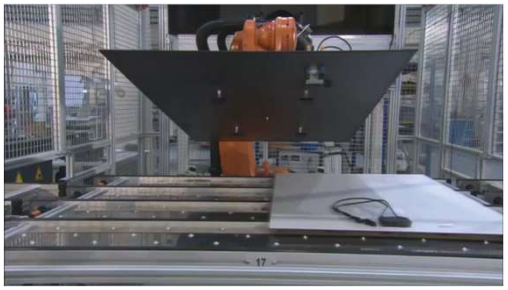
JB positioning and bus bars soldering





3.5. FLASH TESTER





3.5. FLASH TESTER. LABEL PRINTING

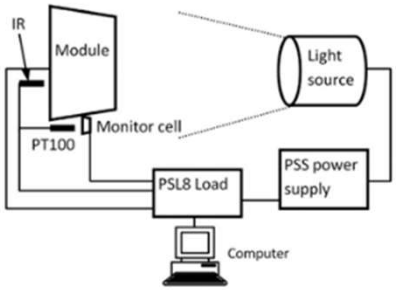
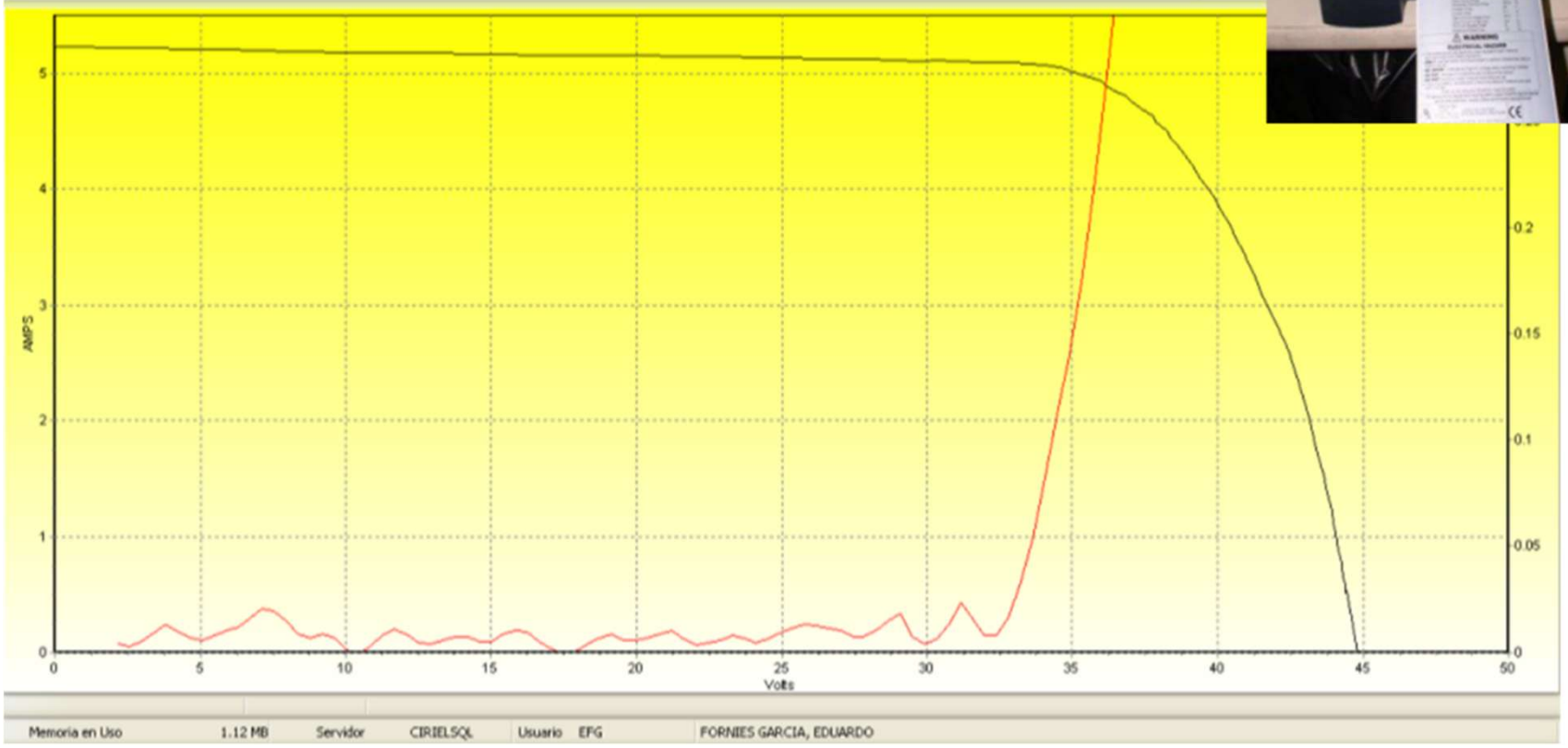
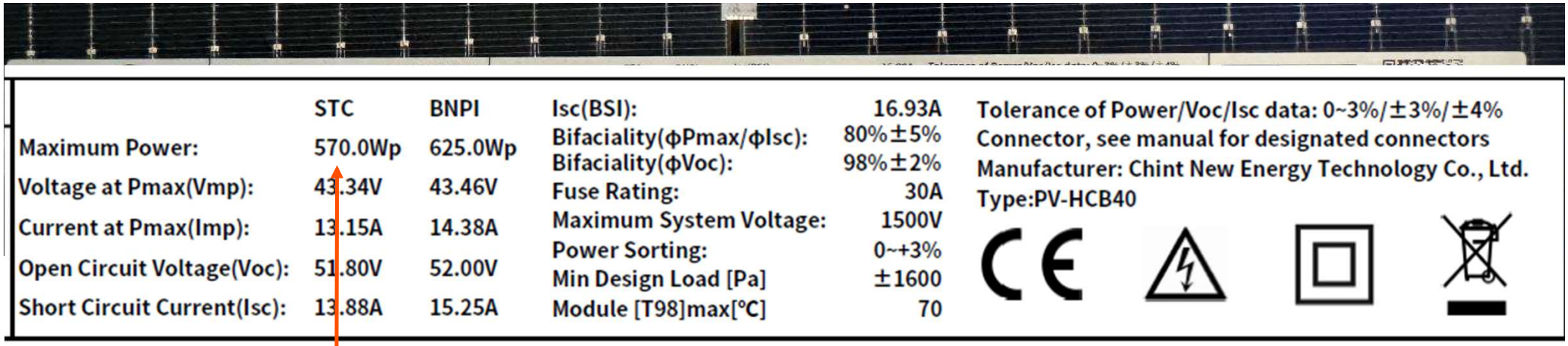


Fig. 2. Sketch of the set up of both commercial Flash tester.

FileName	Class	Fecha	Uoc	Isc	Pmpp	Umpp	Impp	Uop	Iop	Sop	StH	StW	FF	Rs	Rsh
000032784		12/09/08 14:05	44.83	5.23	177.3	36.0	4.9	30.0	5.1	0.0042	0.000	0.00	75.64	0.750	212.1



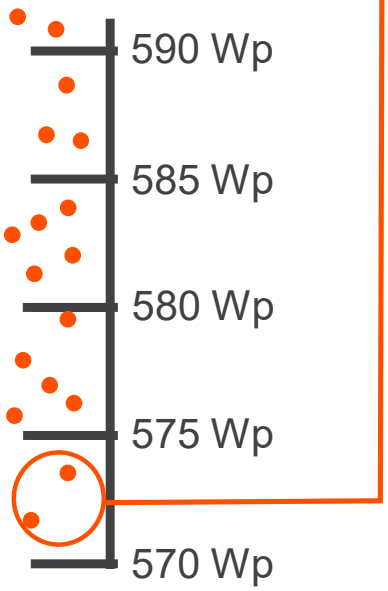
3.5. FLASH TESTER. LABEL PRINTING VS PTL

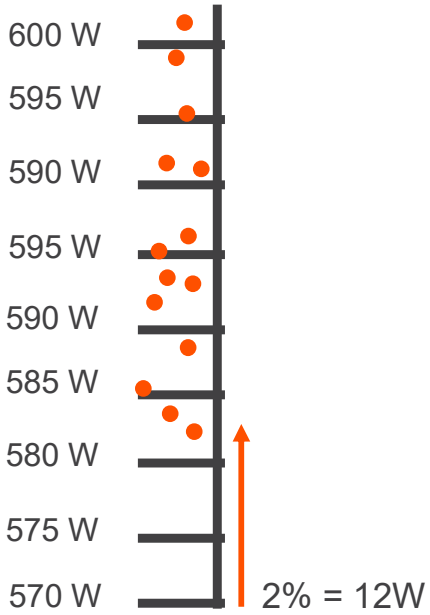


	STC	BNPI
Maximum Power:	570.0Wp	625.0Wp
Voltage at Pmax(Vmp):	43.34V	43.46V
Current at Pmax(Imp):	13.15A	14.38A
Open Circuit Voltage(Voc):	51.80V	52.00V
Short Circuit Current(Isc):	13.88A	15.25A

Isc(BSI):	16.93A
Bifaciality(ϕPmax/ϕIsc):	80%±5%
Bifaciality(ϕVoc):	98%±2%
Fuse Rating:	30A
Maximum System Voltage:	1500V
Power Sorting:	0~+3%
Min Design Load [Pa]	±1600
Module [T98]max[°C]	70

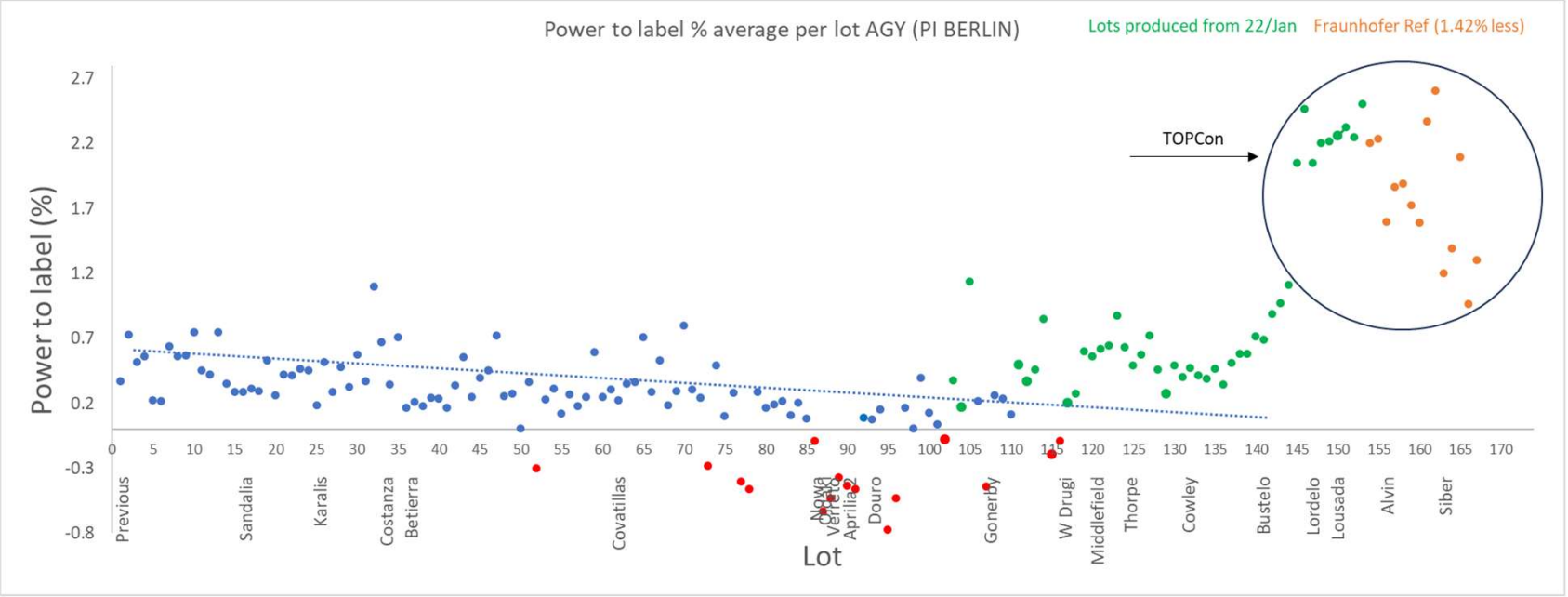
Tolerance of Power/Voc/Isc data: 0~3%/±3%/±4%
Connector, see manual for designated connectors
Manufacturer: Chint New Energy Technology Co., Ltd.
Type:PV-HCB40



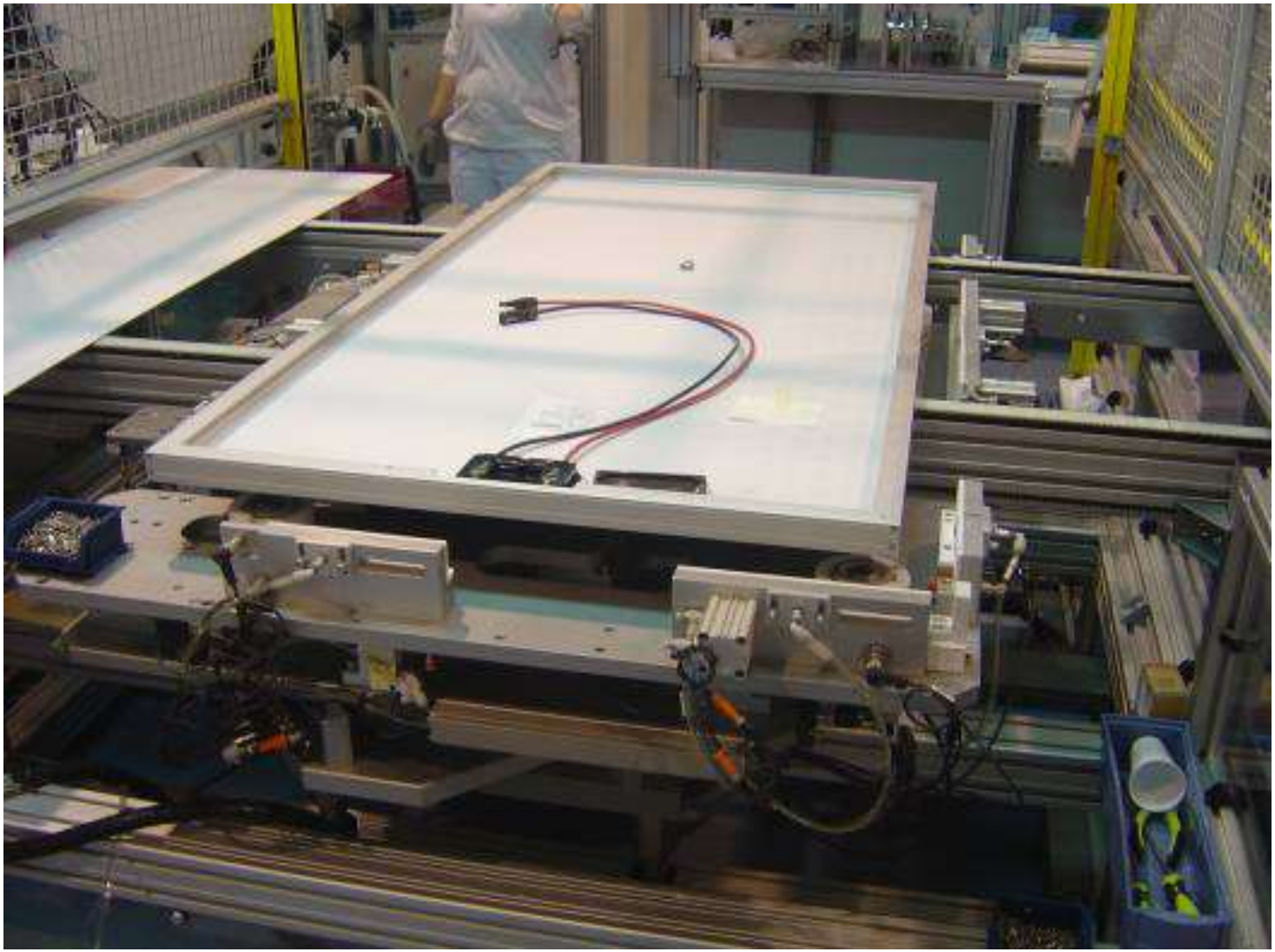


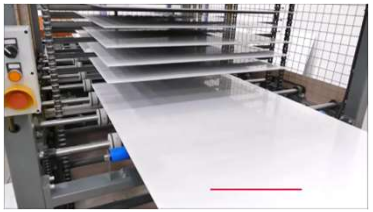
The result of the calibration as below table， and the test report in Fraunhofer see the attached， the test difference is from 11.4W to 13.7W

Module Type	Number	Tested by Astronergy	Tested by Fraunhofer	△
CHSM72N(DG)/F-BH	A212186142c91008254	577.75	566.35	-11.4
	A212186142c91008166	577.75	566.19	-11.6
	A212186142c91008086	575.6	561.95	-13.7
	A212186142c91008092	577.57	564.48	-13.1



3.5. HIGH POTENTIAL TEST. HI-POT

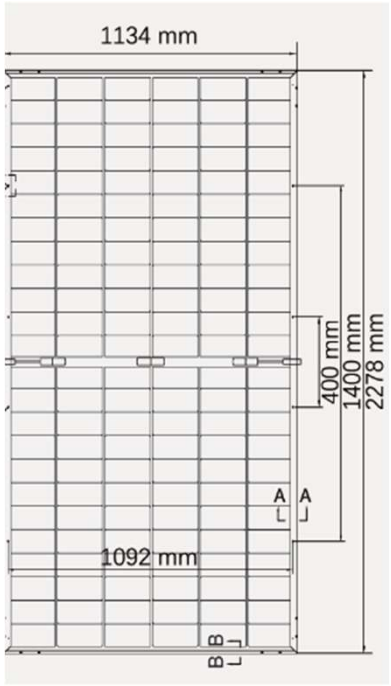




3.5. CRYSTALLINE MODULES. TYPES

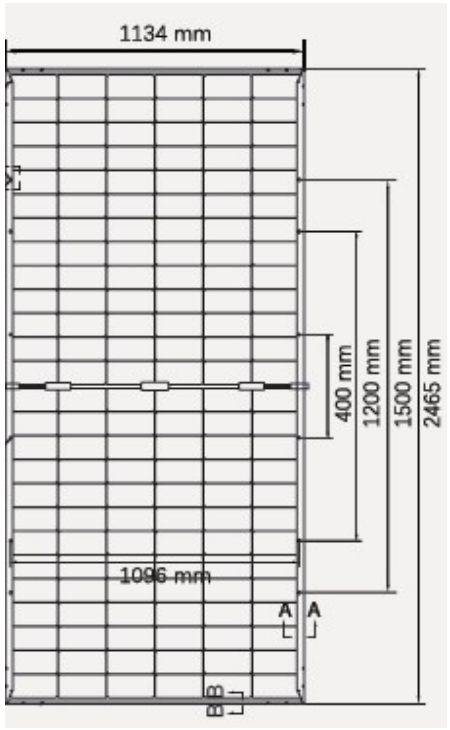
Pmpp = 595W
Voc = 51.90 V
Isc = 14.53 A
Eff = 23%

ASTRO N5
CHSM72N(DG)/F-BH
Bifacial Series
570~595W



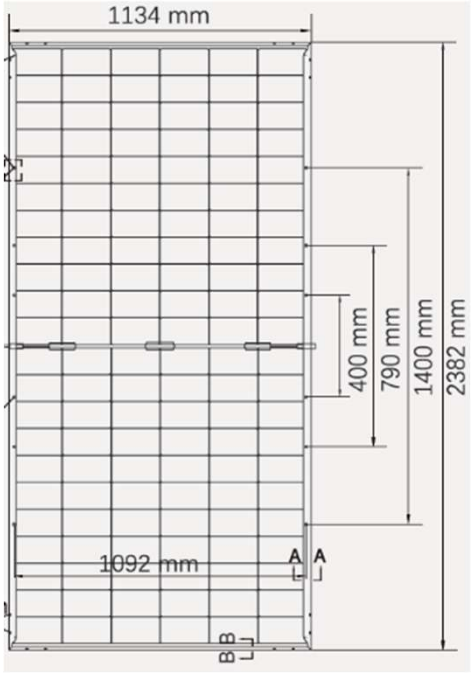
Pmpp = 645W
Voc = 57.08 V
Isc = 14.20 A
Eff = 23.1%

ASTRO N5
CHSM78N(DG)/F-BH
Bifacial Series
620~645W



Pmpp = 615W
Voc = 48.89 V
Isc = 16.02 A
Eff = 22.8%

ASTRO N7
CHSM66RN(DG)/F-BH
Bifacial Series
590~615W



s=33445mm²
8.60W/pcs
(**baseline**)

n-182
ASTRO N5
TOPCon

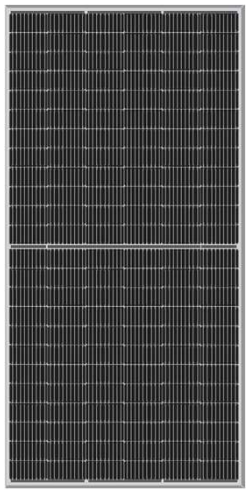
182mm×183.5mm
TOPCon cell eff. : 25.7%

s=38185mm²
9.81W/pcs
The area is increased by more than **14.2%**

n-210R
ASTRO N7
TOPCon 4.0

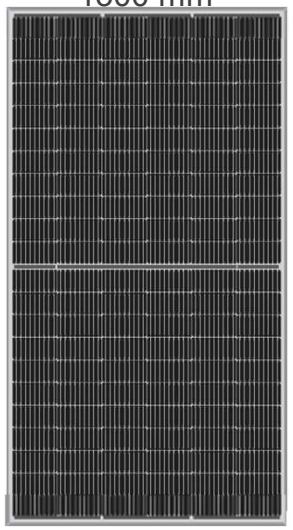
182mm ×210mm Rectangular
TOPCon cell eff. : 25.7%

3.5. CRYSTALLINE MODULES. TYPES



525-545 W (166 mm cell). 6x12x2

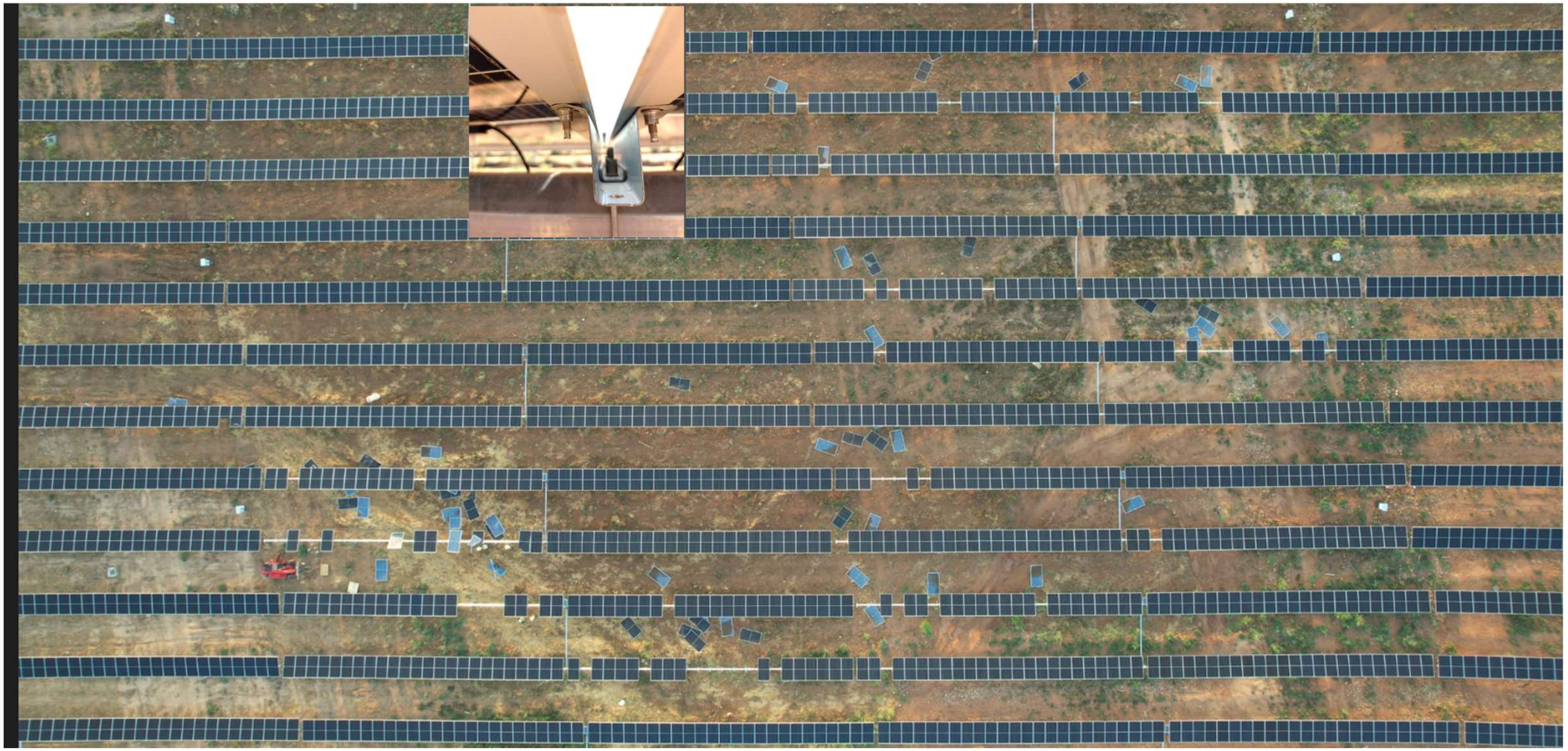
	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT
Maximum Power (Pmax)	525Wp	391Wp	530Wp	394Wp	535Wp	398Wp	540Wp	402Wp	545Wp	405Wp
Maximum Power Voltage (Vmp)	40.80V	37.81V	40.87V	37.88V	40.94V	37.94V	41.13V	38.08V	41.32V	38.25V
Maximum Power Current (Imp)	12.87A	10.33A	12.97A	10.41A	13.07A	10.49A	13.13A	10.55A	13.19A	10.60A
Open-circuit Voltage (Voc)	49.42V	46.65V	49.48V	46.70V	49.54V	46.76V	49.73V	46.94V	49.92V	47.12V
Short-circuit Current (Isc)	13.63A	11.01A	13.73A	11.09A	13.83A	11.17A	13.89A	11.22A	13.95A	11.27A
Module Efficiency STC (%)	20.36%		20.55%		20.75%		20.94%		21.13%	
Operating Temperature(℃)	-40℃~+85℃									
Maximum system voltage	1500VDC (IEC)									
Maximum series fuse rating	30A									
Power tolerance	0~+3%									
Temperature coefficients of Pmax	-0.35%/℃									
Temperature coefficients of Voc	-0.28%/℃									
Temperature coefficients of Isc	0.048%/℃									
Nominal operating cell temperature (NOCT)	45±2℃									
Refer. Bifacial Factor	70±5%									



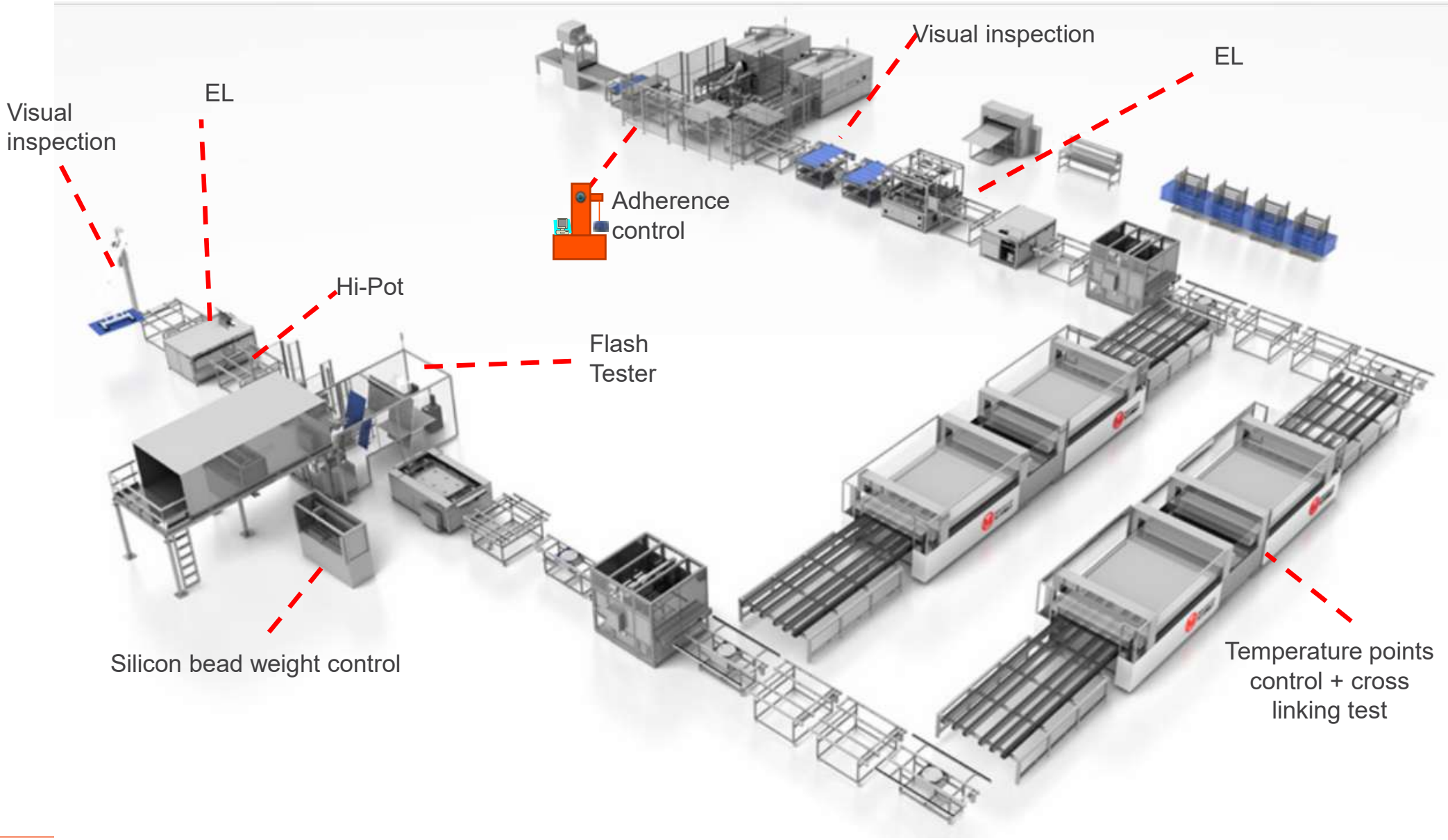
645-660 W (210 mm cell). 6x11x2

STC: Irradiance 1000W/m², Cell Temperature 25° C, AM=1.5		**Bifacial factor:** Average 65%		
Rated output (Pmpp / Wp)	645	650	655	660
Rated voltage (Vmpp / V)	37.48	37.68	37.88	38.08
Rated current (Impp /A)	17.21	17.26	17.30	17.34
Open circuit voltage (Voc / V)	45.29	45.49	45.69	45.89
Short circuit current (Isc /A)	18.27	18.32	18.37	18.42
Module efficiency	20.8%	20.9%	21.1%	21.2%

Temperature Ratings (STC)		Operating Parameters	
Temperature coefficient (Pmpp)	-0.34%/°C	No. of diodes	3
Temperature coefficient (Isc)	+0.04%/°C	Junction box IP rating	IP 68
Temperature coefficient (Voc)	-0.25%/°C	Max. series fuse rating	35 A
Nominal module operating temperature (NMOT)	41±2°C	Max. system voltage (IEC/UL)	1500V_{DC}

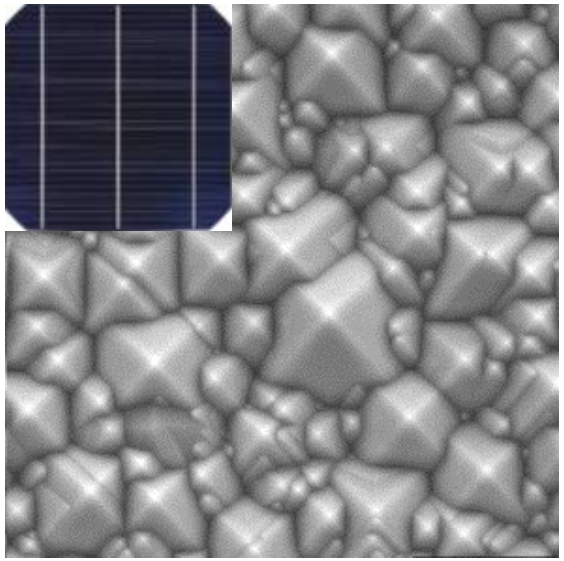


3.5. LINE LAY OUT. QUALITY CONTROLS

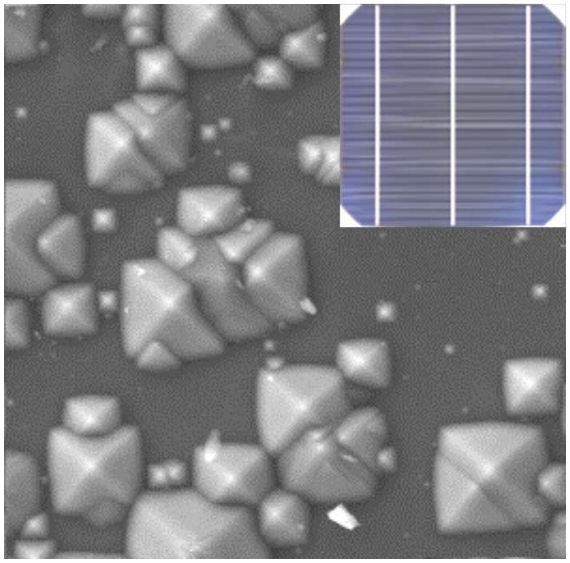


3.6. TYPICAL DEFECTS. SOLAR CELLS. TEXTURE

GOD TEXTURING PROCESS (NAOH). MONO

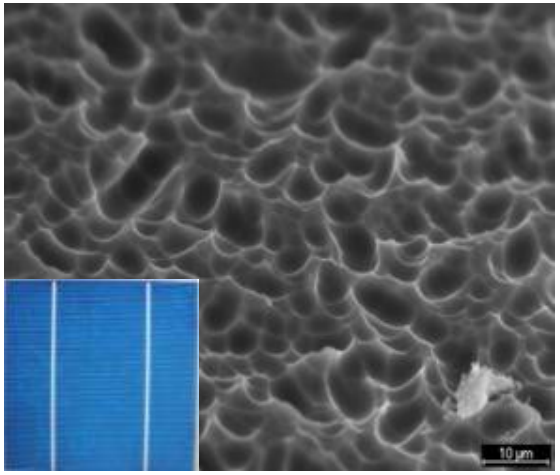


Bad texturing process (NaOH). Mono

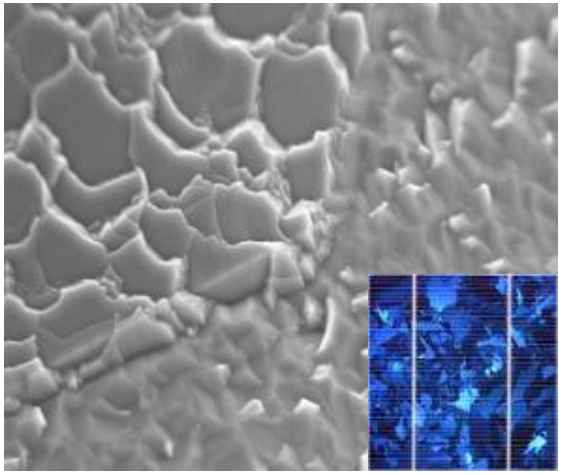


Up to 5% of power loss

God texturing process (NHO_3). Multi

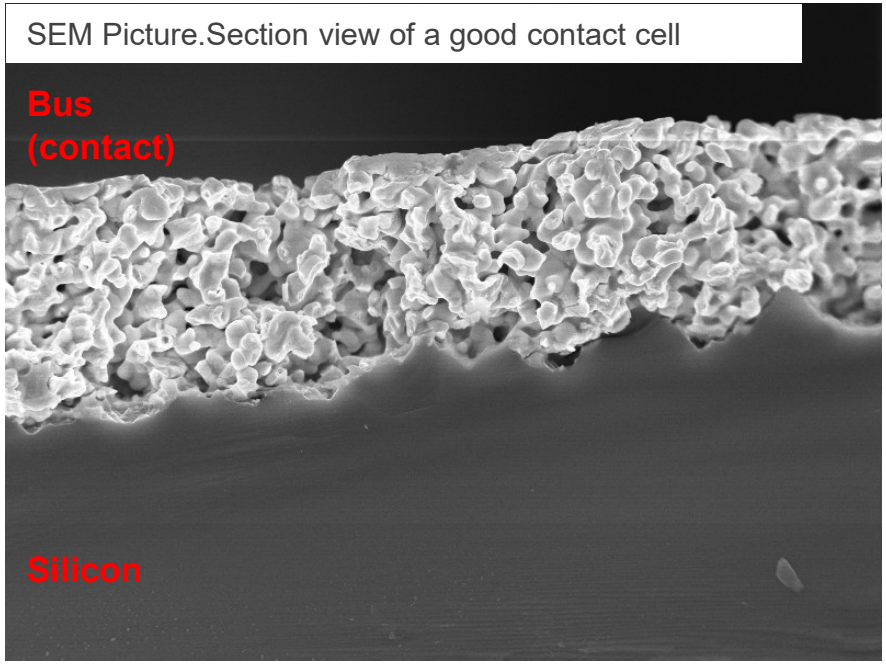


Bad texturing process (NaOH). Multi

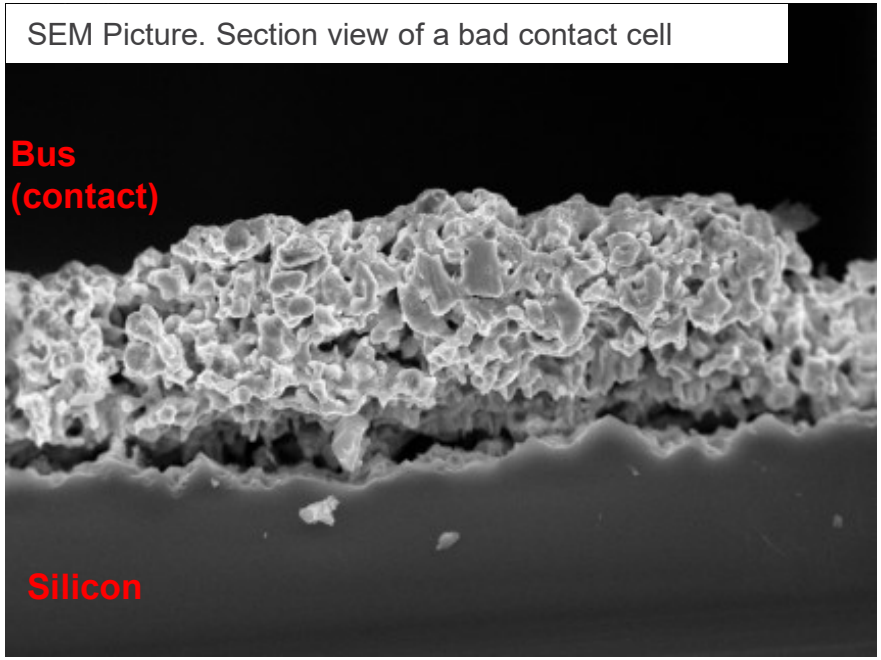


3.6. TYPICAL DEFECTS. SOLAR CELLS. CONTACTS

SEM Picture.Section view of a good contact cell

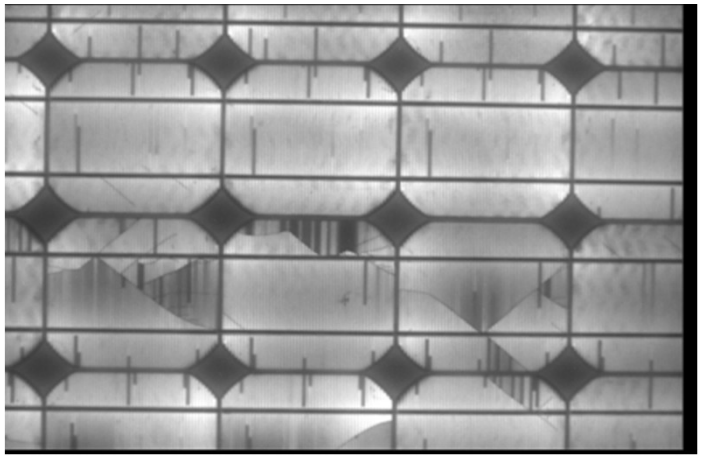


SEM Picture. Section view of a bad contact cell



E. Forniés et al. Cell-to-module losses in standard crystalline PV modules. An industrial approach. 2015. Photovoltaics International. 29. pp 91-100

EL Picture of a module with broken and bad contact cells

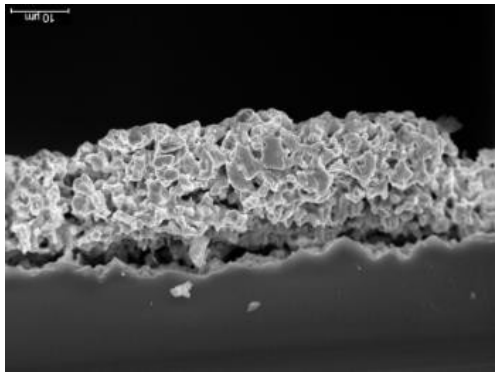


Bad contacts due to a failure in sreen print or firing processes. →

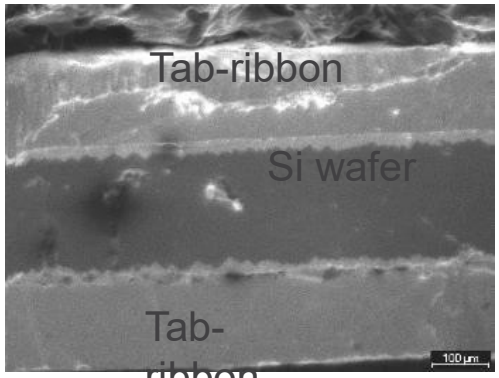
Increase of series resistance, temperature increase→

LOSS OF POWER ALONG THE YEARS!!

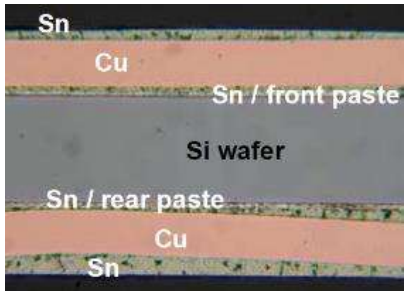
3.6. TYPICAL DEFECTS. SOLAR CELLS. CONTACTS



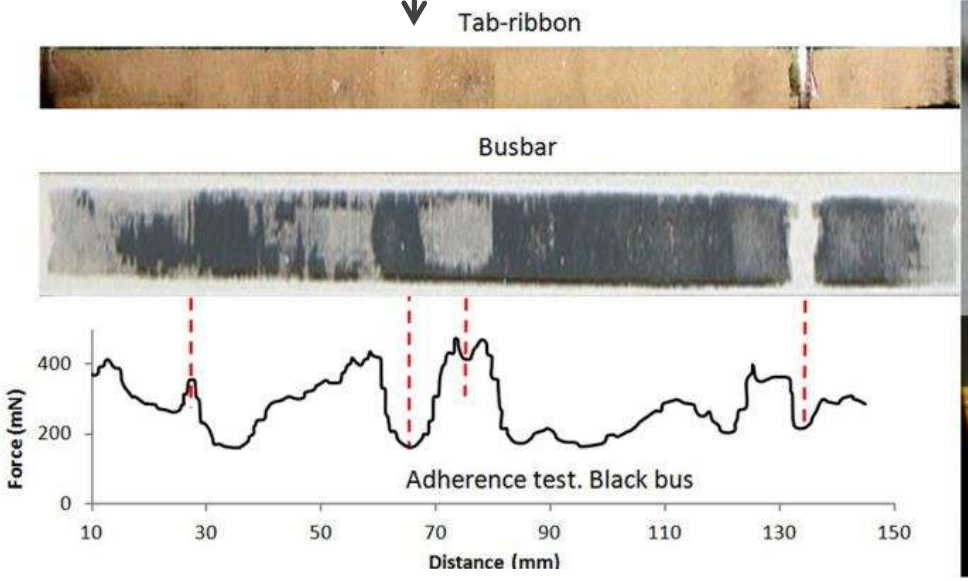
Good soldering

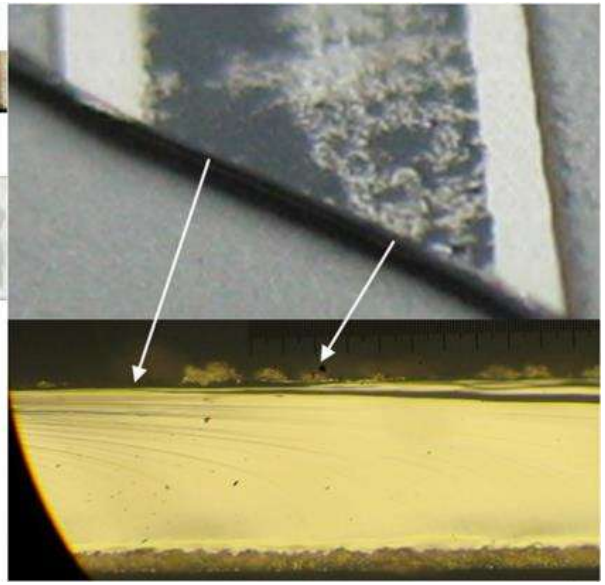


100 µm



D.H. Neuhaus et al. "Loss Analysis of Solar Modules by Comparison of IV Measurements and Prediction from IV Curves of Individual Solar Cells", 20th EUPVSEC, Barcelona, Spain, 2005, pp. 1947-1952.

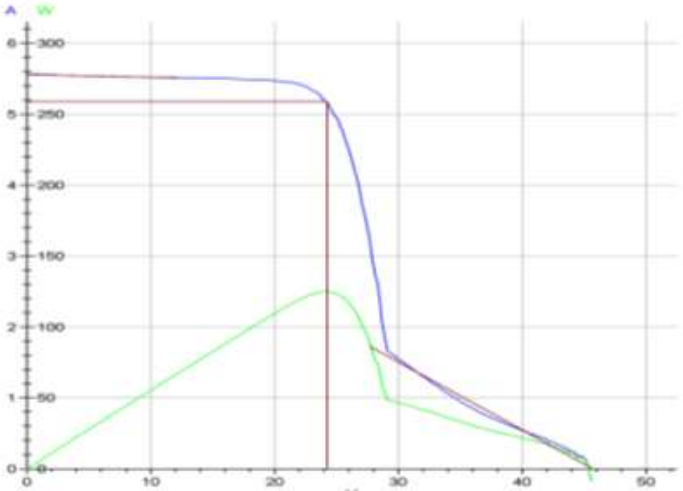


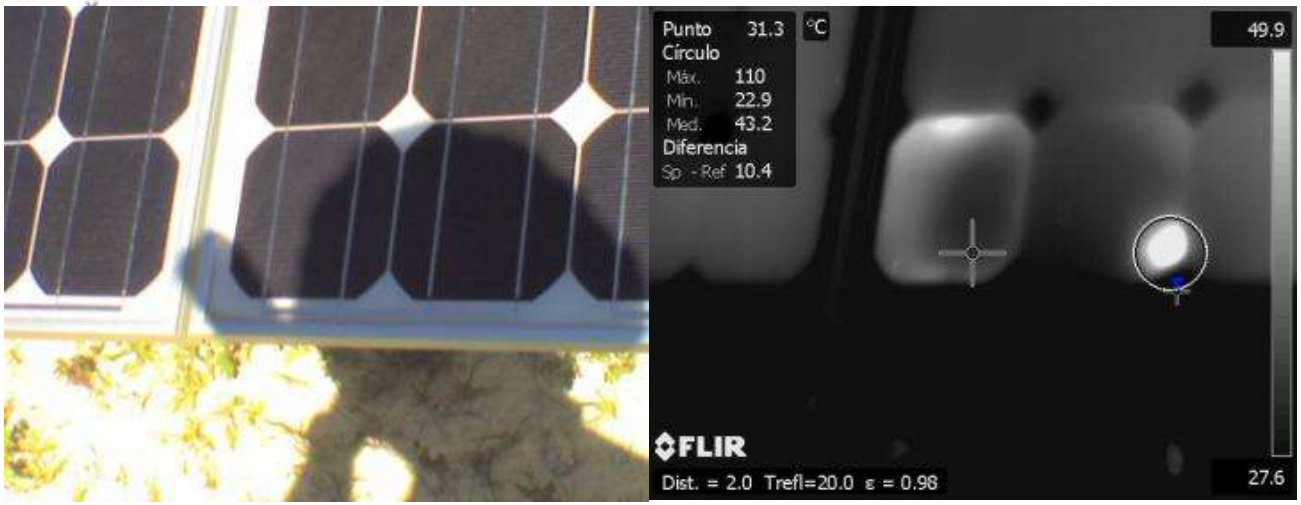


E. Forniés et al. Cell-to-module losses in standard crystalline PV modules. An industrial approach. 2015. Photovoltaics International. 29. pp 91-100

3.6. TYPICAL DEFECTS. RSH

Rsh. Very low cell Rsh can lead to hot spots

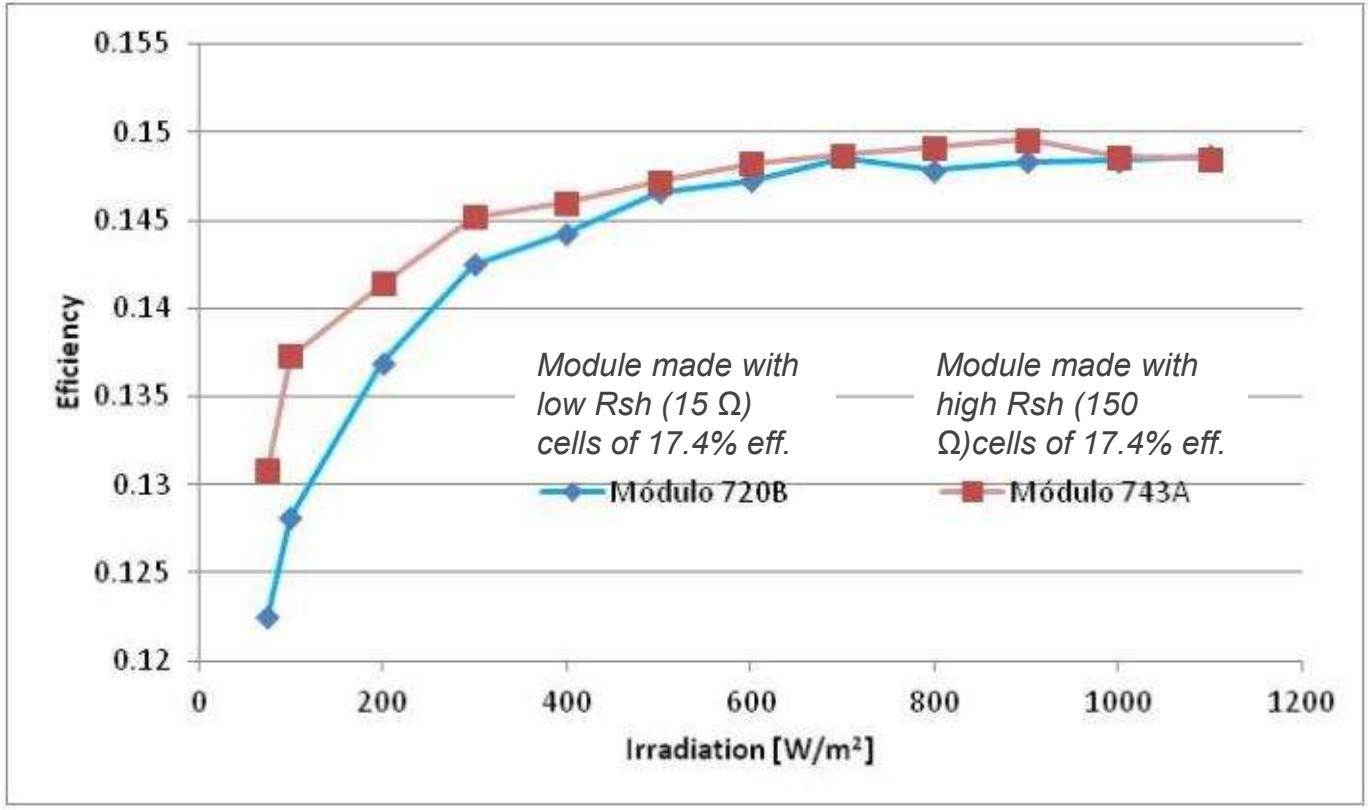




3.6. TYPICAL DEFECTS. RSH

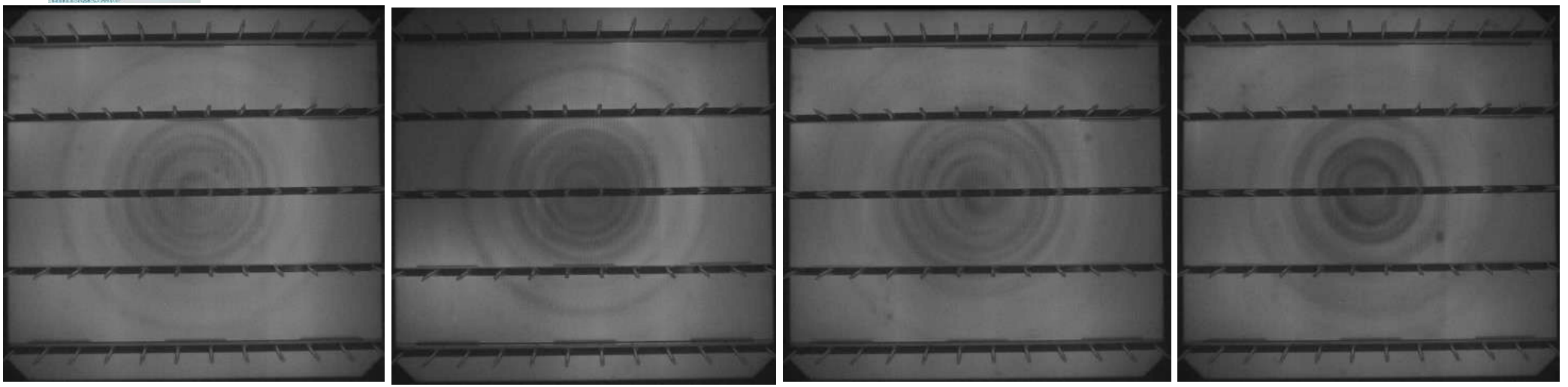
Rsh. Slightly low cell Rsh leads to low module performance at low radiation conditions

E. Forniés et al. Method for module Rsh determination and its comparison with standard methods. 2014. Solar Energy 109. pp 189-199

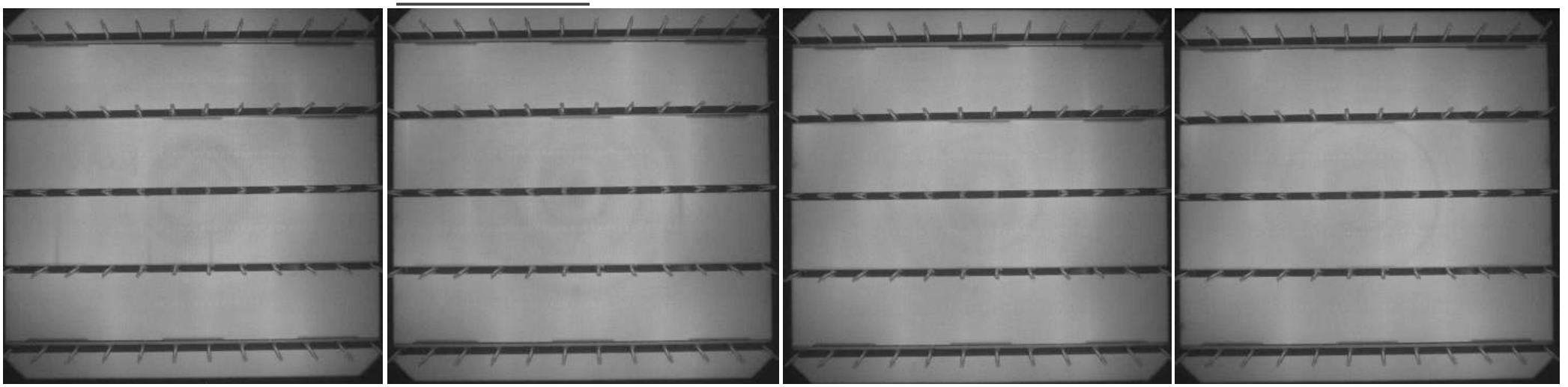


3.6. TIPICAL DEFECTS. SOLAR CELLS/GROWING. TEXTURE

Eff 19.20%. Oxygen precipitates in the top of the mono ingot

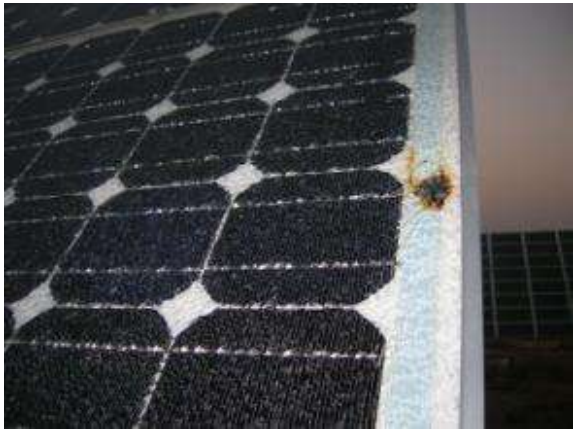


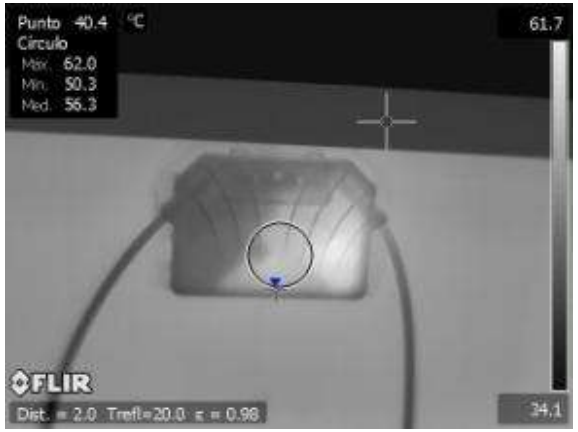
Eff 19.85%



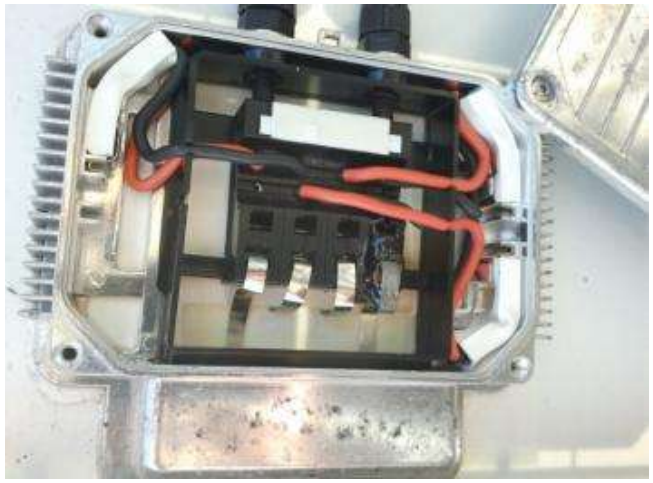
3.6. TYPICAL DEFECTS. DIODES AND JB

Diodes. Bad quality diodes. Unnecessary bypassing and final diode burning





Juntion Box. Wrong mechanical connection.



Juntion Box. Bad soldering

3.6. TYPICAL DEFECTS. LAMINATION PROCESS.

1st problem. (Normal)
EVA (or backsheet?) darkening
→

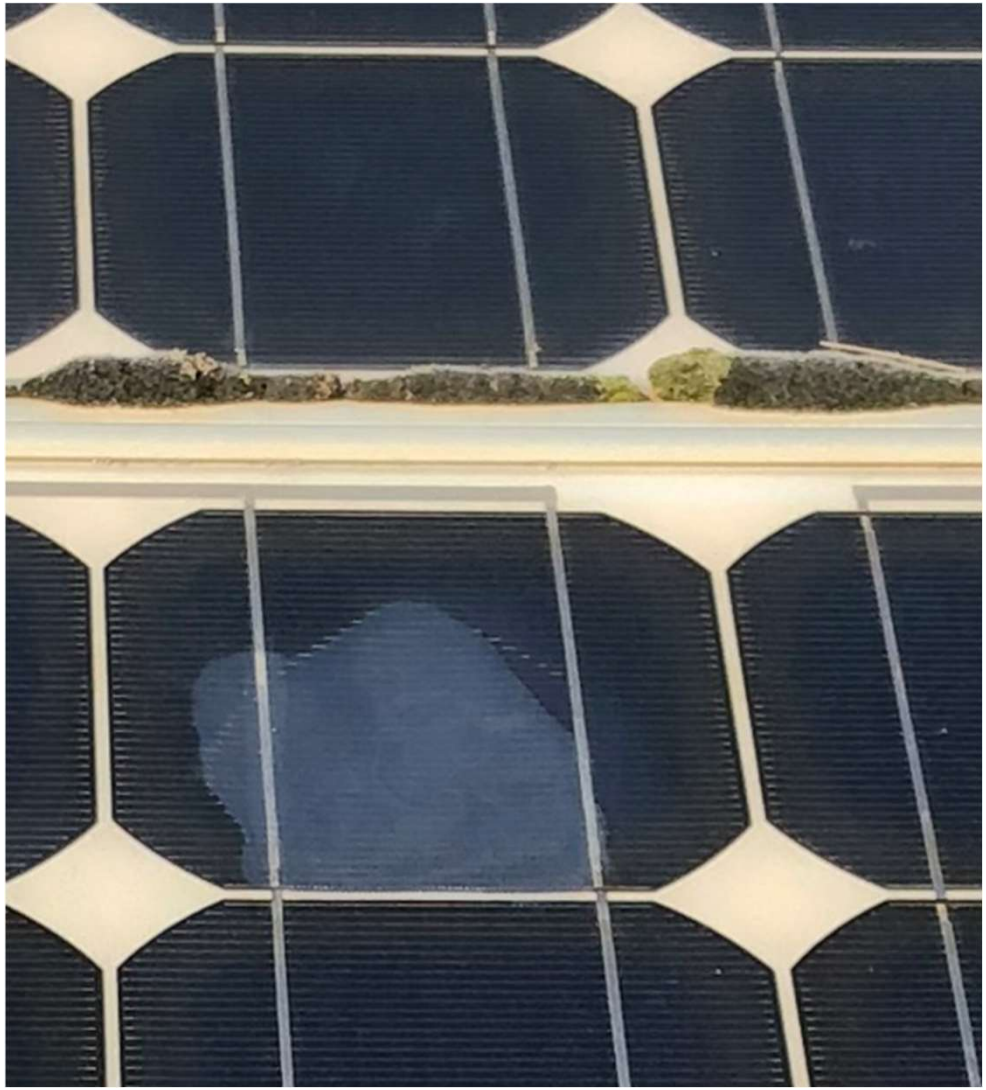
Annually Energy Yield loss due to darkening of the EVA.

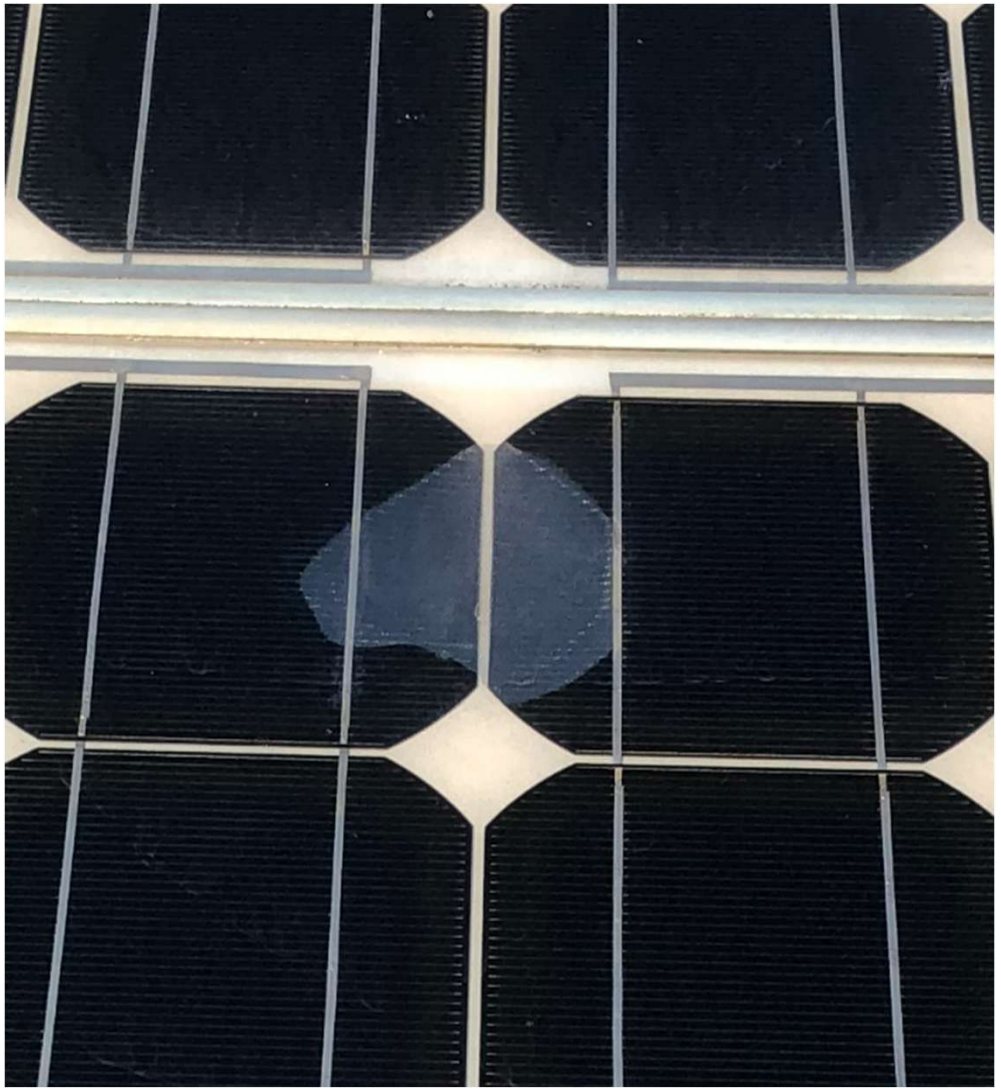
2nd problem (catastrophic)
Bad curing EVA →

Moisture ingress, Electrical isolation failure.

3.6. TYPICAL DEFECTS. LAMINATION PROCESS.

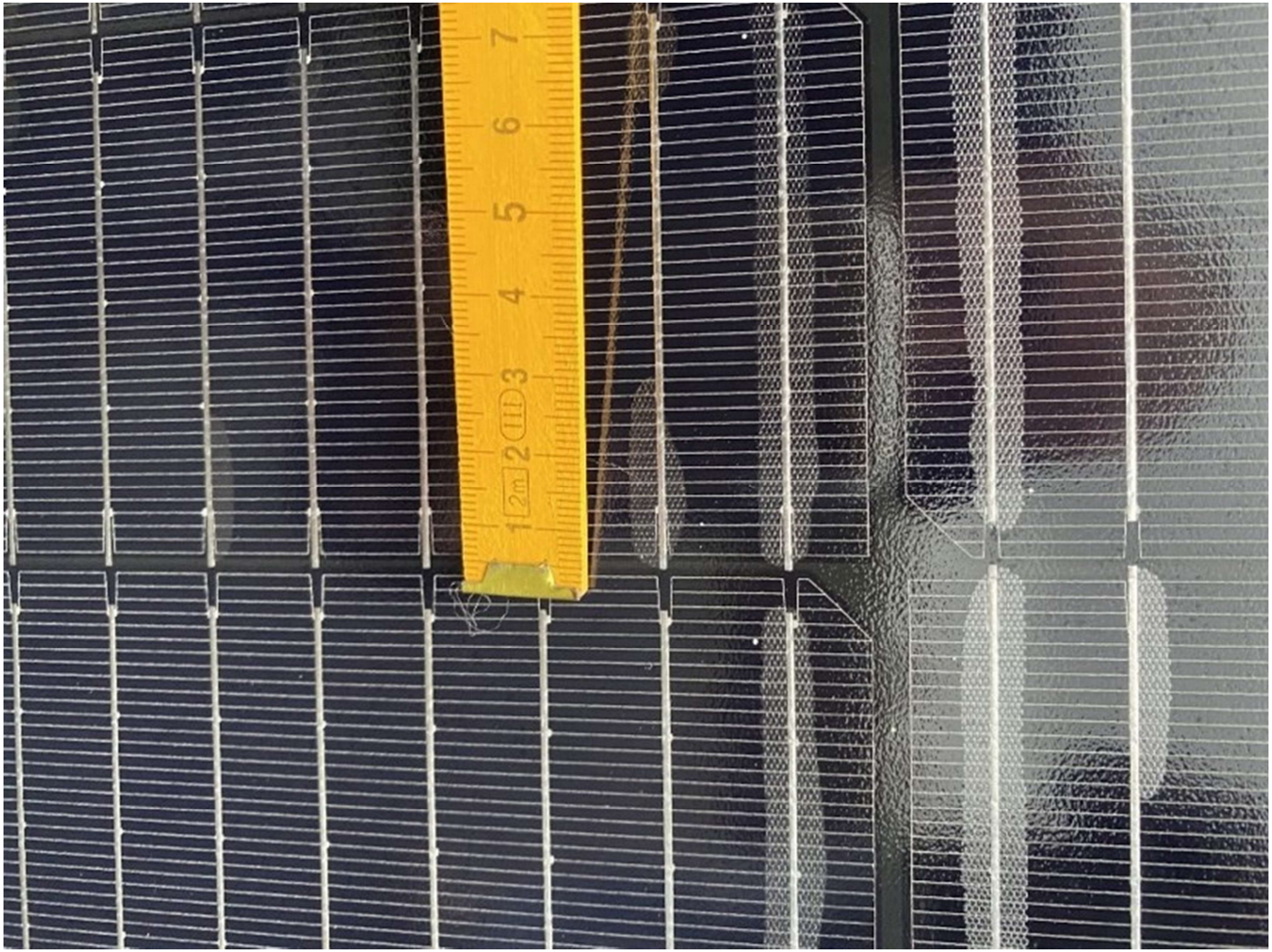
EVA/Cell delamination





3.6. TYPICAL DEFECTS

Delamination



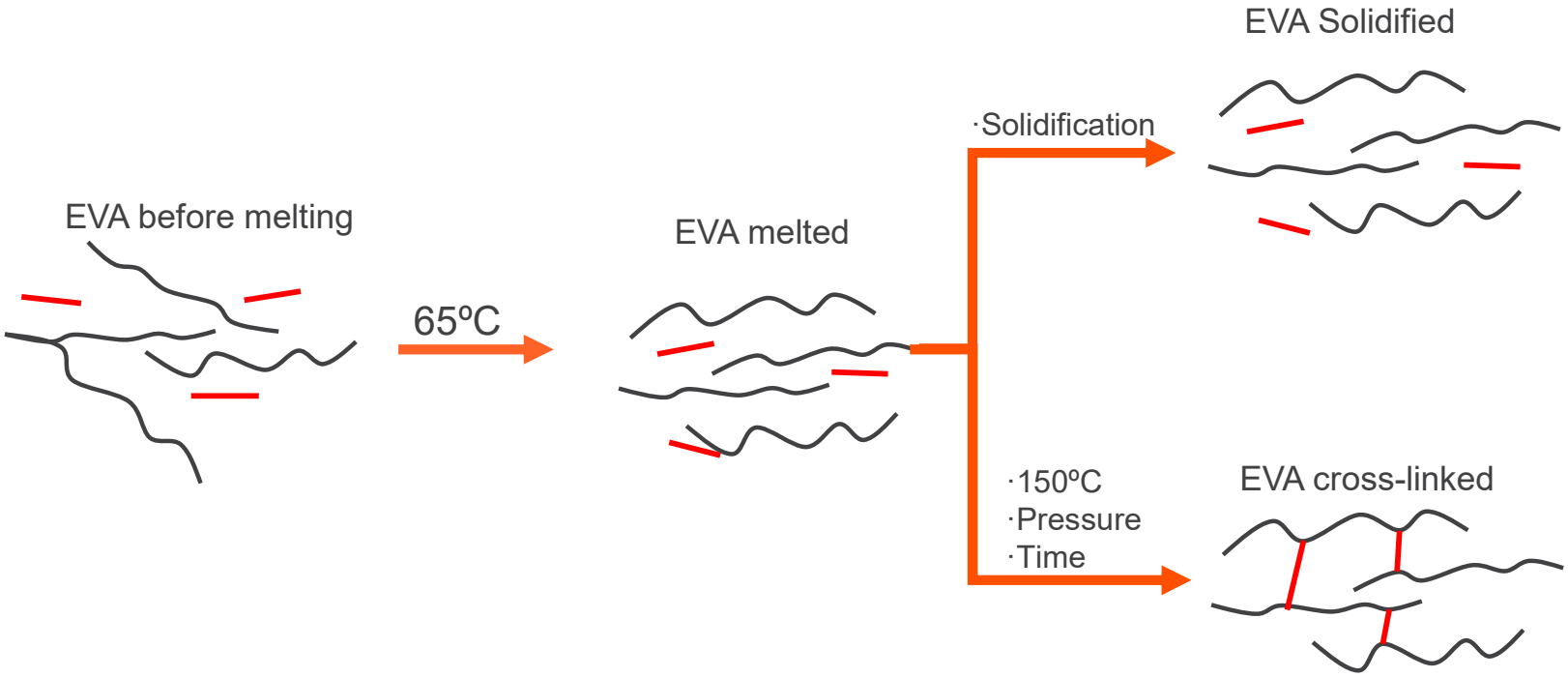
3.6. TYPICAL DEFECTS. LAMINATION PROCESS.

2nd problem
(catastrophic)
Bad curing EVA →

Complete flaking

3.6. TYPICAL DEFECTS. LAMINATION PROCESS.

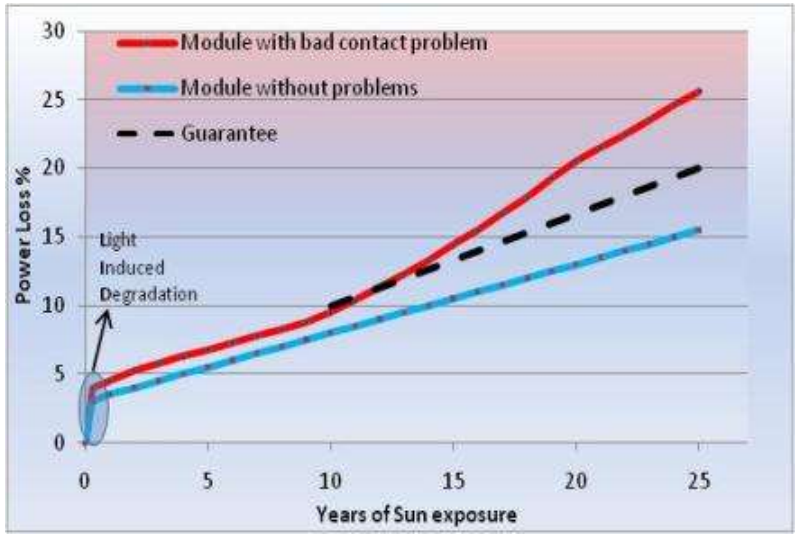
Lamination process. Poor lamination control

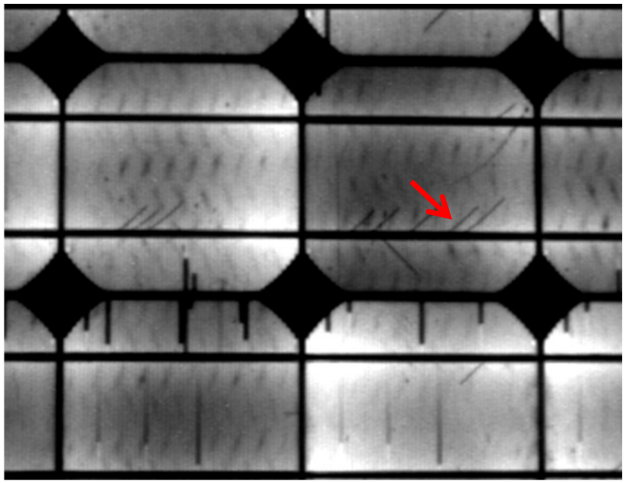


- The properties (transparency, durability, etc) of the EVA depend on the percentage of crosslinked (also called % gel content).
- A deficient lamination process can provoke that the EVA, that has not been crosslinked, has very low properties and it provokes a fast degradation on field.

3.6. TYPICAL DEFECTS. TSM AND INTERCONNECTION.

← Bad (resistive) contact problem, high temperature. It can lead to back-sheet burn and power loss





← Aggressive soldering. The tabb-ribbon breaks the cells below it. Power loss

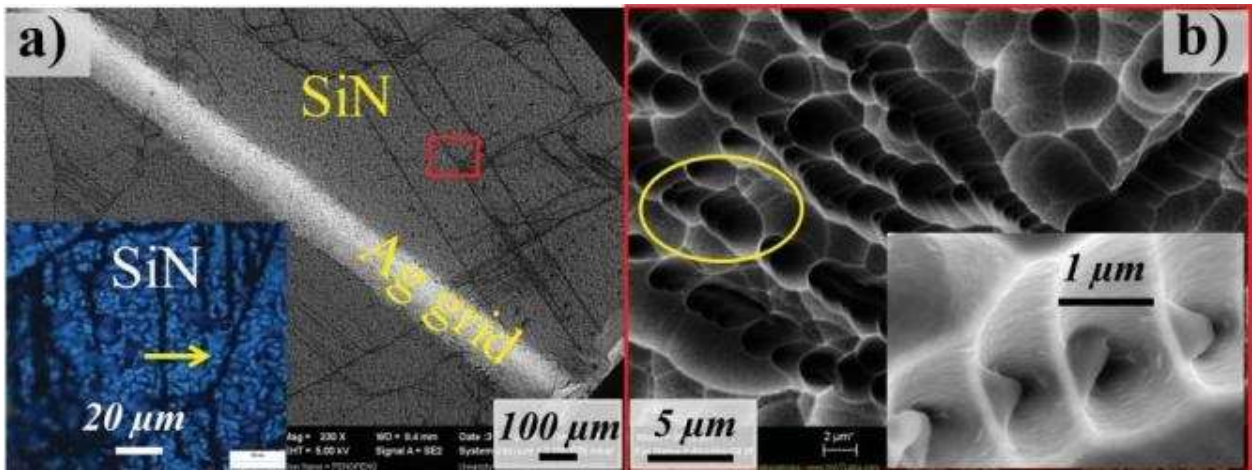
3.6. TYPICAL DEFECTS. TSM AND INTERCONNECTION.

3.6. TYPICAL DEFECTS. SEALANT APPLICATION.

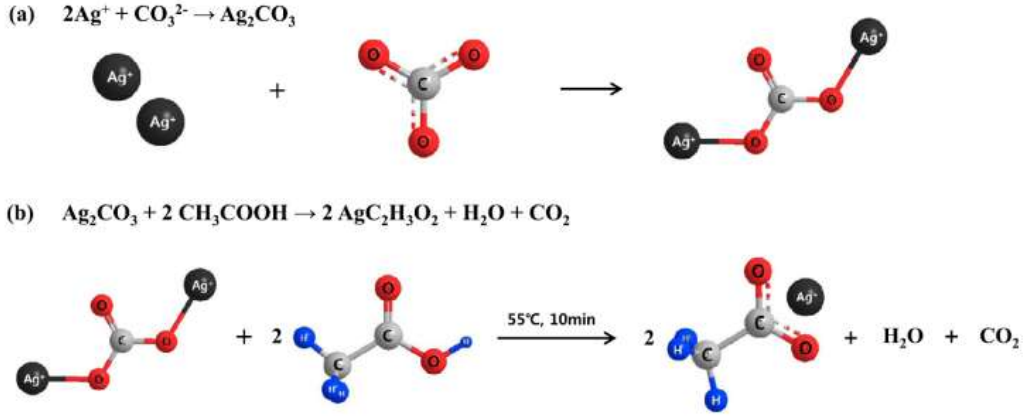
Important.
Failure of the edge sealant due to Bad Quality Product. Electrical isolation is affected. Module must be replaced.

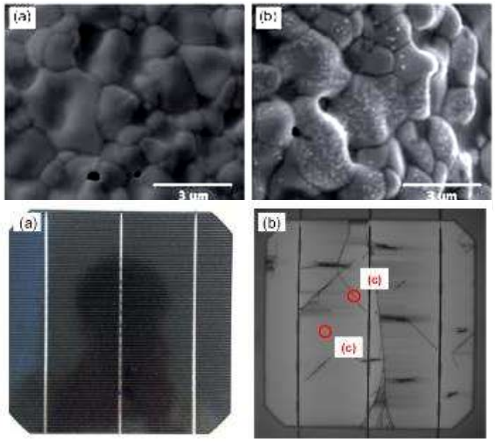
3.6. TYPICAL DEFECTS. SNAIL TRAILS

1 explanation (less likely to happen): Discoloration (darkening due to higher light trapping) of silicon nitride due to the presence of micropores.
Peng Peng et al. Microscopy study of snail trail phenomenon on PV modules. RSC Advances 2012,2, 11359-11365

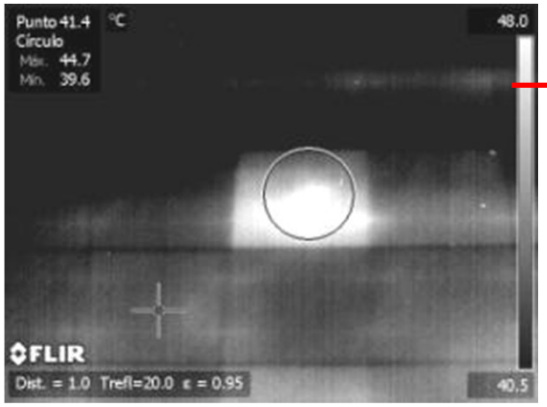


2 explanation (most likely to happen): Discoloration of grid due to the formation of nanoparticles of Ag_2CO_3 (that results from the reaction of CO_3^{2-} and Ag ions) that reacts with acetic acid from EVA forming $AgC_2H_3O_2$. CO_3^{2-} comes from atmospheric CO_2 and dissolved in water vapor that penetrates the cell up to the surface through cracks.
Namsu KIM et al. Analysis and reproduction of snail trails on silver grid lines in crystalline silicon photovoltaic modules. Solar Energy 124 (2016) 153-162

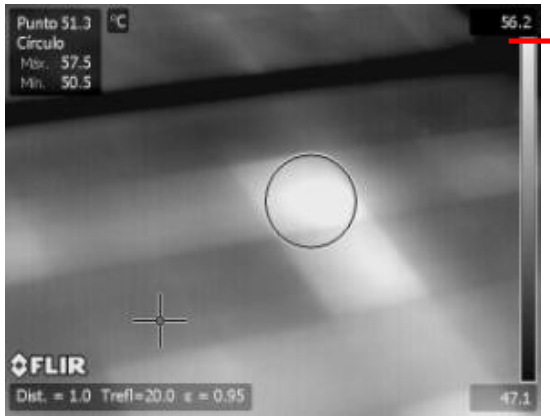




3.6. TYPICAL DEFECTS. SNAIL TRAILS



IR image reveals evidences of cell failure. Probably breakage

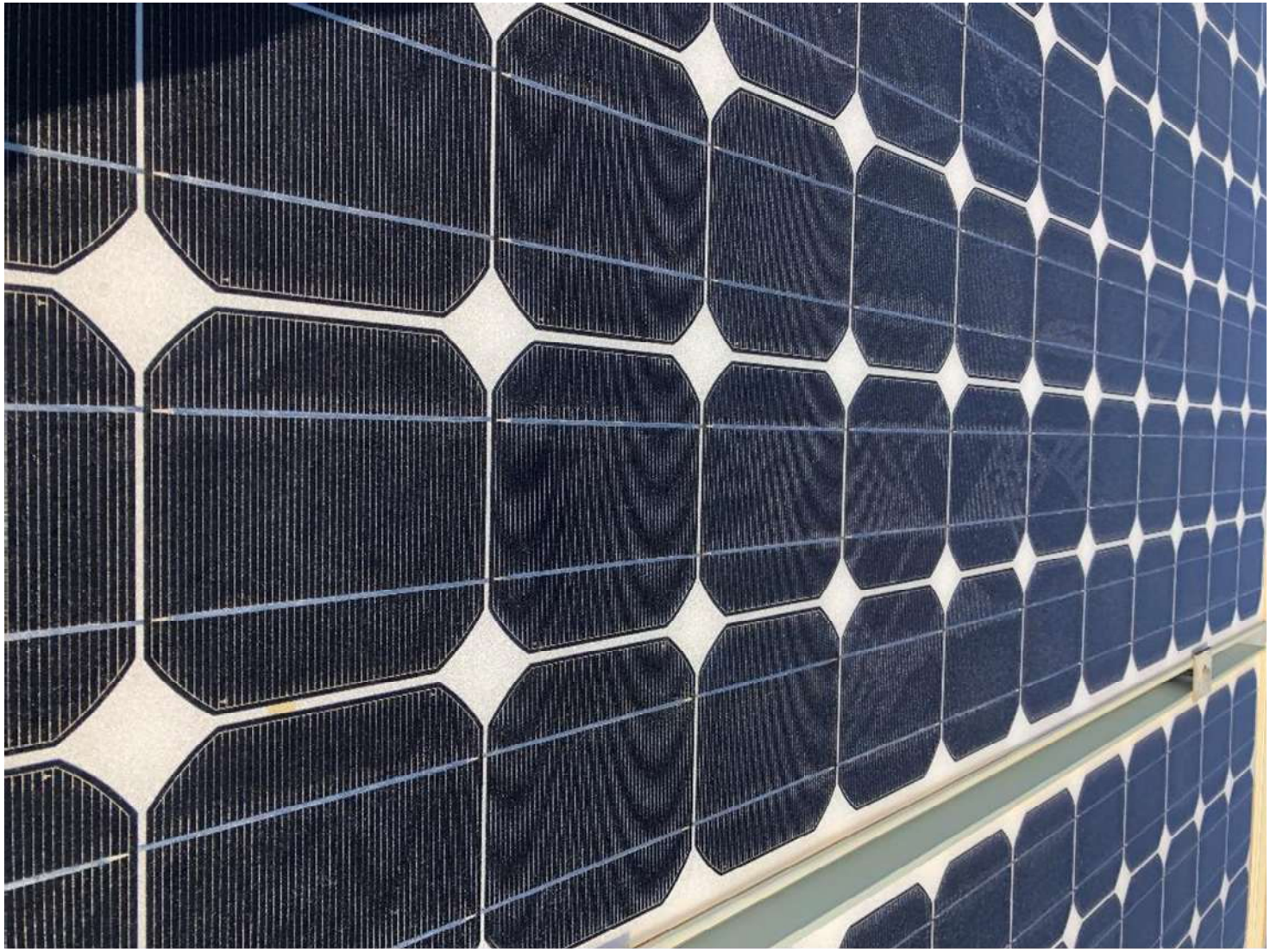


Massive affectation of modules in a certain installation

After visual inspection of the backside of the modules the snail trail coincides with scratches and impacts → 2º explanation = Cell breakage due to improper handling during installation

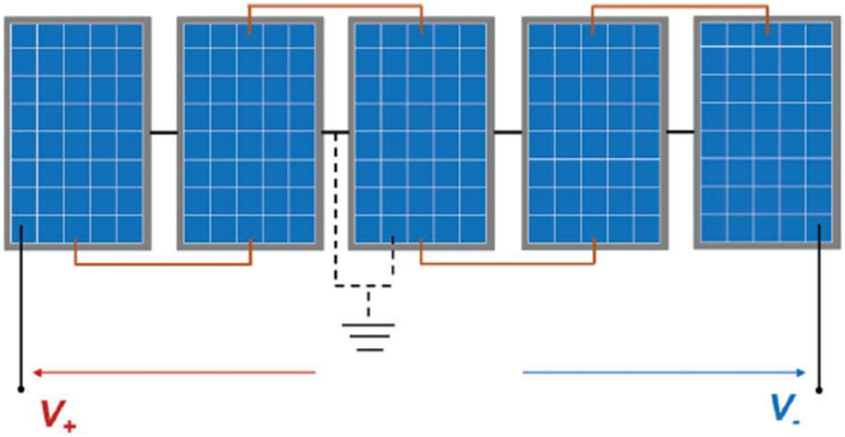
SNAIL TRAILS

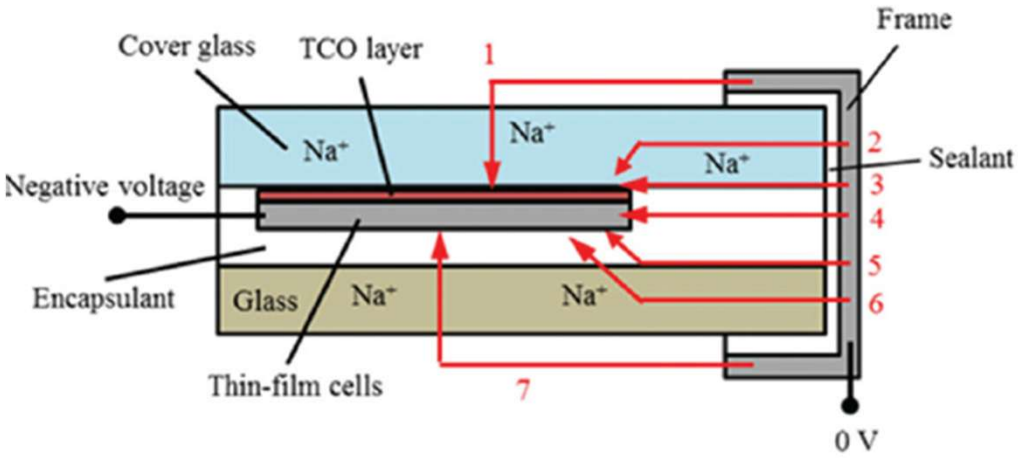
Breakages with the mono pattern

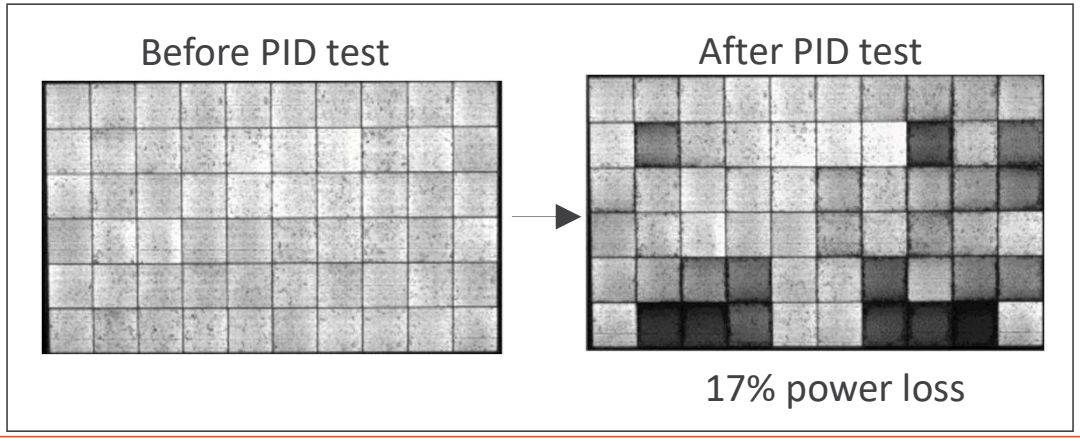


3.6. POTENTIAL INDUCED DEGRADATION (PID_{SH})

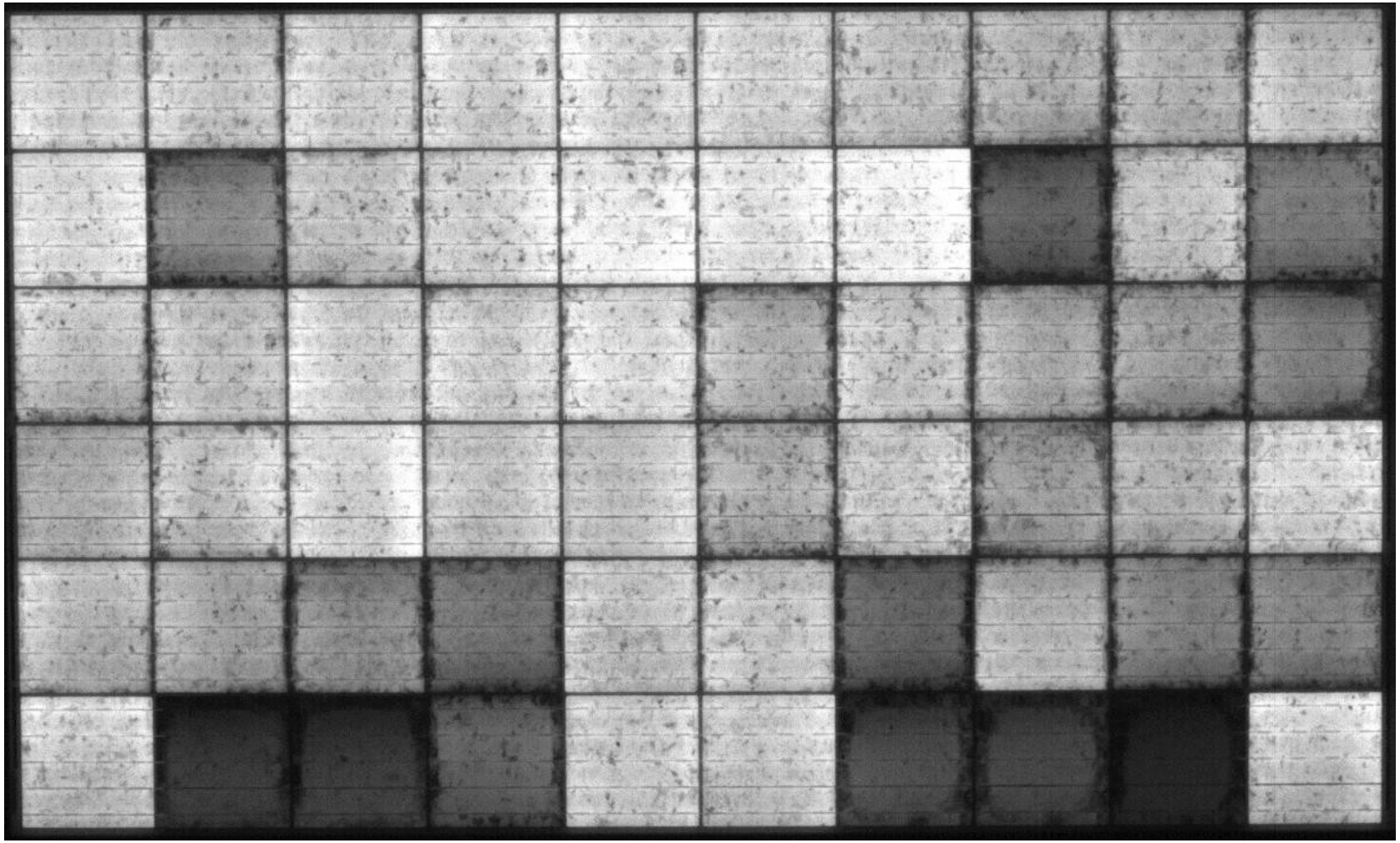
- PID is a major performance issue which can cause performance loss up to 90%.
- PID = Insufficient durability of the cells at the system voltage
- PID must be controlled in cell production.
- Affects to mono, multi, monofacial, bifacial, P-Type, N-Type and HJT.



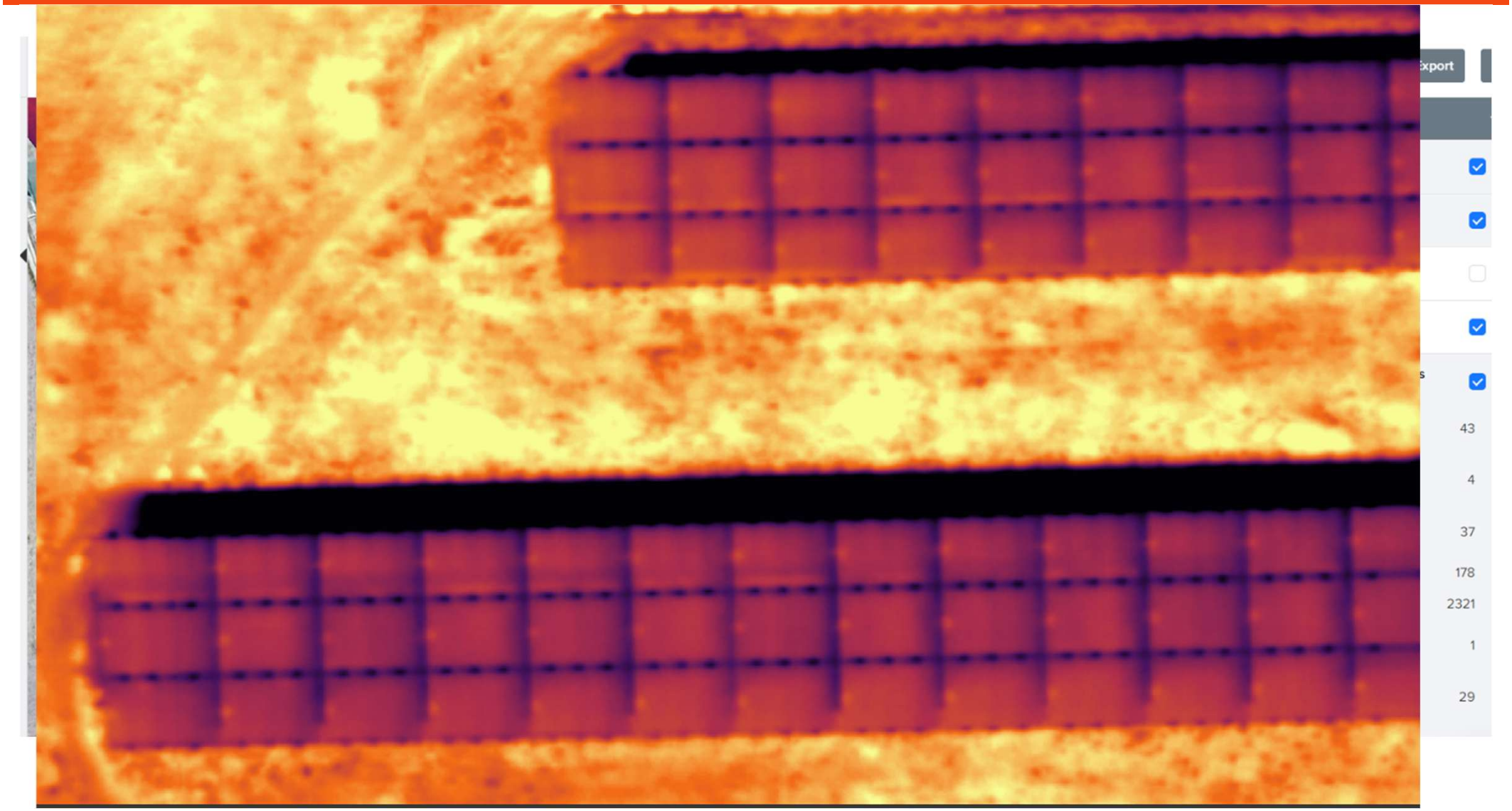




3.6. POTENTIAL INDUCED DEGRADATION (PID)

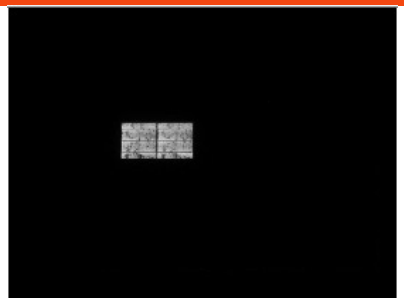


3.6. POTENTIAL INDUCED DEGRADATION (PID)



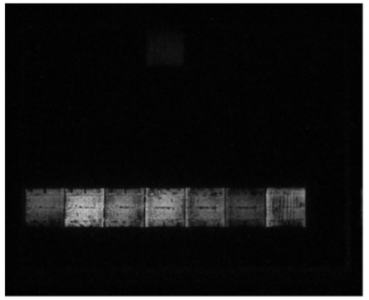
3.6. POTENTIAL INDUCED DEGRADATION (PID)

1

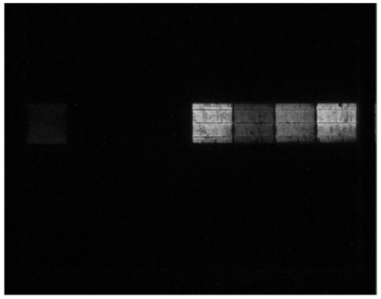


2

3



4



Geres. Results after PID test:

- -1000 V
  80% RH
  60 ºC
- 8 and 9 @ +1000 V

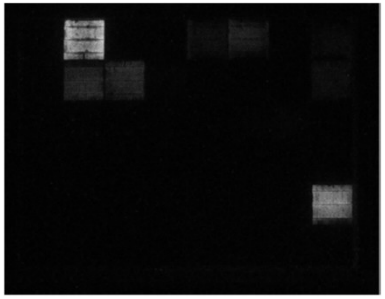
12.2.2.1. SUMMARY OF RESULTS OF DEGRADATION TEST

TESTING SUMMARY								
ID MODULE	SERIAL NUMBER	APPLIED BIAS	PID AMBIENT CONDITIONS	BATCH	MQT01 MAJOR VISUAL DEFECTS?	MQT06.1 P_{MAX} VARIATION	EL.02 SYMPTOMS OF PID DEGRADATION?	MQT15 INSULATION RESISTANCE CRIT. > 30.84 MΩ
30.4419.0-001	121303051848	Negative	60ºC @ 85% RH	1	YES [1]	-90.3%	YES	OK
30.4419.0-002	121303051847	Negative	60ºC @ 85% RH	1	YES [1]	-85.1%	YES	OK
30.4419.0-003	121303052427	Negative	60ºC @ 85% RH	1	YES [1]	-87.6%	YES	OK
30.4419.0-004	121303051853	Negative	60ºC @ 85% RH	1	YES [1]	-89.9%	YES	OK
30.4419.0-005	121303052421	Negative	60ºC @ 85% RH	1	YES [1]	-85.1%	YES	OK
30.4419.0-006	121502059393	Negative	85ºC @ 85% RH	2	YES [1]	-87.9%	YES	OK
30.4419.0-007	121502059681	Negative	85ºC @ 85% RH	2	YES [1]	-95.9%	YES	OK
30.4419.0-008	121502059371	Positive	85ºC @ 85% RH	2	YES [1]	-0.1%	NO	OK
30.4419.0-009	121502059061	Positive	85ºC @ 85% RH	2	YES [1]	0.0%	NO	OK

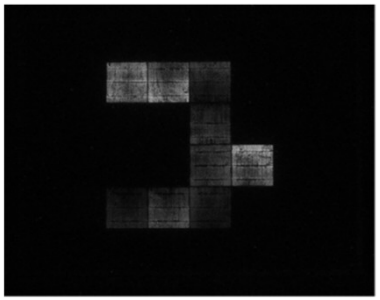
[1] The major visual defects are not related with the PID test and were detected in the initial visual inspection.

Table 12.42. Test results summary after Detection of PID test.

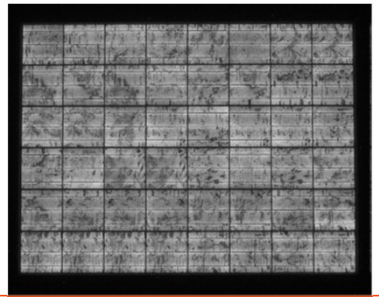
5



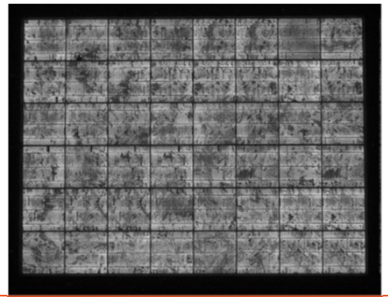
6



9



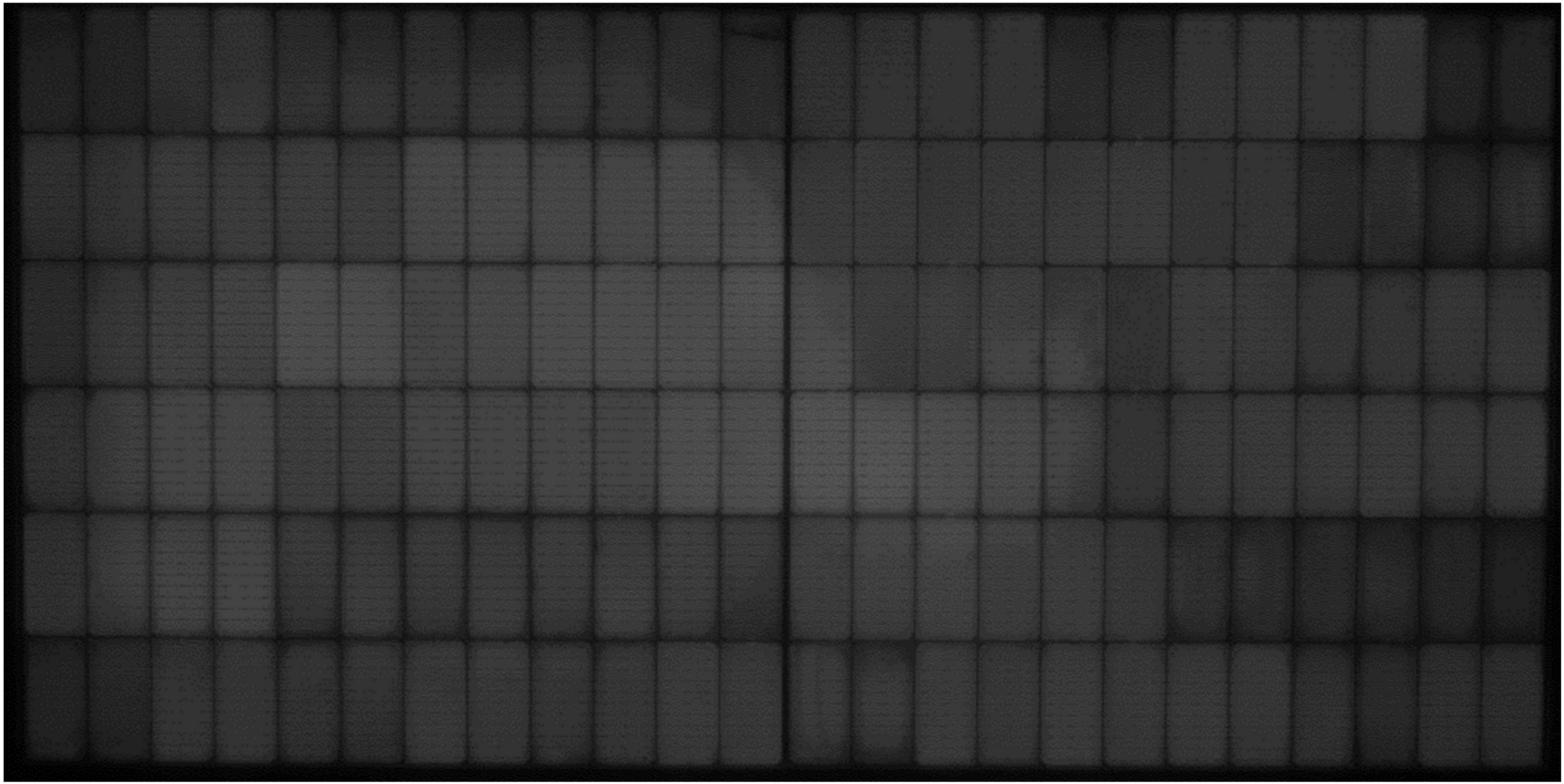
8



7

3.6. POTENTIAL INDUCED DEGRADATION (PID)

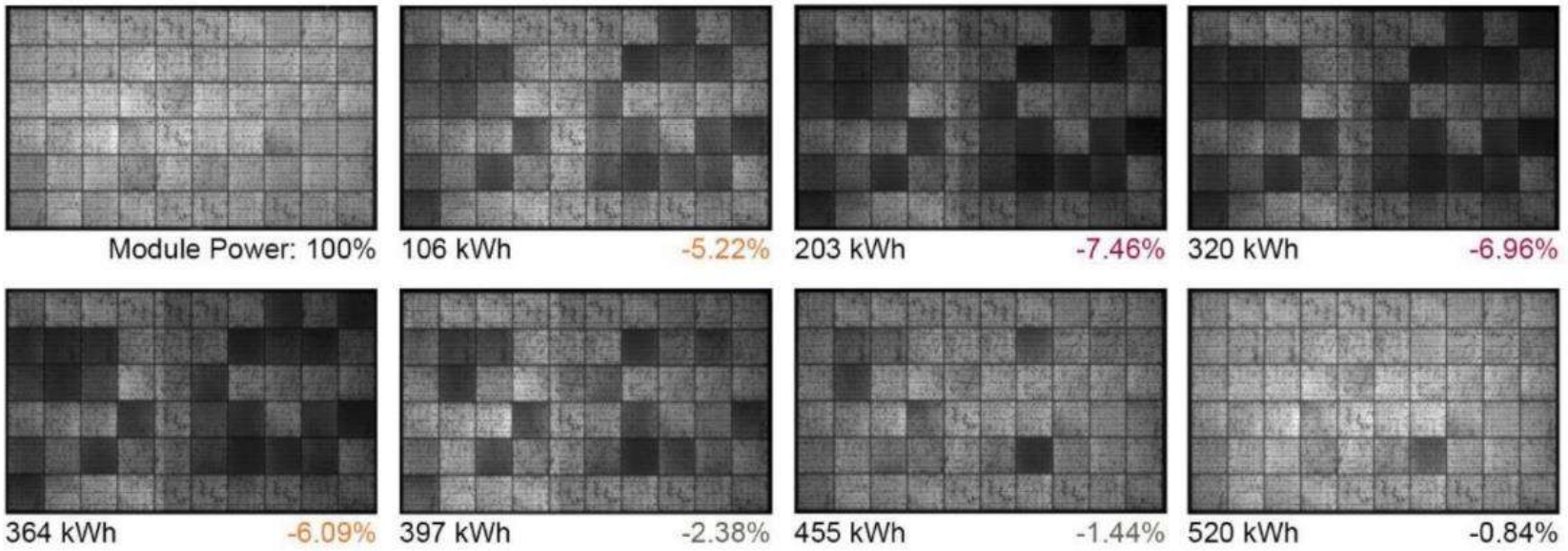
Modules from 2022. PERC. PID degradation at -1500V = 7.2%



3.6. LID AND LETID

LID = Light Induced Degradation.
Complexes B-O and Fe-B are the causes.
Around 2% power loss after 30 hours of light exposure
Hydrogen needed for regeneration

LeTID = Light and elevetaed Temperature Induced Degradation.
Unknown causes.
Around 4% power loss after 30 hours of light exposure.
Multi more susceptible to suffer LeTID



Regeneration takes up to 6 months

"A New Light Induced Volume Degradation Effect of mc-Si Solar Cells and Modules".F. Kersten et al. 31st EUPVSEC

Fast regeneration using temperature and illumination

3.6. BACKSHEET DEGRADATION

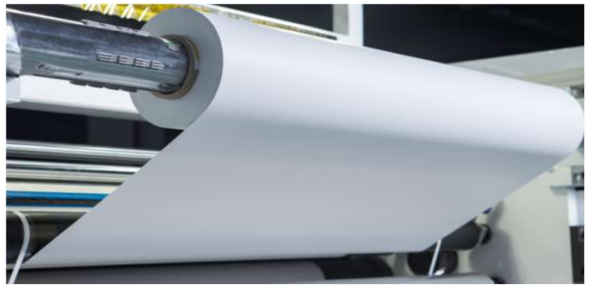
Common back-sheet compositions.

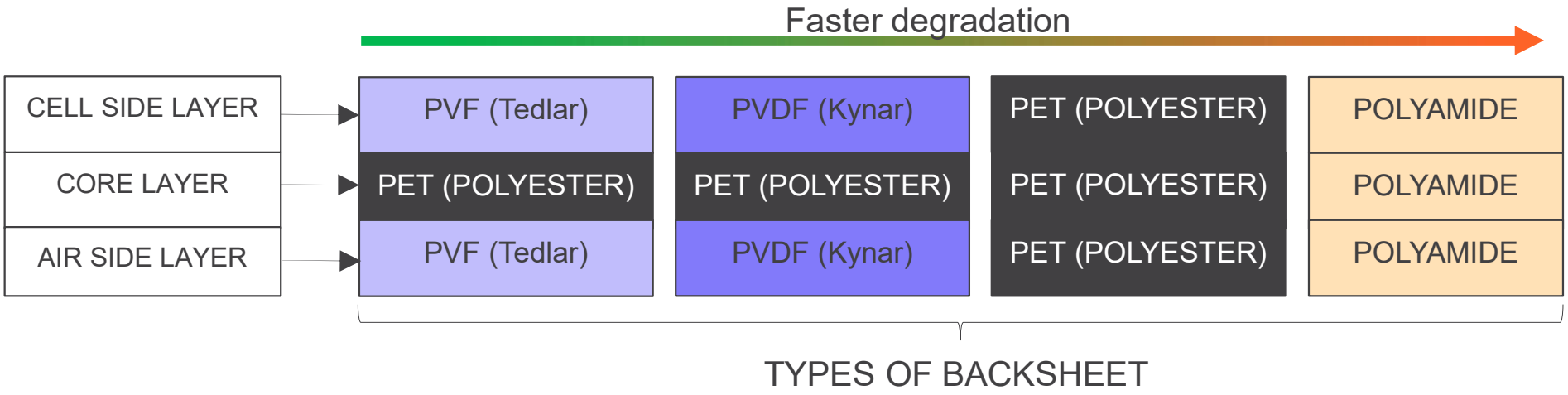
Backsheet functions:

1) Provide electrical insulation

2) Block moisture ingression

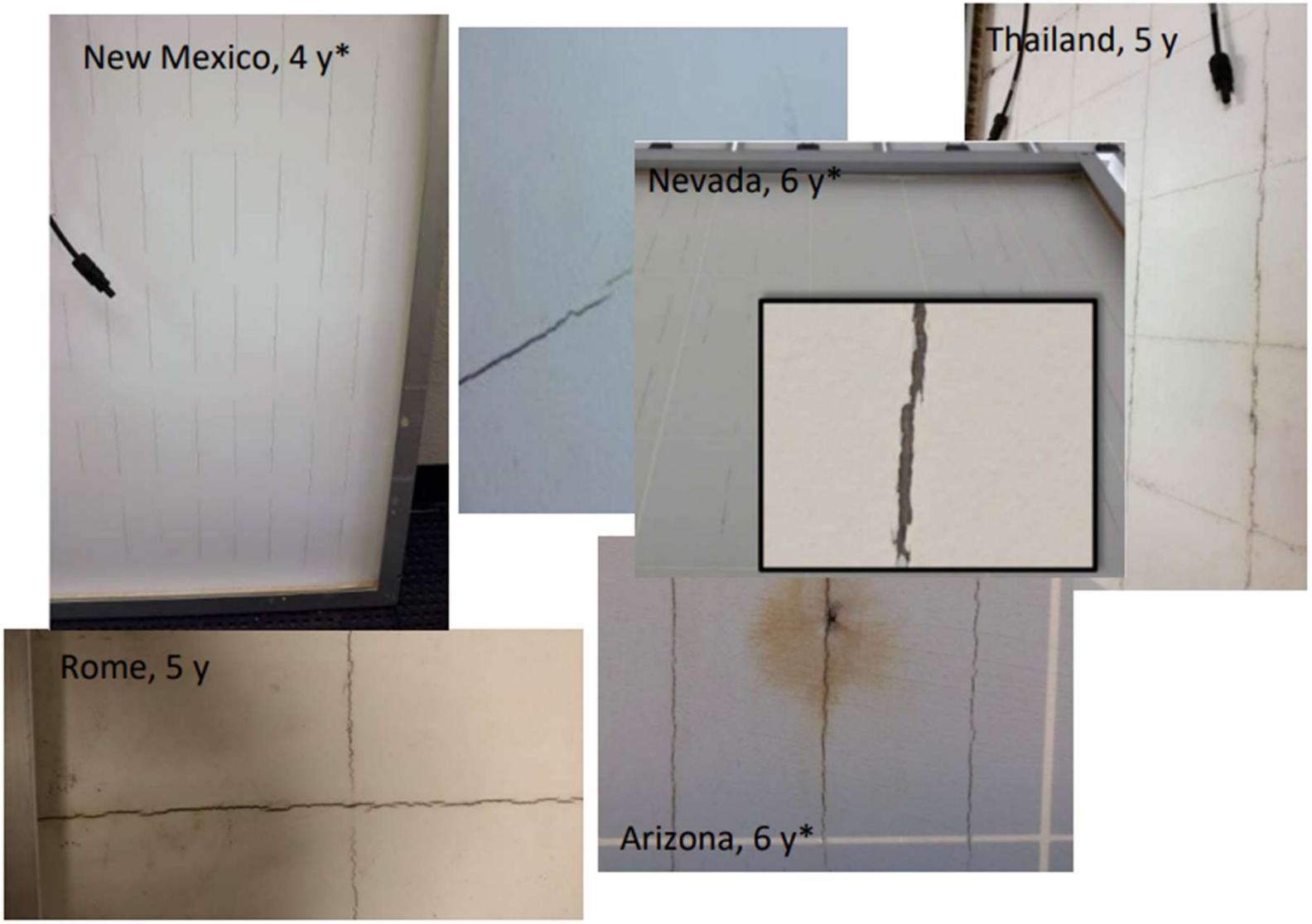
- The backsheet is normally formed by 3 layers.
- The material of those layers, especially the outer layer, determines the durability of the backsheet





3.6. BACKSHEET DEGRADATION

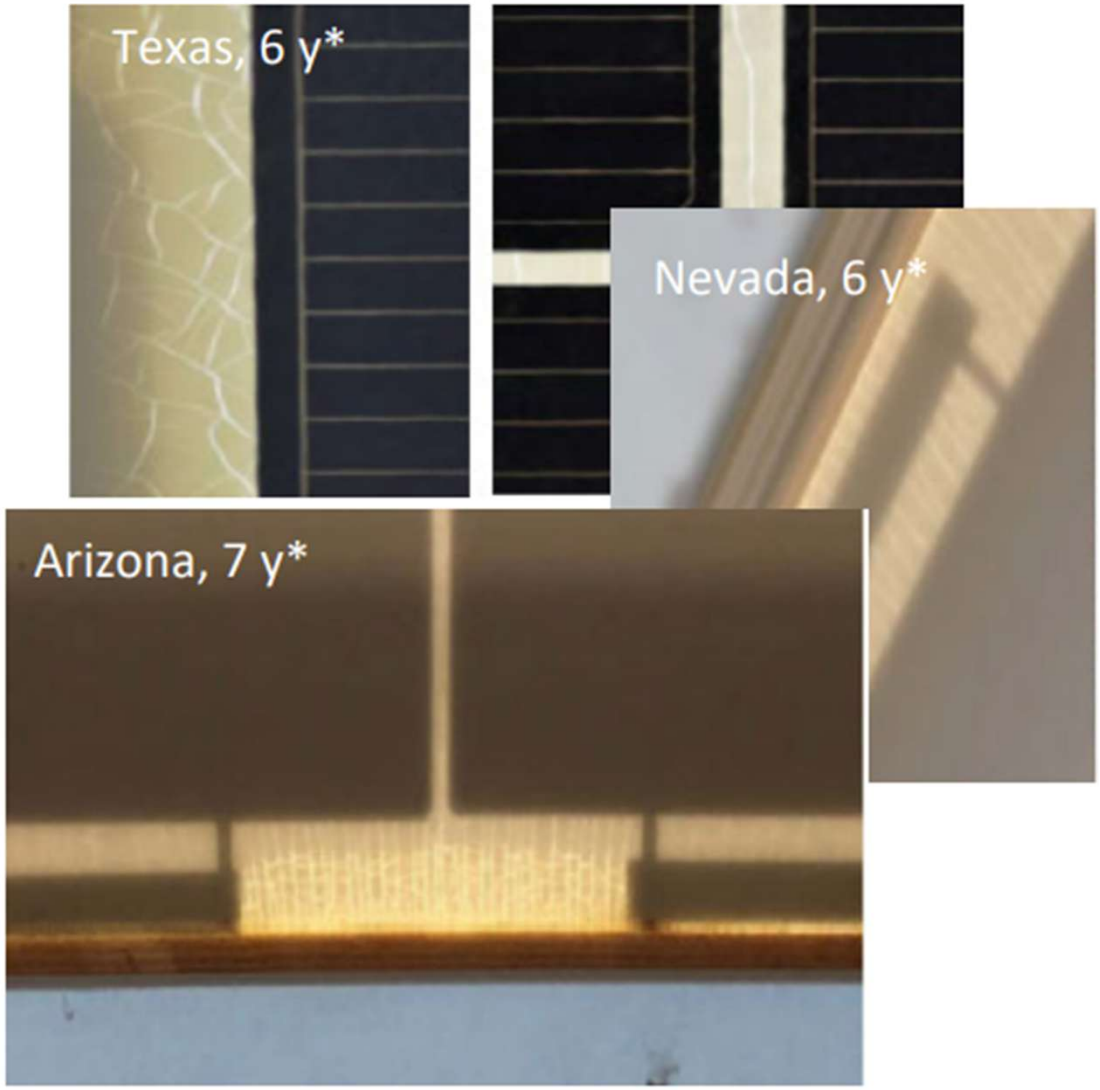
Back-sheet degradation: Cracking of Polyamide based back-sheets



3.6. BACKSHEET DEGRADATION

Back-sheet degradation. Failure modes

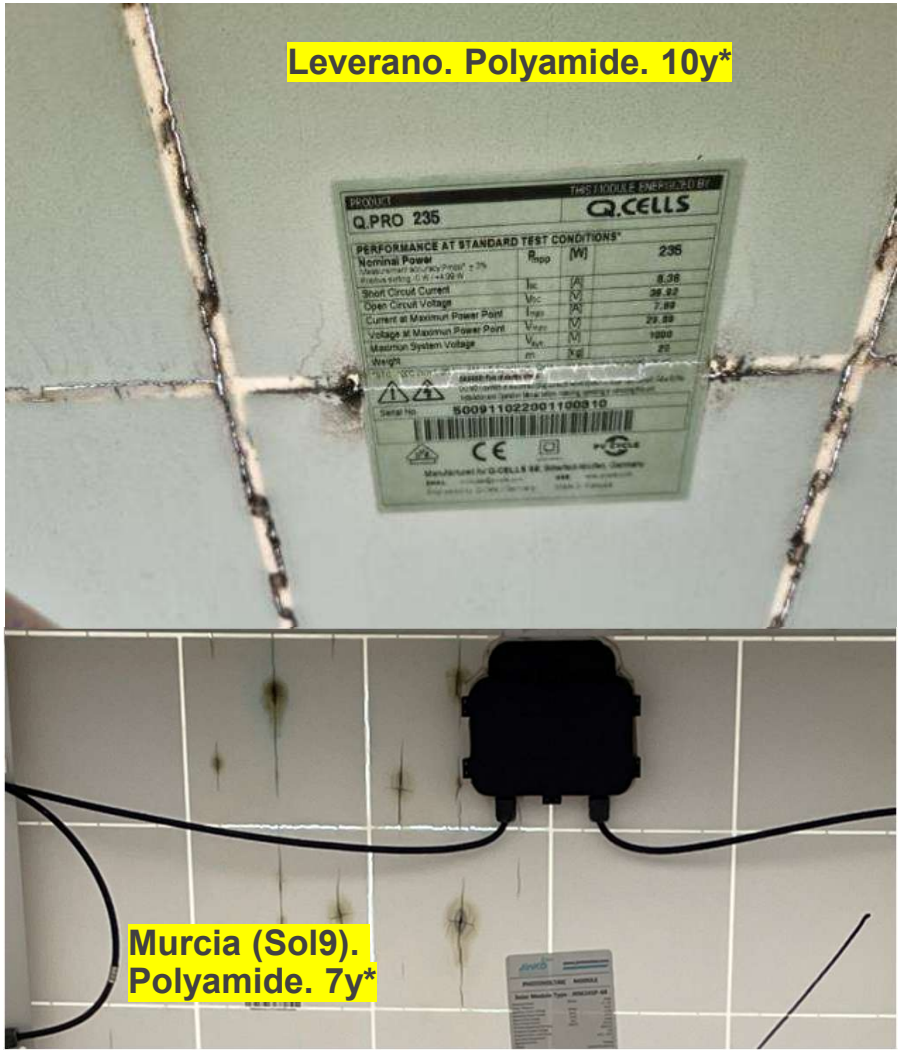
Polyester



- Cracking of an inner layer
- Accelerates corrosion
- Leads to short-circuiting and inverter shutdown

3.6. BACKSHEET DEGRADATION

Back-sheet degradation. Failure modes



3.6. BACKSHEET DEGRADATION

Back-sheet degradation. Failure modes

Villena. <14y*

3.6. BACKSHEET DEGRADATION

Two different backsheets used → Different degradation

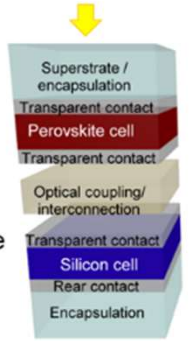
4.1. SOLAR CELL TECHNOLOGIES. WHAT´S NEXT?

TECHNOLOGY DEVELOPMENTS CELLS

PERC will lose market share with decreasing production cost for n-type cell concepts

According to ITRPV PERC solar cells will maintain a notable market share in the future. Specially for p-type mono PERC. This is mainly for the low cost of production and the needed future amortization of the current capex destined to PERC capacity nowadays.

TANDEM (Perovskite):*
- Perovskite on Silicon
- CIGS - Perovskite
- Perovskite –Perovskite
- Organic - Perovskite

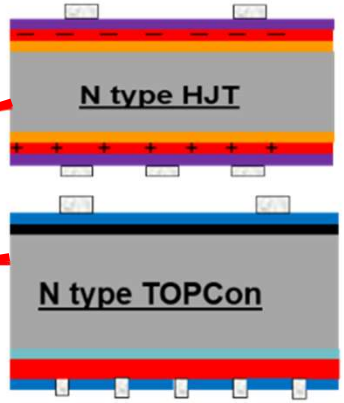


Theoretical maximum efficiency

Eff < 38%

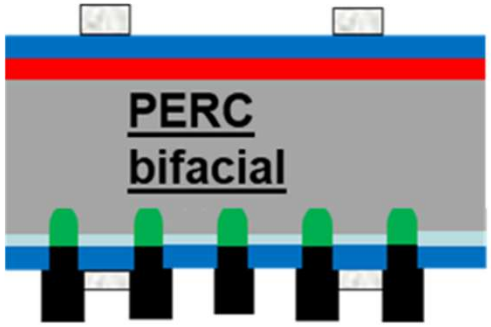
*** Industrial maximum efficiency

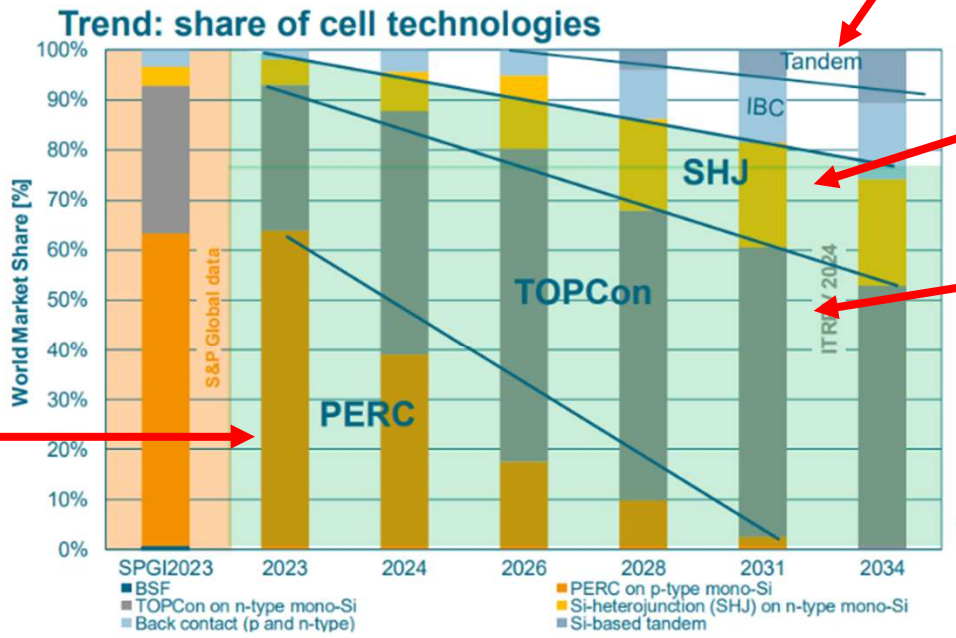
Eff < 26 %

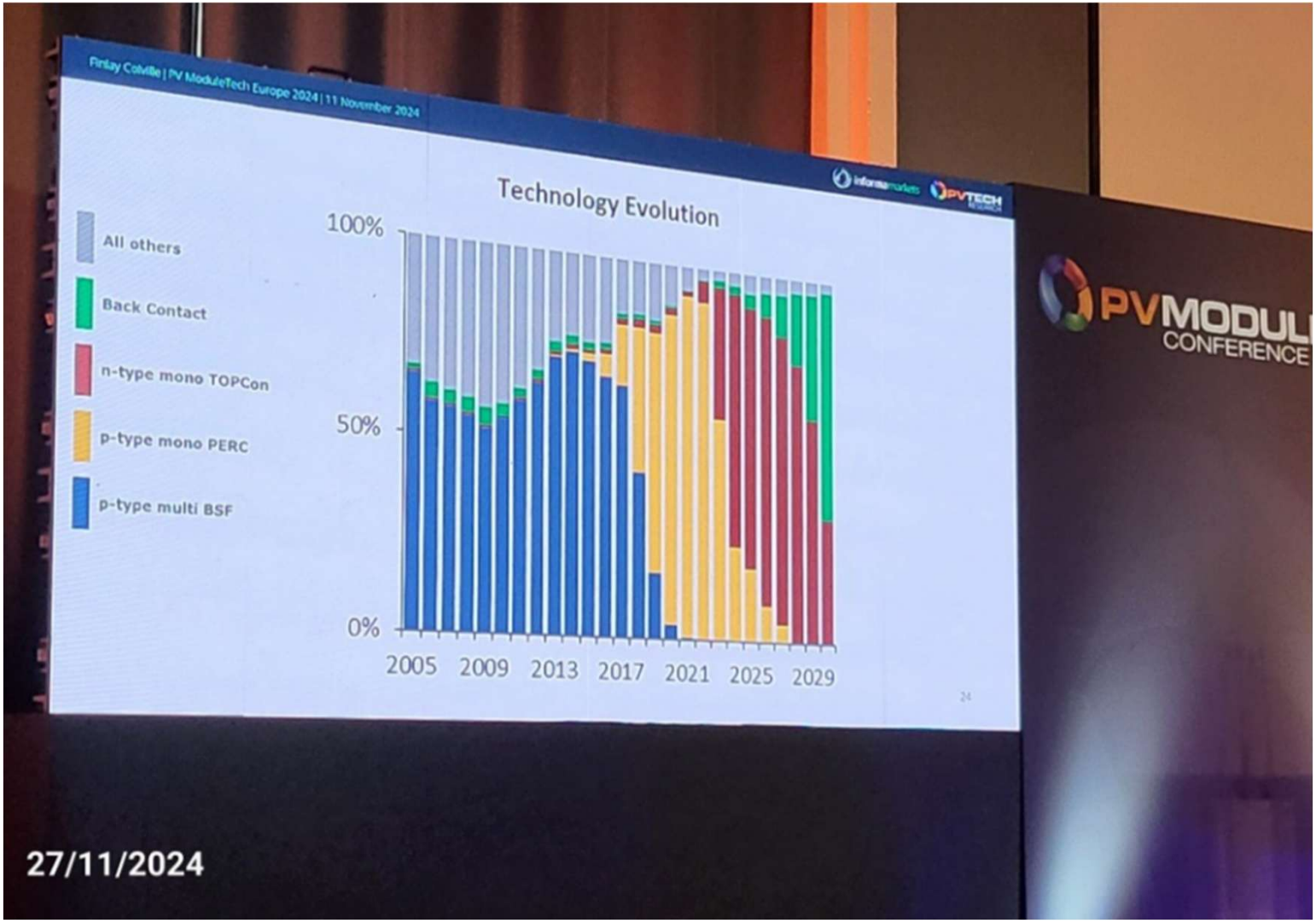


Eff < 26 %

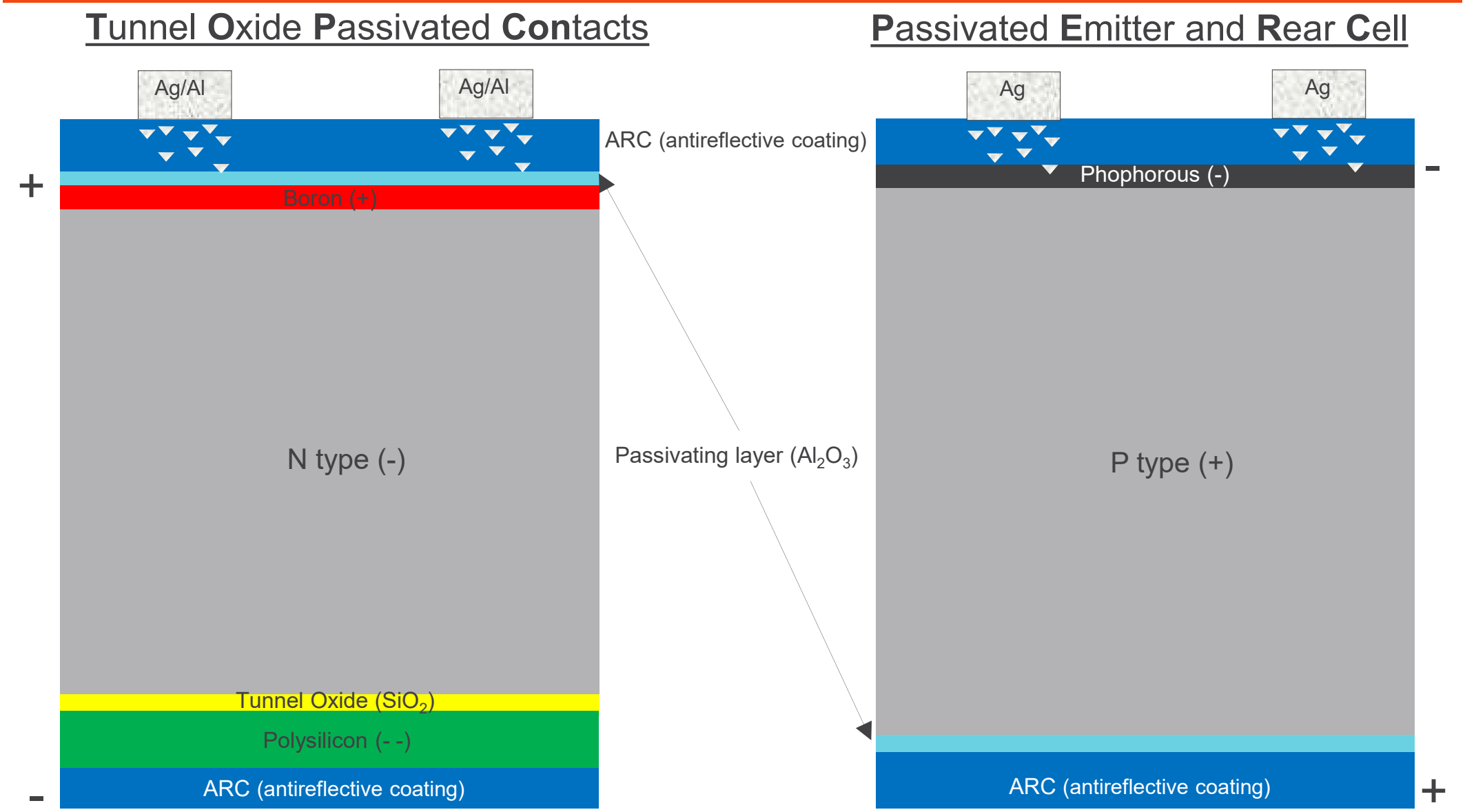
Eff < 24%



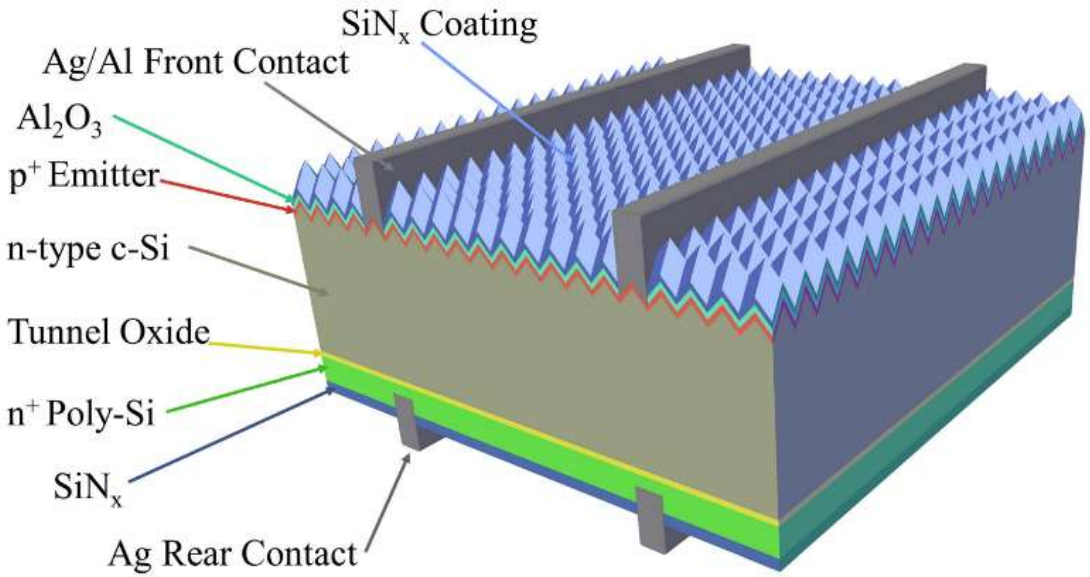


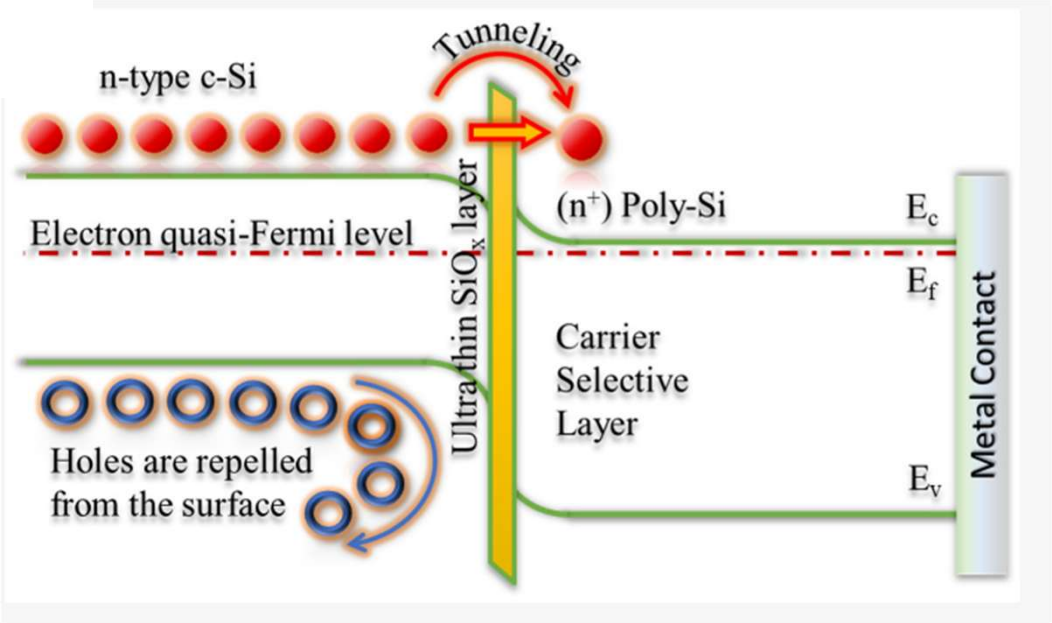

Finlay Colville | PV ModuleTech Europe 2024 | 11 November 2024
Technology Evolution
All others
Back Contact
n-type mono TOPCon
p-type mono PERC
p-type multi BSF
100%
50%
0%
2005
2009
2013
2017
2021
2025
2029
PVMODUL
CONFERENCE
27/11/2024

TOPCon VS PERC



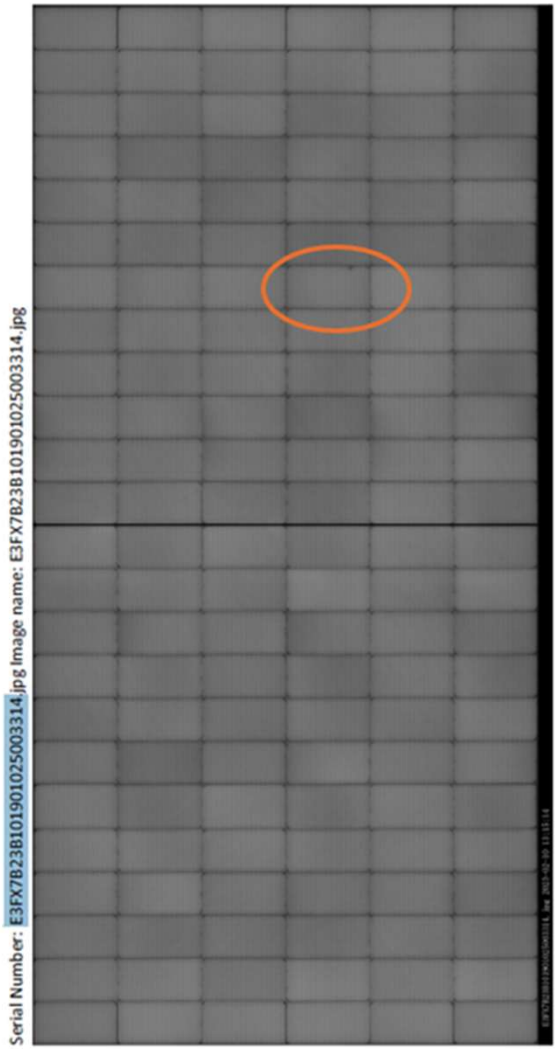
4.1. SOLAR CELL TECHNOLOGIES. TOPCON



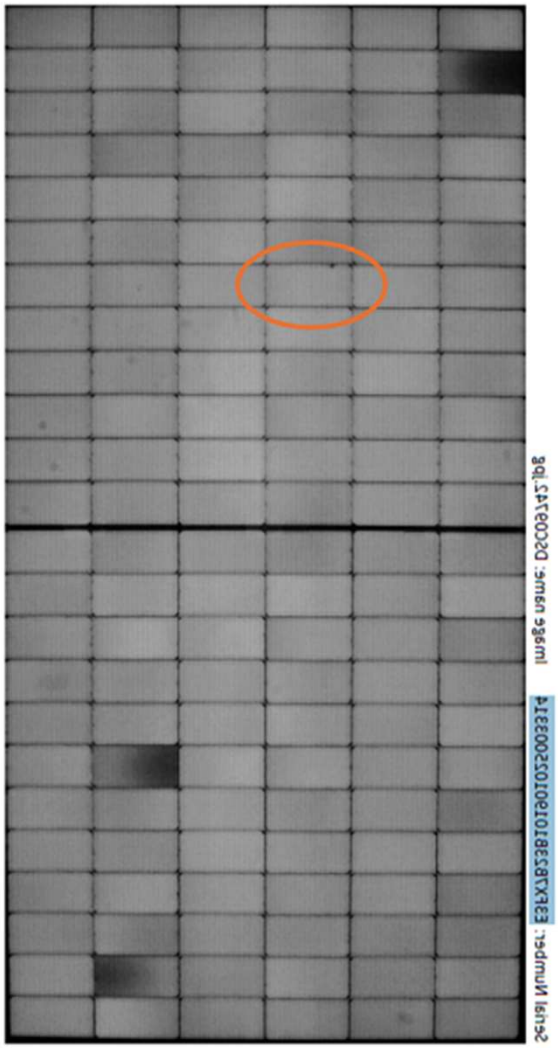


4.2. TOPCON UV DEGRADATION

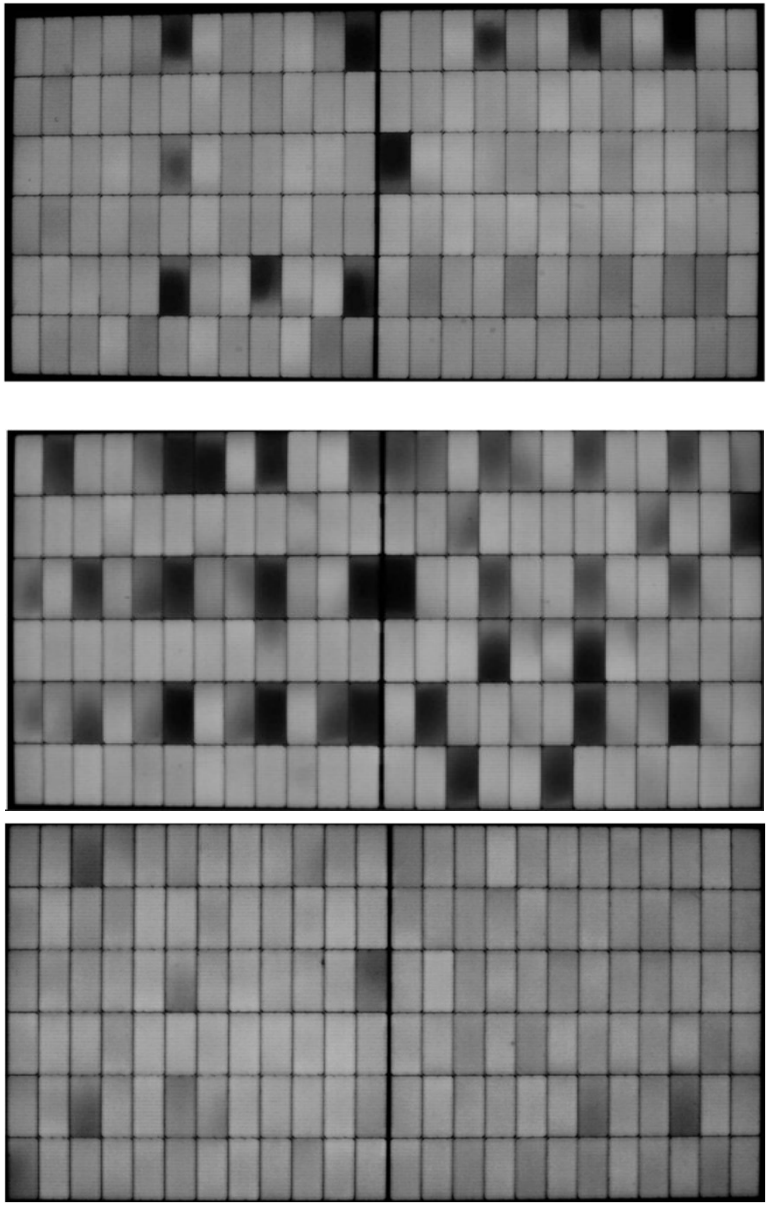
TOPCon. UV degradation. Module 2022



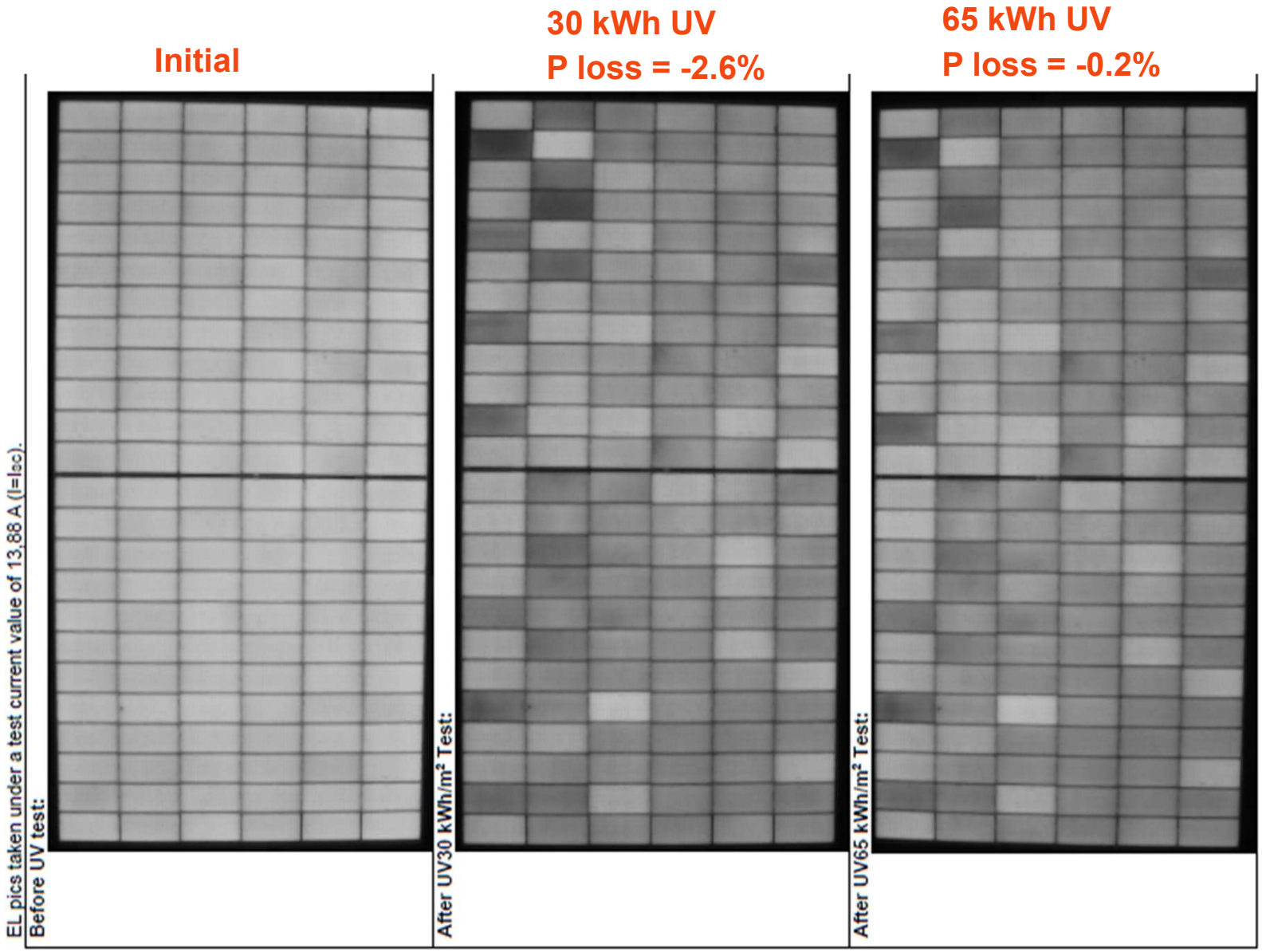
Factory EL



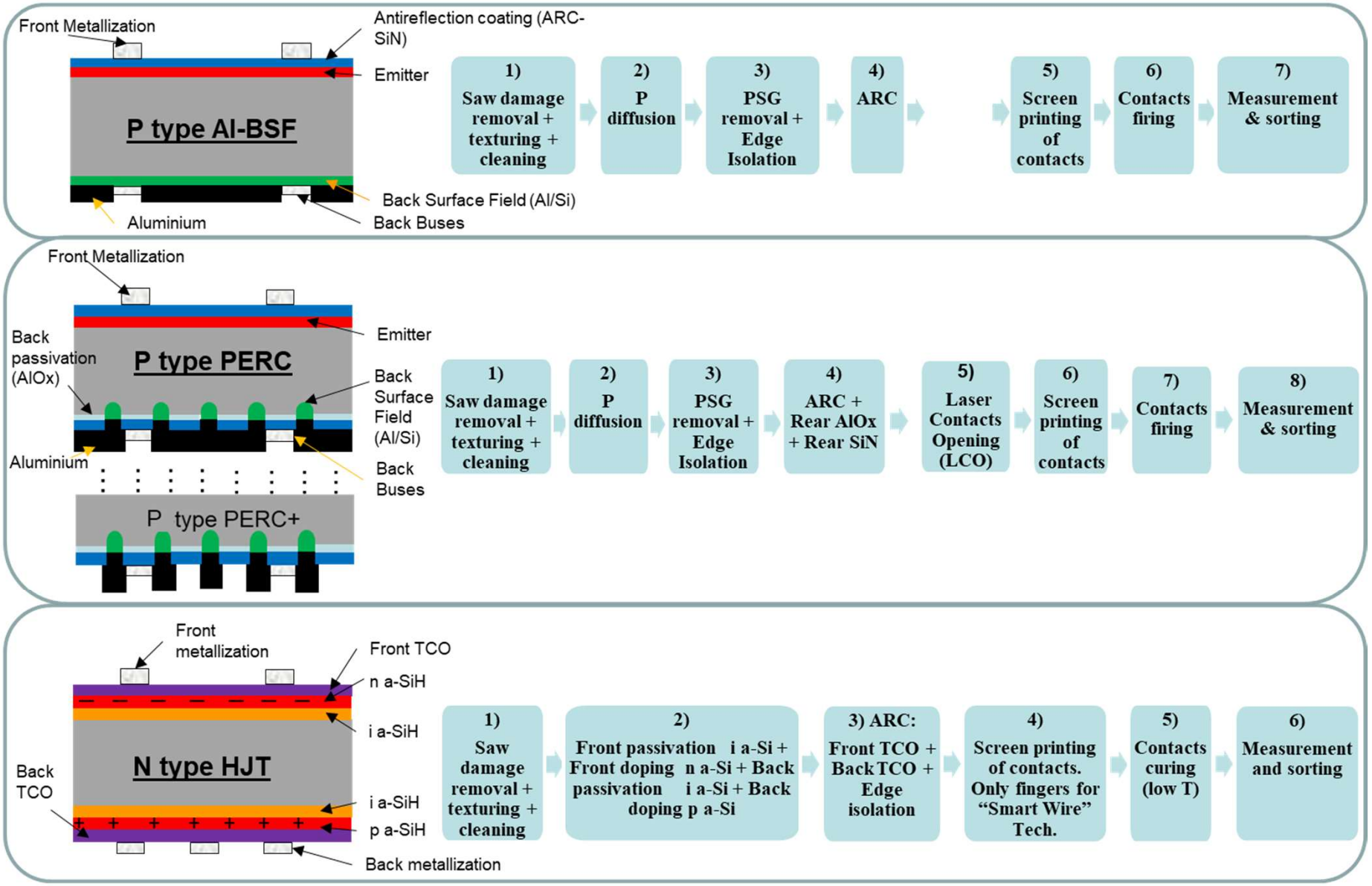
Field EL



4.2. TOPCON UV DEGRADATION

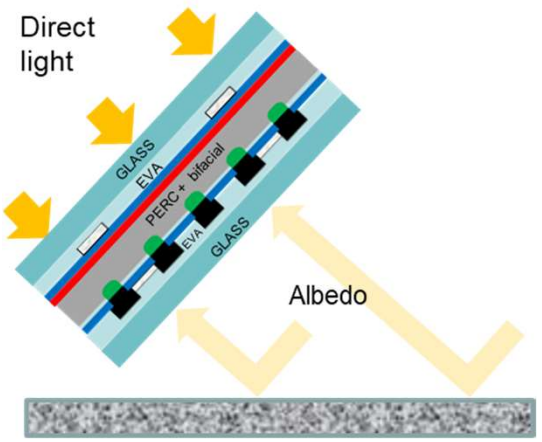


4.2. SOLAR CELL TECHNOLOGIES.



4.3. MAIN MODULE TECHNOLOGIES. HALF CELLS, BIFACIAL, SHINGLE, MBB

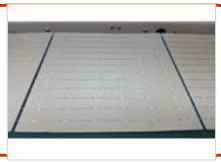
1



BIFACIAL: For slanted systems 5%-30% energy gain compare to monofacial depending on: Orientation, Albedo, Tilt

2

MBB (Multi Bus Bar): Up to 12 bus bars.
Energy gain due to less series resistance



3

Laser cutting

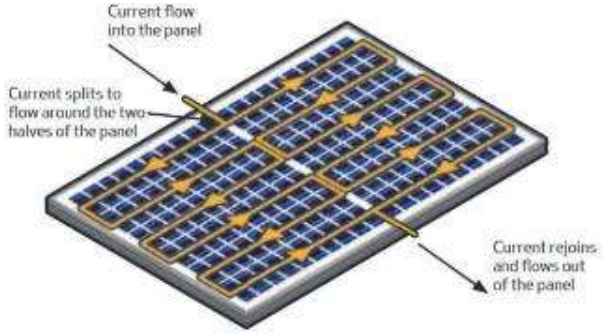
4

Half cut cells: Energy gain reasons:

1) The reduced cell current leads to a reduction in series resistance losses (3% more power)

2) Less power los due to shading

https://www.gses.com.au/how-do-half-cut-solar-cells-work-while-shaded/



5

Shingled modules

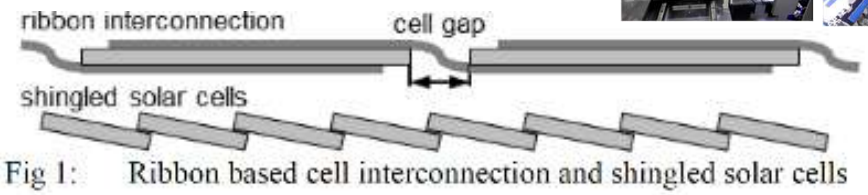
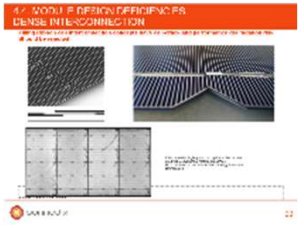


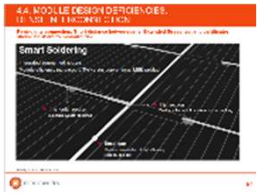
Fig 1: Ribbon based cell interconnection and shingled solar cells

Cell-to-Module (CTM) Analysis for Photovoltaic Modules with Shingled Solar Cells
June 2017
Conference: 44th IEEE Photovoltaic Specialist Conference, Washington D.C

Tilling Ribbon

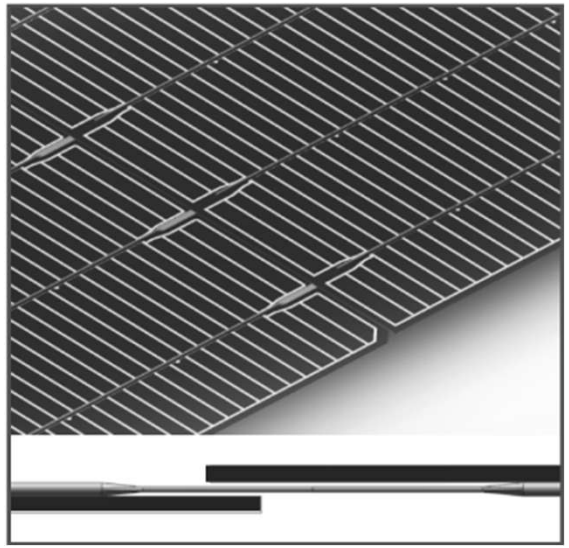


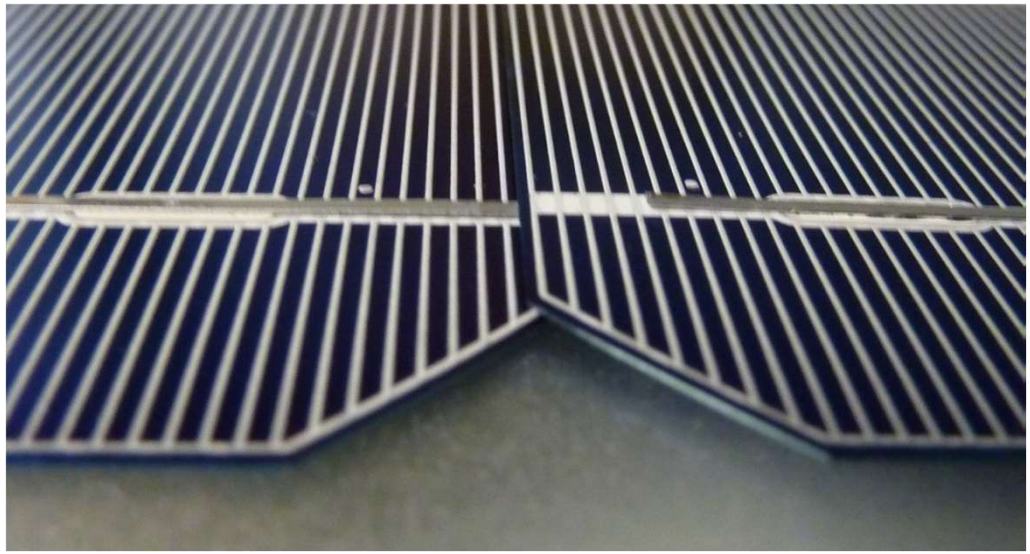
Paving interconnection

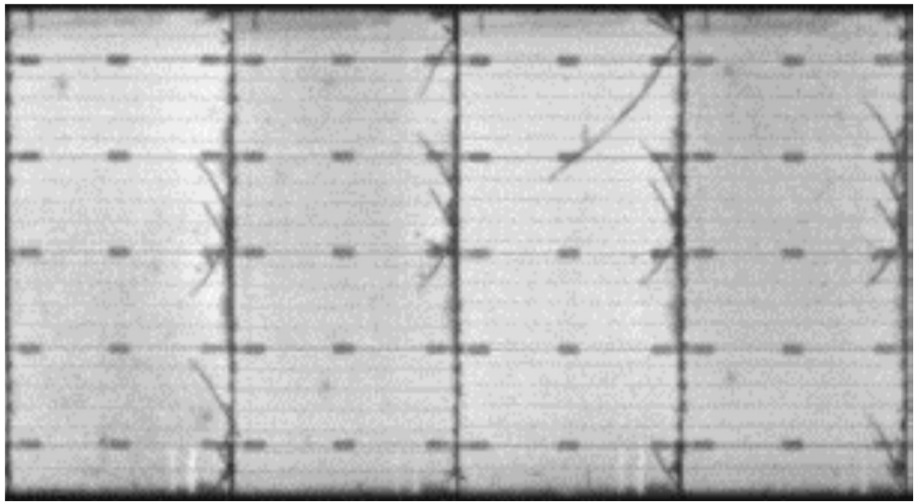


4.4. MODULE DESIGN DEFICIENCIES. DENSE INTERCONNECTION

Tilling Ribbon cell interconnection concepts have cell-crack and performance degradation risk. Should be rejected



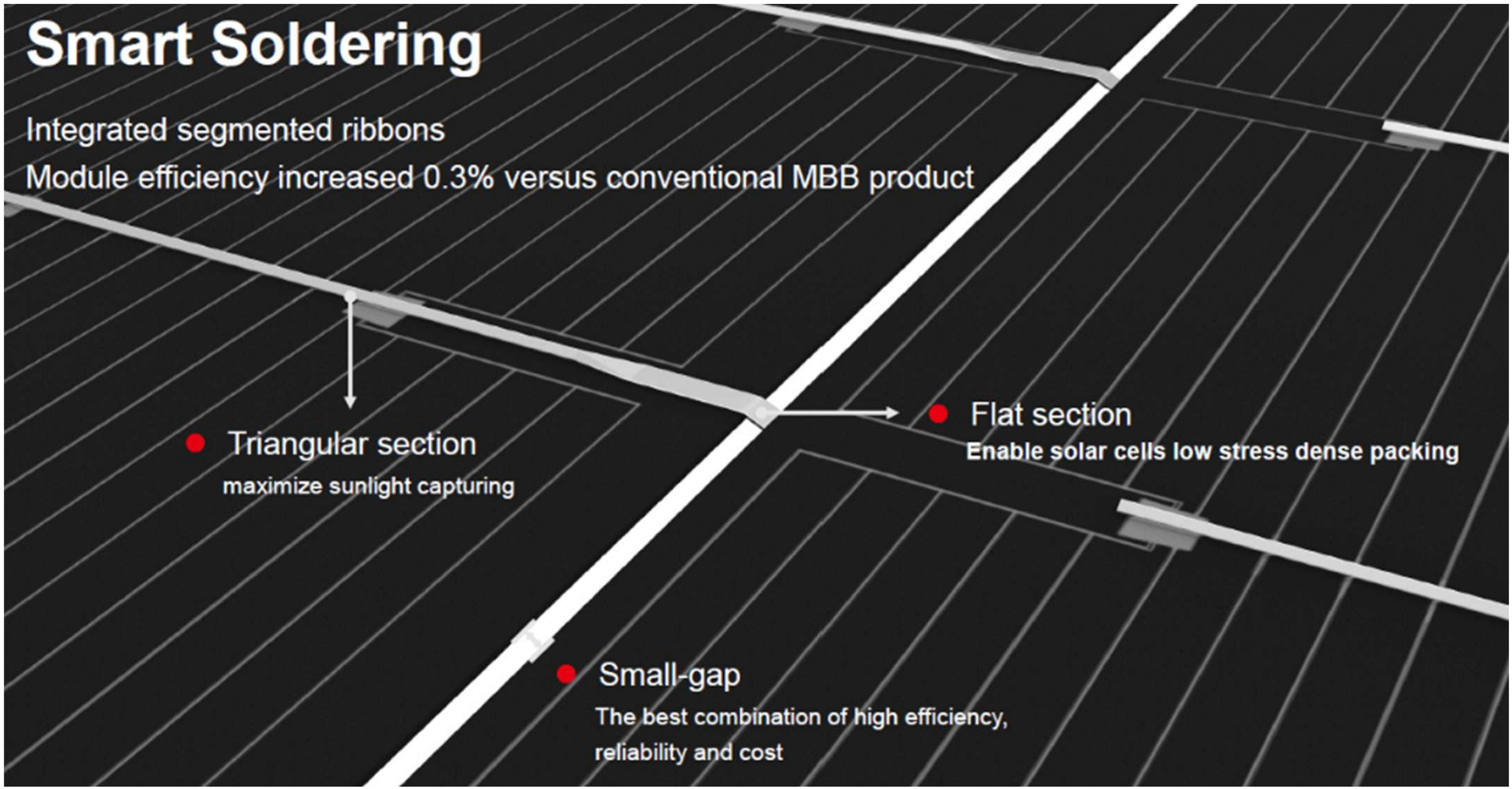




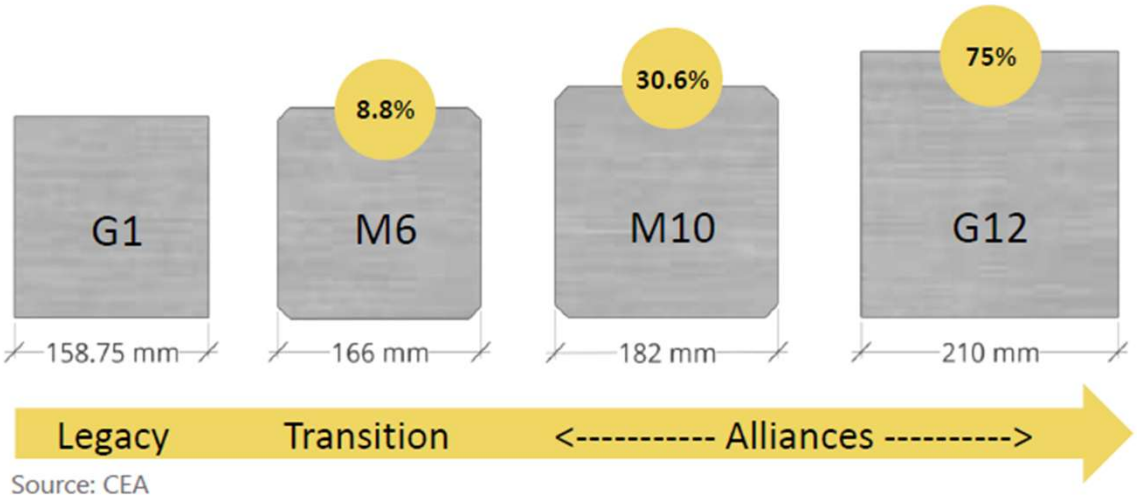
Interconnect-shingling: maximizing the active module area with conventional module processes
H. Schulte-Huxel. Institute for Solar Energy Research Hamelin (ISFH)

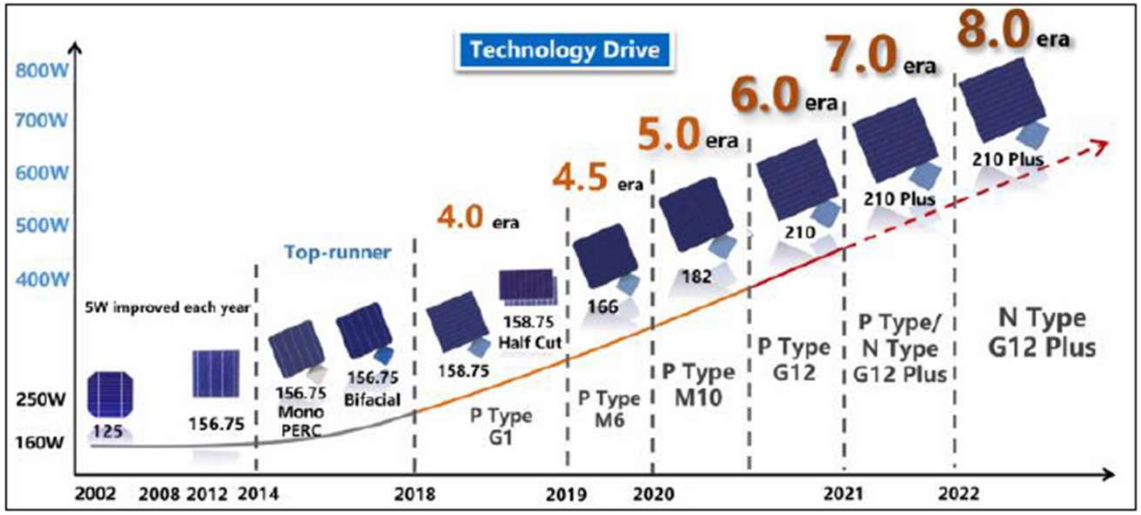
4.4. MODULE DESIGN DEFICIENCIES. DENSE INTERCONNECTION

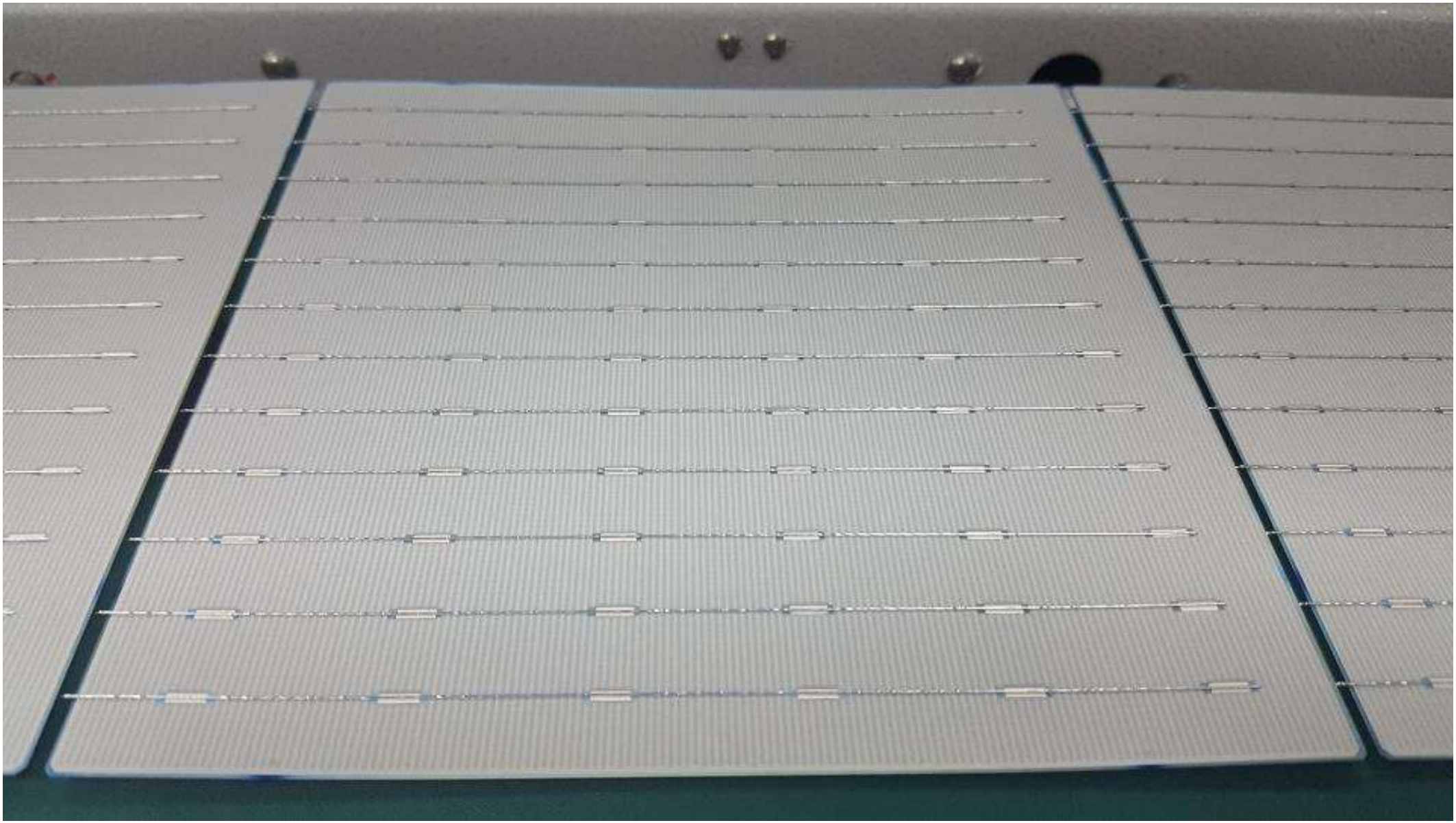
Paving interconnection. Short distance between cells. Extended Stress test and certificates should be share to minimize risk



4.5. DIFFERENT CELL SIZES

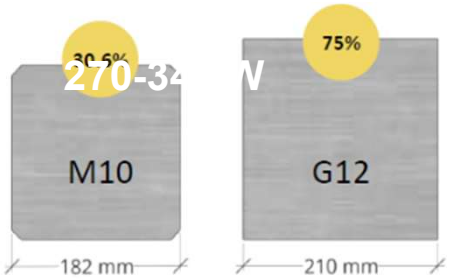
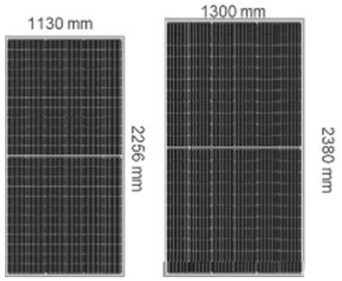
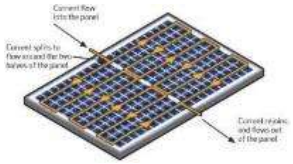






Multi Bus Bar (MBB) + Half cut cell

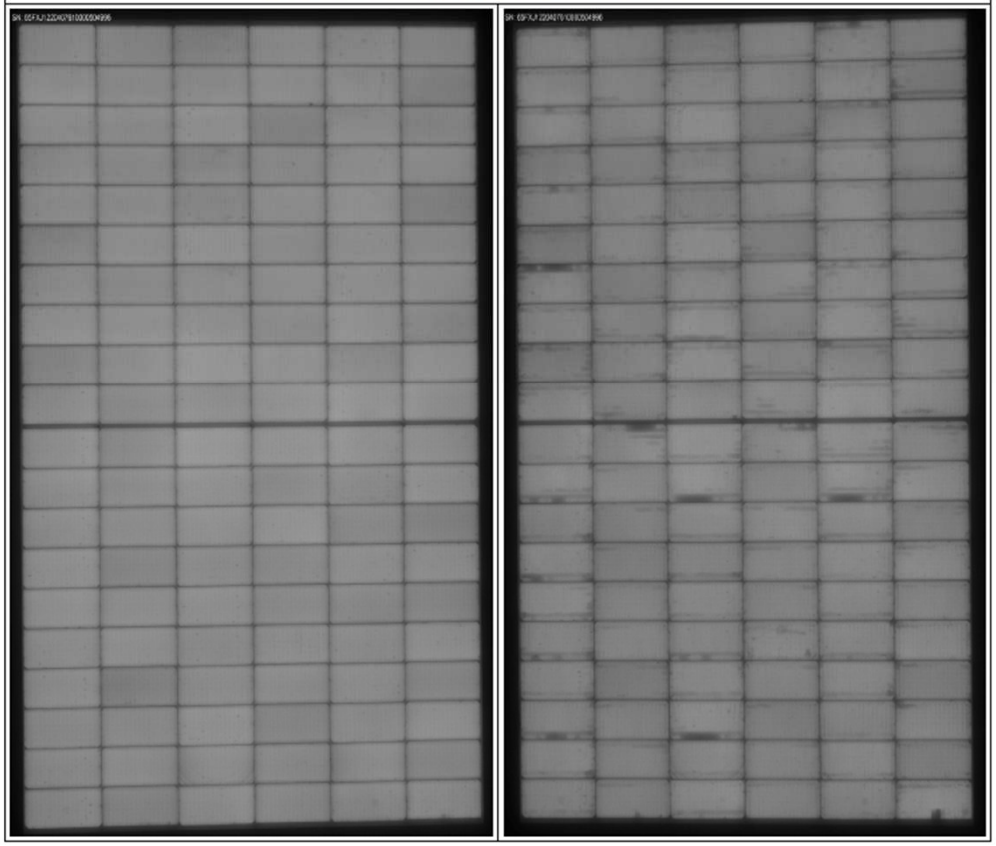
4.6. MODULE SELECTION

	OR
Monofacial VS Bifacial	OR
	OR
	OR
	OR
TOPCon, HJT, PERC	OR

?

JINKO N-TYPE. POMBAL (OASIS)

Vor HF10 Test	*Nach HF10 Test*
65FXJ1220407610000504996 / 2022-1473-072	



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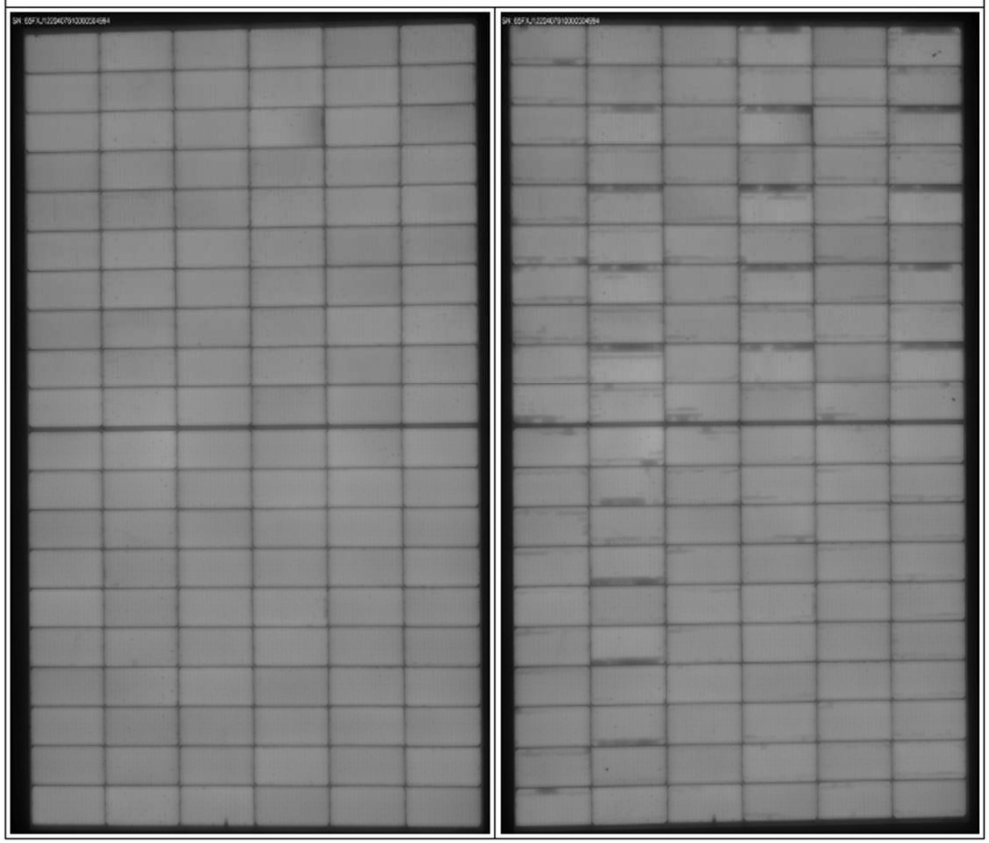
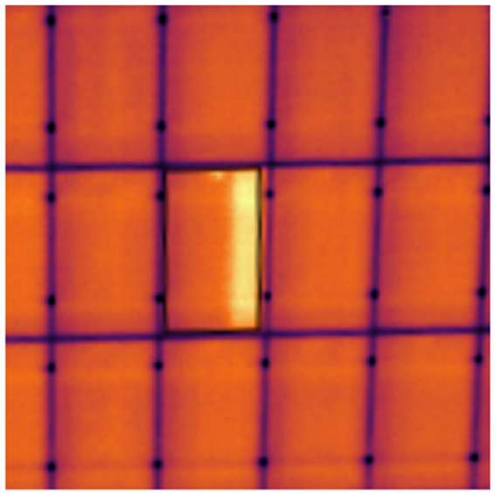
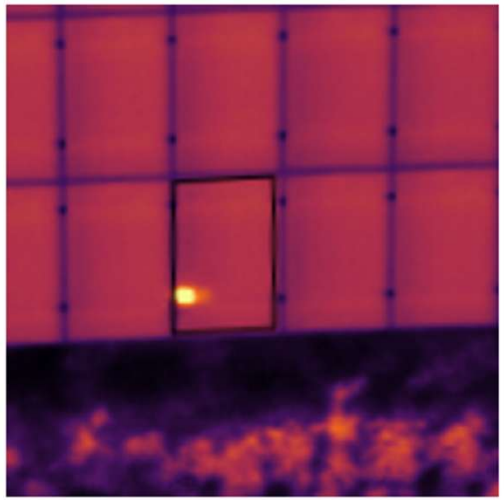
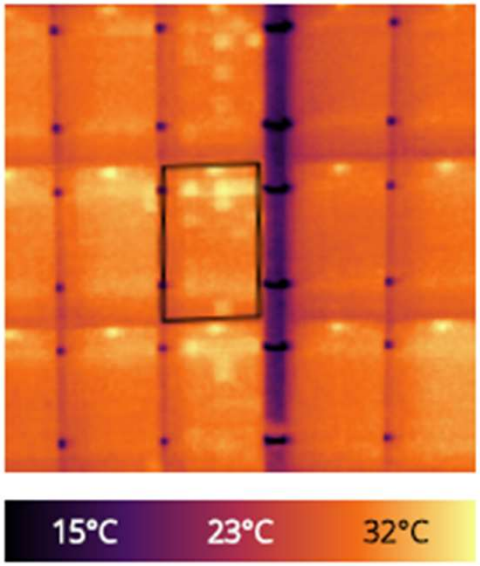


IR MEASUREMENT EXAMPLES







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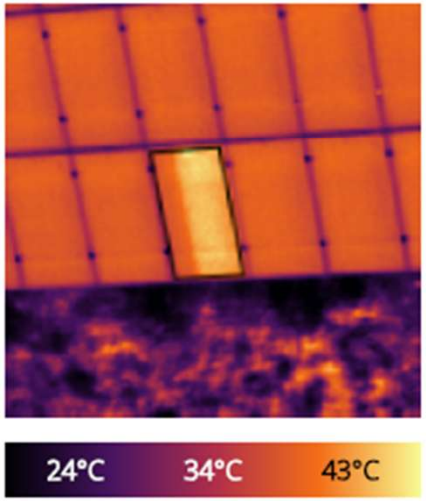

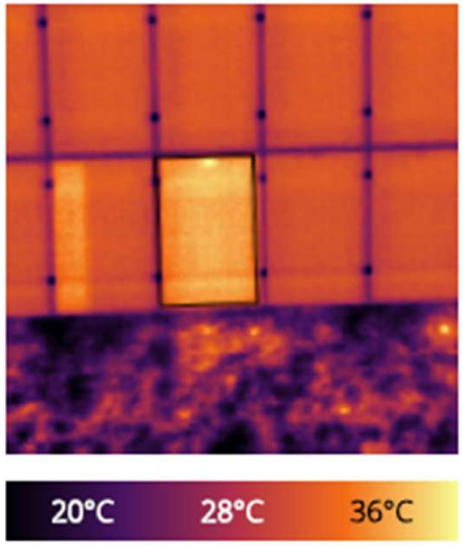

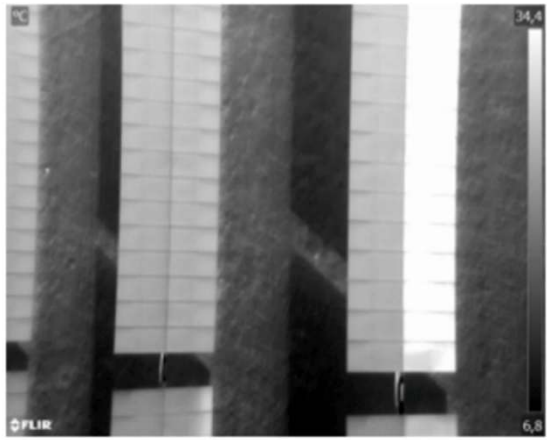

Don
Benito

Thank you!

Dr. Eduardo Forniés
eduardo.fornies@sonnedix.com
sonnedix.com